Euclidean Condensate Theory: A Narrative Companion

From the Φ-Medium to Spacetime, Gravity,
Quantum Mechanics, and Observational Predictions

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Abstract

This article is a comprehensive narrative companion to the technical preprint *"Euclidean Condensate Theory (ECT): Emergence of Spacetime, Quantum Mechanics, and Gravity from Spontaneous $O(4)$ Symmetry Breaking."* ECT is presented as a single condensate-based framework in which Lorentzian spacetime, quantum structure, gravity, and the phenomenology of the cosmological and galactic dark sector emerge from spontaneous symmetry breaking of one scalar condensate on a four-dimensional Euclidean manifold. Time, the causal cone, the effective metric, and the split between quantum and classical dynamics arise only on the ordered branch, rather than being independent postulates.

The companion traces this single architecture across all sectors: the emergence of Lorentzian signature and the hierarchy of effective equations (Klein–Gordon, Schrödinger, Dirac); three characteristic regimes (Planck, electroweak, galactic) organised by one underlying condensate architecture; the S_0 action-scale route and its Level B matching to $\hbar$, together with the arrow of time; a structural route toward the Standard Model gauge content and particle spectrum; generalised Einstein equations with an orientation-stress tensor; a retained five-probe cosmological consistency band combining the Hubble tension, JWST early-galaxy observations, cosmic chronometers, growth of structure, and the Integrated Sachs–Wolfe effect; a reinterpretation of the galactic dark sector with a single-parameter fit of 165 SPARC rotation curves and the baryonic Tully–Fisher slope of four as an algebraic consequence of the ordered-branch closure; a reinterpretation of the cosmological-constant problem in which the direct vacuum-baseline catastrophe is blocked in the leading ordered-branch sector (Level A) while the observed value of ρ_Λ remains a separate Level C infrared question; the Principle of Euclidean Stationarity and a conditional Born-rule route; decoherence and the quantum–classical boundary; a structural perspective on gravity-induced decoherence and entanglement, mediator taxonomy, and the islands/quantum-extremal-surface comparison in black-hole physics; and an explicit list of falsifiers.

Derived results, effective descriptions conditional on closure assumptions, and explicitly open problems are kept separate: every result is classified by its rigour level (A/B/C/Open), as in the technical preprint. The text uses primarily narrative explanations with only the essential formulas retained; no intermediate mathematical derivations are included, so the logical flow, physical reasoning, and architectural scope of ECT can be read without working through the full technical apparatus.

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Companion to: *"Euclidean Condensate Theory (ECT): Emergence of Spacetime, Quantum Mechanics, and Gravity from Spontaneous $O(4)$ Symmetry Breaking"* (doi:10.5281/zenodo.18917930).

Keywords: Euclidean Condensate Theory; emergent spacetime; emergent gravity; emergent time; spontaneous symmetry breaking; $O(4) \to O(3)$; ordered condensate; quantum foundations; decoherence; galaxy rotation curves; baryonic Tully–Fisher relation; Hubble tension; dark energy; dark matter phenomenology; cosmological constant problem; cosmological-constant direct-channel blocking; emergent unimodular gravity; vacuum-offset decoupling; IR dark-energy scaling; pseudo-Goldstone dark energy; black-hole information problem; Principle of Euclidean Stationarity; fine-structure constant; lensing–growth sector; gravitational slip; Pauli exclusion; holographic entropy bound; area-law scaling; entanglement entropy; Born rule; neutrino masses; seesaw mechanism; neutrinoless double-beta decay; vacuum stability; false vacuum; Coleman vacuum decay; electroweak metastability; RG running; Hopf-fibered phase–orientation locking; layered electroweak programme; Big Bang singularity; pre-Lorentzian regime; cosmological ordering transition; retained-band cosmological analysis; five-probe cross-consistency; uniform-ε diagnostic layer; cosmic chronometers; Integrated Sachs–Wolfe effect; $f\sigma_8$ redshift-space distortions; JWST age-viability; Planck–SH0ES tension; gravity-induced entanglement; BMV experiment; post-quantum classical gravity; hybrid-consistency problem; reduced-state architecture; two-level ontology; mediator taxonomy; BMS asymptotic symmetries; gravitational memory; soft gravitons; islands formula; quantum extremal surface; SU(3) colour completion; postulate P7; internal colour module; reverse-analysis closure layer; obstruction theorems; background-level triviality; strong CP reopened; anomaly cancellation; Witten–Veneziano relation; Yang–Mills mass gap (Clay problem reformulated, not solved)

Glossary of abbreviations and symbols

This companion article uses a compact set of abbreviations and symbols; the fully technical glossary is in the preprint (Tables of Symbols and Abbreviations). The items below cover the companion itself.

Abbreviations.

ECT	Euclidean Condensate Theory (this work)
SSB	Spontaneous symmetry breaking
EFT	Effective field theory
GR	General Relativity
QFT	Quantum field theory
QM	Quantum mechanics
QED	Quantum electrodynamics
QCD	Quantum chromodynamics
SM	Standard Model of particle physics
EW	Electroweak (sector)
RG	Renormalisation group
GEE	Generalised Einstein equations
PES	Principle of Euclidean Stationarity
BR1	Branch rule 1 (coherent-branch selection rule)
OP	Open problem (labelled OP-*name* in the preprint)
DP	Dynamical principle
LIV	Lorentz-invariance violation
CPT	Charge–Parity–Time (symmetry theorem)
WKB	Wentzel–Kramers–Brillouin (semi-classical approximation)
OS	Osterwalder–Schrader (axioms / reconstruction theorem)
RP	Reflection positivity
KG	Klein–Gordon (equation)
CP	Charge–Parity (symmetry)
CKM	Cabibbo–Kobayashi–Maskawa (quark-mixing matrix)
PMNS	Pontecorvo–Maki–Nakagawa–Sakata (lepton-mixing matrix)
MSW	Mikheyev–Smirnov–Wolfenstein (matter effect in neutrino oscillations)
VEV	Vacuum expectation value
UV	Ultraviolet (high-energy regime)
NLO	Next-to-leading order
PPN	Parametrised post-Newtonian (formalism)
WEP	Weak equivalence principle
BH	Black hole
GW	Gravitational wave
CMB	Cosmic microwave background
BAO	Baryon acoustic oscillations
CDM	Cold dark matter (as in ΛCDM)
DM	Dark matter
DE	Dark energy
JWST	James Webb Space Telescope
ISW	Integrated Sachs–Wolfe (effect / cross-correlation)

Abbreviations (continued).

CC	Cosmic chronometers (differential-ages $H(z)$ probe)
RSD	Redshift-space distortions
SH0ES	Supernovae and H_0 for the Equation of State (local H_0 programme)
SPARC	Spitzer Photometry and Accurate Rotation Curves (galaxy sample)
BTFR	Baryonic Tully–Fisher relation
RAR	Radial acceleration relation
MOND	Modified Newtonian dynamics
NFW	Navarro–Frenk–White (dark-matter halo profile)
FLRW	Friedmann–Lemaître–Robertson–Walker (metric)
DESI	Dark Energy Spectroscopic Instrument
LISA	Laser Interferometer Space Antenna
LIGO	Laser Interferometer Gravitational-Wave Observatory
LSST	Legacy Survey of Space and Time (Vera C. Rubin Observatory)
LHC	Large Hadron Collider
GUT	Grand unified theory
BMV	Bose–Mazumdar–Morley / Marletto–Vedral (gravity-induced entanglement proposal)
BMS	Bondi–Metzner–Sachs (asymptotic symmetry group at null infinity)
GIE	Gravity-Induced Entanglement
HPS	Hawking–Perry–Strominger (soft-hair proposal)
LOCC	Local Operations and Classical Communication
PCCG	Post-quantum Classical Gravity (Oppenheim hybrid framework)
QES	Quantum Extremal Surface

Principal symbols.

Φ	Fundamental scalar field on the 4D Euclidean manifold (the Φ-medium).
X^A, n_A	Euclidean coordinates / orientation variable selected by gradient condensation.
μ^2, λ	Bare-potential mass and self-coupling parameters ($V = -\frac{\mu^2}{2}\Phi^2 + \frac{\lambda}{4}\Phi^4$).
ϕ_0	Bare-potential minimum $\phi_0 = \sqrt{\mu^2/\lambda} \sim \bar{M}_{\rm Pl}$.
u_0	Gradient-condensate amplitude $\lvert\langle\partial_A\Phi\rangle\rvert$, GeV^2.
α, β	Anisotropic/isotropic stiffness couplings; canonical benchmark $(\alpha, \beta) = (2, 1)$.
c_*	Ordered-branch causal speed: $c_* = \sqrt{\beta/(\alpha - \beta)}$; its empirical identification with the observed vacuum light speed c is a matching step.
v_2	Electroweak scale $\approx 246\,\mathrm{GeV}$ (second condensate scale).
$\mathcal{I}_{\rm EW}$	Logarithmic measure of the electroweak hierarchy: $\mathcal{I}_{\rm EW} = \ln(\phi_0/v_2) \approx 36.8$; central reduction quantity for the OP-EW-scale programme (technical preprint).
$M_G, \bar{M}_{\rm Pl}$	Induced gravitational-stiffness scale / reduced Planck mass $\approx 2.435 \times 10^{18}\,\mathrm{GeV}$.
$G_N, \hbar, c$	Standard gravitational, quantum, and luminal constants (matching-level outputs in ECT).
$G_{\rm eff}(z)$	Position- / redshift-dependent effective gravitational coupling.
$\Theta_{\mu\nu}$	Orientation-stress tensor from n^A dynamics (generalised Einstein equations).

Principal symbols (continued).

ε	Uniform late-time effective drift parameter of the retained-band diagnostic layer; $G_{\rm eff}(z)/G_N = (1+z)^{2\varepsilon}$. Retained five-probe joint band $\varepsilon \in [0.0296, 0.0376]$ at 1σ.
$f\sigma_8(z)$, r_s	RSD growth observable / sound horizon; retained-band probes.
$t_0^{\rm ECT}$	Age of the Lorentzian branch; retained-band indicator $\sim 13.5\,$Gyr baseline-H_0 / $\sim 12.5\,$Gyr inferred-H_0.
$g_\dagger$	Critical acceleration $\propto cH_0$; galactic-rotation baseline scale.
ΛCDM	Standard cosmological model: cold dark matter + cosmological constant.
ρ_Λ	Observed late-time cosmological vacuum density $\sim (10^{-3}\,\text{eV})^4$; in ECT its natural scaling is $\sim M_{\rm Pl}^2 H_0^2$ at order-of-magnitude, with coefficient open (Level C).
n^A	Ordered direction in 4D Euclidean space (unit vector, $n^A n_A = 1$); the kinematic constraint underlying the Level A blocking of the direct vacuum-baseline channel in the cosmological-constant catastrophe.
$m_{\rm guide}$	Heuristic scalar-sector orienting mass scale for mesoscopic gravity-induced-entanglement experiments (order-of-magnitude guide only, not a first-principles entangling-channel prediction).
Δn_A	Persistent asymptotic orientation shift left behind by a bulk emission event; ECT reading of gravitational memory.

Contents

Introduction

This article is a concise narrative summary of the technical preprint *"Euclidean Condensate Theory (ECT): Emergence of Spacetime, Quantum Mechanics, and Gravity from Spontaneous $O(4)$ Symmetry Breaking"* (doi:10.5281/zenodo.18917930). ECT is a theoretical framework in which Lorentzian spacetime, quantum mechanics, and gravity emerge from spontaneous $O(4) \to O(3)$ symmetry breaking of a single scalar condensate on a four-dimensional Euclidean manifold. The present article covers the same material in a narrative form accessible to a broader audience, retaining only the essential formulas and key results. For all derivations, proofs, and technical details, the reader is referred to the full preprint.

ECT is organised into four principal Parts plus a Summary Part. Part I (Foundations) introduces the Φ-medium, its six postulates, the mechanism of gradient condensation, and the emergence of Lorentzian signature, wave equations, gauge symmetries, the arrow of time, and quantisation. Part II (Macroscopic Physics) develops emergent gravity, cosmology, galactic rotation curves, cluster lensing, the dark sector, and observational predictions. Part III (Quantum Sector) develops the coherent branch: Hilbert-space reconstruction, the Principle of Euclidean Stationarity, entanglement, tunnelling, decoherence, quantum gravity, and the black-hole information problem. Part IV (Reverse Analysis, SU(3) Completion, and Structural Constraints) is a reverse-analysis closure layer: it does not replace the forward programme of Parts I–III, but clarifies which exact-colour completions of the medium are compatible with it, formalises a reverse-engineered postulate P7 for the colour sector, and honestly reframes what is now structurally closed, sharpened, or still open. Part V (Summary) consolidates comparisons, key equations, predictions, and open problems.

Throughout this article, each result is classified by its derivation level: Level A (structural consequence of the postulates), Level B (requiring additional closure assumptions), Level C (phenomenological estimates, benchmarks, stress tests, or candidate mechanisms, matched rather than derived), or Open (stated explicitly as an unsolved problem). This classification follows the technical preprint exactly.

Derivation levels used in this article:
Level A — structural consequence of the postulates P1–P6 and BR1; no additional assumptions required.
Level B — requires stated closure assumptions, effective approximations, or matching identifications.
Level C — phenomenological estimate, benchmark, stress test, or candidate mechanism; matched to data or plausibility through closure parameters rather than derived; weaker than a Level B closure.
Open — stated explicitly as an unsolved problem or a programme target.

Layered physical picture. ECT is easiest to read as a layered theory. At the deepest level there is a four-dimensional Euclidean Φ-medium with no time, no causal cones, no particles, and no observer-independent split into space and time. At the next level the medium orders: the gradient condensate selects a direction n_A, and the Lorentzian branch appears for perturbations around that ordered background. Time, causal cones, relativistic wave equations, and the universal signal speed belong to this ordered branch.

At the next level, particles, fields, observers, and measuring devices are stable reduced patterns inside the ordered branch. A particle is not a tiny classical bead placed into spacetime; it is a stable excitation pattern of the condensate. A quantum state is not an extra substance floating above the medium; it is a reconstructed coherent description of a reduced sector. A horizon is real for ordered-branch observers, but it is not a primitive boundary of the Euclidean medium.

Many apparent paradoxes arise when a higher-layer concept is treated as fundamental. ECT's organising claim is that these higher-layer structures are physically real and operationally meaningful, but not primitive. The primitive object is the Euclidean condensate; ordinary spacetime physics is the effective language of its ordered branch.

What is real at each level?
The Euclidean Φ-medium is primitive. The Lorentzian branch is emergent but physically real for ordered-branch observers. Particles and fields are stable reduced excitations of that branch. Quantum states are reconstructed coherent descriptions of reduced sectors. Classical events are decohered records. Horizons, thermality, and cosmic expansion are real within the ordered branch, but they are not primitive features of the Euclidean substrate.

Part I

Foundations: From the Φ-Medium to Emergent Laws

1. Motivation: why question the foundations?

Modern physics rests on two extraordinarily successful pillars: General Relativity, which describes gravity as the curvature of a smooth four-dimensional spacetime, and quantum field theory, which describes the electromagnetic, weak, and strong interactions as the exchange of quanta on a fixed spacetime background. Despite their remarkable empirical success, these two frameworks remain fundamentally incompatible. Quantising gravity in the standard way produces uncontrollable infinities; treating spacetime classically while matter is quantum leads to conceptual paradoxes.

Both theories share one curious and rarely questioned assumption: the distinction between space and time is *fundamental*. The spacetime interval

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2 \tag{1}$$

contains one coordinate with a minus sign and three with a plus. This Lorentzian signature $(-,+,+,+)$ is the origin of causality, light cones, and the fact that time flows irreversibly while space does not. But *why* does this minus sign exist?

Neither General Relativity nor the Standard Model answers this question. They assume it. String theory inherits it. Loop quantum gravity inherits it. Even causal set theory, which tries to build spacetime from scratch, puts the causal structure in by hand. The minus sign is one of the most rarely questioned assumptions in physics.

There is a deep theoretical reason to suspect that the minus sign is not fundamental. The Euclidean signature $(+,+,+,+)$ possesses a higher symmetry — the rotation group $O(4)$, with $4 \times 3/2 = 6$ independent rotation generators — compared to the Lorentzian $O(3,1)$, which has only three rotations and three boosts. Throughout physics, the fundamental laws have *more* symmetry than the observed states: a ferromagnet's Hamiltonian is rotationally symmetric, but the magnetised state picks a direction; the Higgs potential is gauge-symmetric, but the vacuum selects a specific phase. It is natural to ask whether spacetime itself follows the same pattern: a higher-symmetry Euclidean phase, spontaneously broken to produce the Lorentzian world we inhabit.

Furthermore, the path-integral formulation of quantum mechanics is naturally connected to a Euclidean description through the analytic continuation $t \to -i\tau$. Many of the most powerful computational tools in quantum field theory — instantons, lattice gauge theory, non-perturbative renormalisation — operate in Euclidean space. This suggests that the Euclidean description may be more fundamental than a mere computational trick.

Euclidean Condensate Theory (ECT) explores precisely this possibility. It proposes that the deepest level of reality is Euclidean, and that the Lorentzian spacetime we observe is an emergent property of an ordered phase of a more symmetric vacuum. If correct, this single idea has sweeping consequences: the Lorentzian metric, the universal causal cone, the hierarchy of effective wave equations, gravitational response, quantum kinematics, thermodynamic irreversibility, the observed constants, and particle-sector structures are no longer independent primitive inputs. Some are derived as strict structural consequences; others are reconstructed, matched, or treated as explicit Level B/Open closure targets of the Φ-condensate programme.

2. The Φ-medium: what it is and what it is not

At the foundation of ECT lies a single real scalar field Φ, defined on a four-dimensional Euclidean manifold $\mathcal{M}^4$. This manifold has no built-in time: all four coordinates $X^A = (x, y, z, w)$ enter with the same positive-definite Euclidean metric δ_{AB}. There is no preferred direction, no causal structure, no light cone.

The field Φ is the sole fundamental degree of freedom. Its dynamics is governed by an $O(4)$-invariant action with a self-interaction potential — a double-well potential familiar from the Higgs mechanism and superfluidity. In many ways Φ is analogous to the order parameter of a superfluid or ferromagnet, but operating at a much deeper level: it describes the medium from which spacetime itself is made.

It is essential to understand what Φ is *not*. It is not a quantum field defined on a pre-existing spacetime — that would presuppose the very thing ECT aims to derive. It is not a metric or gravitational field; those will emerge later as collective properties. It is not an inflaton, Higgs field, or dark matter particle, though some of its modes will play roles analogous to these at specific energy scales. And it is not a "hidden variable" in the quantum-mechanical sense. Φ is a *pre-spacetime, pre-quantum, pre-gravitational* entity.

A subtle but critical point: the "physics" of Φ at the fundamental Euclidean level cannot be described using energy, force, or potential in their usual sense. Those concepts presuppose Lorentzian time, which does not yet exist. The Euclidean action $S_E[\Phi]$ is a functional of field configurations on the four-dimensional manifold — not a time integral of a Lagrangian. Importing energy language into the pre-time Euclidean domain is a common source of confusion and must be avoided.

The Φ-medium has specific ontological properties, explicitly stated in the technical preprint (S1–S10): it is continuous and differentiable (S1); it is scalar — no internal spin or index structure at the fundamental level (S2); it fills the entire four-dimensional Euclidean space (S3); its dynamics is purely relational — no external clock, no preferred reference frame (S4); it admits ordered and disordered configurations (S5); no single configuration is privileged a priori (S6); it supports both smooth and singular (defect) configurations (S7); pointwise internal coordinates are descriptive redundancies, not observables (S8); ordering need not be globally uniform (S9); all topological sectors compatible with the vacuum manifold are admissible (S10). These properties characterise a specific type of physical medium whose mathematical formalisation leads directly to the postulates.

3. The six postulates: formalising the Φ-medium

The postulates of ECT formalise the ontological properties of the Φ-medium described above. Each postulate captures a specific physical property and translates it into a precise mathematical statement.

P1 — Four-dimensional Euclidean arena (formalises S1, S3). The fundamental arena is a connected four-dimensional Euclidean manifold $\mathcal{M}^4$ with metric δ_{AB}. All four directions and all points are initially equivalent. There is no distinguished time coordinate at the fundamental level. This is the maximal symmetry of a four-dimensional space with positive-definite metric.

P2 — $O(4)$-invariant action (formalises the symmetry of S3, S4). The dynamics respects the full rotational symmetry of the four-dimensional Euclidean space: $S_E[\Phi] = S_E[\Lambda \cdot \Phi]$ for all $\Lambda \in O(4)$. This is the mathematical expression of the equivalence of all four directions in the unbroken phase.

P3 — Minimal scalar microdynamics (formalises S1, S2). The fundamental field is a single real scalar $\Phi(X)$ with a minimal local action:

$$S_E[\Phi] = \int d^4X \left\{ \frac{1}{2}(\partial_A \Phi)^2 + V(\Phi) \right\}, \qquad V(\Phi) = -\frac{\mu^2}{2}\Phi^2 + \frac{\lambda}{4}\Phi^4, \quad \mu^2 > 0. \tag{2}$$

The double-well potential has two minima at $\pm\phi_0$, where $\phi_0 = \sqrt{\mu^2/\lambda}$. Crucially, the orientation variable n_A does not appear in this bare action — it arises only after vacuum ordering.

P4 — Ordered vacuum branch (formalises S5, S9). The physically relevant vacuum belongs to an ordered branch characterised by a nonzero *gradient condensate*: $\langle \partial_A \Phi \rangle = u_0 \, \delta_{Aw}$, $u_0 > 0$. This spontaneously breaks $O(4) \to O(3)$, selecting a preferred Euclidean direction w. The direction is not imposed externally but selected spontaneously — just as a ferromagnet selects a magnetisation direction below the Curie temperature.

P5 — Medium character of the ordered branch (formalises S8). Pointwise internal coordinates of the effective order parameter — specifically the compact phase $\theta(X)$ and the local orientation $n_A(X)$ — are descriptive redundancies rather than independently observable degrees of freedom. Physical observables depend only on *gradient* and *holonomy* (loop-integral) data.

This is directly analogous to the physics of known media: in a superfluid, the absolute phase at a single point is not observable — only phase gradients and circulation integrals carry physical information. In a nematic liquid crystal, the pointwise director angle is unobservable — only director gradients and topological defects are physical. P5 asserts that the Φ-medium has this same continuum-medium character.

The consequence of P5 is profound: since pointwise internal coordinates are redundancies, local redefinitions of $\theta(X)$ and $n_A(X)$ must be treated as local gauge redundancies. This requires compensating gauge connections and leads to the emergence of gauge symmetries (Section 11.1).

P6 — Multiplicity of configurations and non-uniform ordering (formalises S6, S9, S10). No single configuration is singled out a priori. Ordered configurations need not be globally uniform: spatially varying condensate backgrounds, domain structures, and transition regions are all admissible.

In addition to these six postulates, the theory identifies a derived structural rule:

BR1 — Smooth coherent sector (derived from P3 + P4). In the smooth infrared sector of the ordered branch, the effective order parameter admits the polar form $\Phi_{\text{eff}} = \rho\, e^{i\theta}$, and single-valuedness of the phase on closed loops gives the winding condition: $\oint \partial_A\theta\, dX^A = 2\pi n$, $n \in \mathbb{Z}$. This is not a separate postulate but a *topological consequence* of smooth ordering. It introduces the pre-quantum action scale S_0 and the loop-sector structure that will later give rise to quantum behaviour (Section 9).

What is *not* postulated deserves emphasis. ECT does not postulate: external time; Lorentzian signature; a Hilbert space; the Born rule; the Schrödinger equation; Newton's gravitational equation; a rigid fundamental cosmological constant; a particle-DM carrier for the galactic mass-discrepancy channel; or the speed of light as a fundamental constant. These entries do not all have the same logical status. External time and Lorentzian signature are replaced by the ordered-branch construction; the wave-equation hierarchy and Hilbert-space route are structurally reconstructed; Newtonian gravity and the galactic sector arise through effective gravitational closures; the Born rule is a conditional reconstruction target rather than a completed theorem; and the dark-sector entries are not derived as new substances, but reinterpreted as ordered-branch condensate-response phenomena.

4. Spontaneous symmetry breaking and the emergence of time

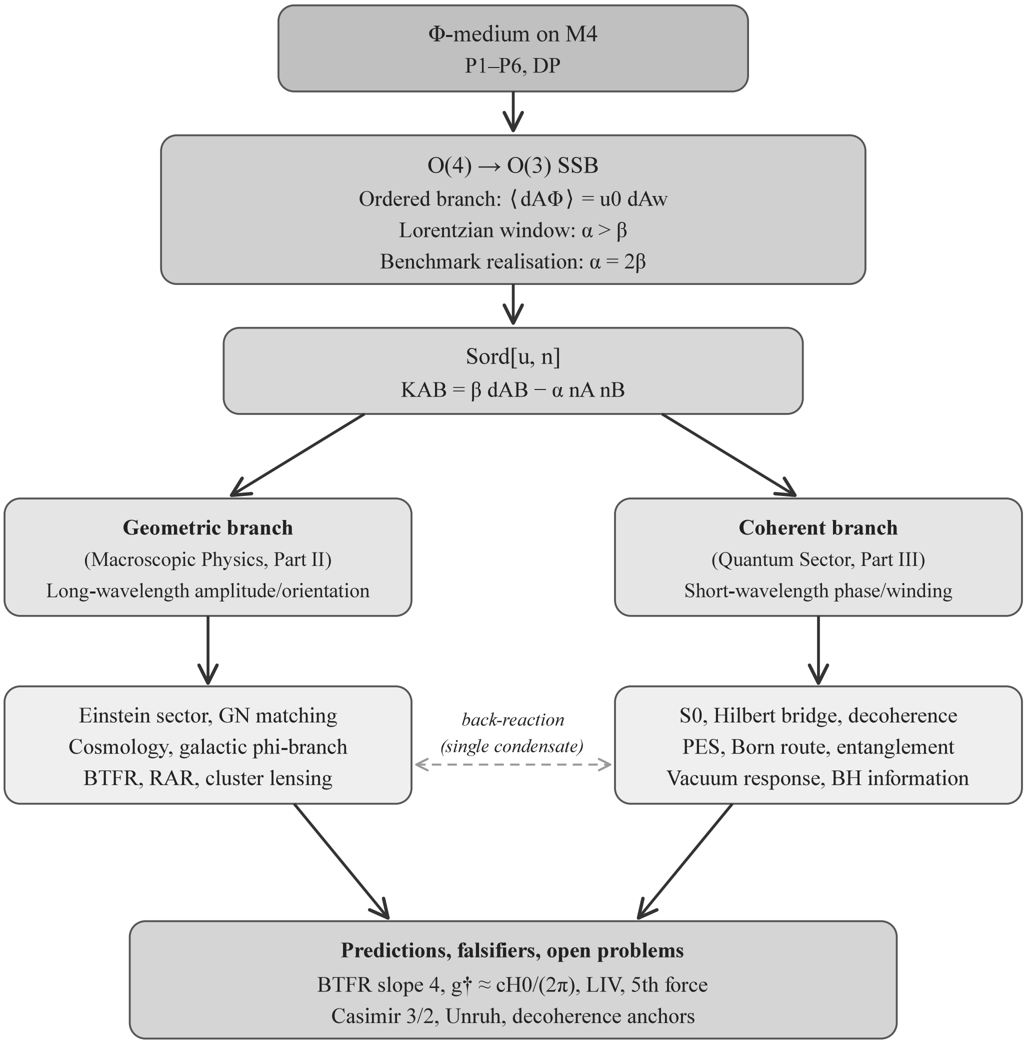


Figure 1: Architectural overview of ECT. A single Euclidean Φ-field undergoes $O(4) \to O(3)$ symmetry breaking through gradient condensation. Two complementary branches emerge: the geometric branch (gravity, cosmology, galactic dynamics) and the coherent branch (quantum mechanics, decoherence, vacuum effects). The major sectors addressed in this companion are placed within this single architecture. Level A applies to the branching architecture itself; individual arrows and sector-specific closures carry their own status labels in the preprint.

Spontaneous symmetry breaking (SSB) is one of the most powerful organising principles in physics. A spherical magnet has no preferred direction, until it cools below the Curie temperature and all spins align. A perfectly round ball on the tip of a Mexican hat has no preferred position, until it rolls down and selects one valley. In both cases, the underlying laws are symmetric, but the realised state is not.

In ECT, the Φ-medium undergoes an analogous transition: the $O(4)$ symmetry of the four-dimensional Euclidean arena is broken to $O(3)$ (Fig. 1). Three directions remain equivalent (they become "space"); one (w) is singled out (it becomes the seed of "time").

This transition is not merely a technical symmetry breaking. It is the event that makes physics in the familiar sense possible at all. Before ordering, the Euclidean condensate distinguishes no time, no causal order, no preferred class of propagating excitations. After ordering, all of the following arise simultaneously: the causal cone, Lorentzian kinematics, the excitation spectrum, and the decoherence hierarchy. Time is not added to an already prepared physics; it arises together with it as a property of the ordered phase. In this sense the $O(4) \to O(3)$ transition plays the role of a "crystallisation of the physical world."

The mechanism behind this transition is *gradient condensation.* In the ordered phase, not the field Φ itself but its *gradient* $\partial_A \Phi$ acquires a nonzero expectation value pointing in a specific direction: $\langle \partial_A \Phi \rangle = u_0 \, \delta_{Aw}$. This distinction is crucial: a nonzero $\langle \Phi \rangle$ would be $O(4)$-invariant (since Φ is a scalar under rotations, a scalar VEV preserves all rotational symmetries) and would produce no symmetry breaking. The gradient $\partial_A \Phi$, by contrast, transforms as a *vector* under $O(4)$. Its nonzero expectation value breaks the symmetry by selecting a direction — just as the magnetisation $\langle \mathbf{M} \rangle$ in a ferromagnet is a vector that picks a direction and breaks $O(3) \to O(2)$.

The gradient field can be decomposed into an amplitude and a direction:

$$\partial_A \Phi = u \cdot n_A, \qquad u = |\partial \Phi| \geq 0, \qquad n^A n_A = 1. \tag{3}$$

In the ordered vacuum, $\langle u \rangle = u_0$ and $\langle n^A \rangle = \delta^A{}_w$. This decomposition is the key to the entire theory. Each component plays a distinct physical role:

The **amplitude mode** (u; fluctuations $\sigma = u - u_0$): massive, short-ranged ($\xi_{\text{cond}} \sim m_\sigma^{-1} \sim \ell_{\text{Pl}}$), connected to gravitational stiffness and the Planck scale. It plays a role analogous to the Higgs boson, but at the Planck scale rather than the electroweak scale.

The **orientation modes** (δn^A): the symmetry breaking $O(4)/O(3) \simeq S^3$ naively admits three Goldstone directions, but integrability of the scalar-only basis reduces the propagating soft content to a single surviving longitudinal scalar mode χ (see below). The induced effective-metric sector provides a route to a massless spin-2 graviton in the completed linear Fierz–Pauli closure (*B/Open*); exact TT projection, ghost-freedom, and non-perturbative masslessness remain verification tasks.

The **phase mode** (θ, in the coherent sector $\Phi_{\text{eff}} = \rho \, e^{i\theta}$): connected to gauge fields and quantum dynamics.

The dynamical origin of gradient condensation from the bare action P3 is an open theoretical programme (OP-grad in the preprint). A generalised kinetic action $P(X)$, $X = (\partial_A \Phi)^2$, with a minimum at $X = u_*^2 > 0$, provides the natural dynamical setting for such condensation — analogous to how a superfluid develops nonzero superflow when the free energy favours a finite condensate velocity. The theory establishes that *given* the ordered branch (P4), a rich and self-consistent effective physics follows. Recent sharpening in the preprint decomposes this programme into three recorded subproblems (OP3.1a–c) and localises the remaining OP3.1a gap to a single constraint-relaxation/coarse-graining bridge: the candidate shortcut bridges — the EFT-level orientation route in the strict integrable regime and the potential-sector route — have been computed and excluded as completions.

An important structural result (the "$P(X)$ no-orientation theorem", proved in the preprint) shows that a purely kinetic action $P(X)$ can generate gradient condensation (nonzero u_0) but *cannot* generate orientation stiffness — the rigidity that resists bending of the condensate direction. Orientation stiffness must come from higher-order terms or from the NLO sector of the EFT, and it is this stiffness that ultimately gives rise to gravity.

5. How the Lorentzian signature emerges

Once the condensate has selected a direction, the physics of small excitations around the ordered background changes profoundly. The key quantity is the effective kinetic tensor K^{AB} governing fluctuations. The technical preprint proves a uniqueness theorem: the most general quadratic action for fluctuations φ around the ordered branch, compatible with the residual $O(3)$ symmetry, takes the form (*Level A*):

$$S_{\text{eff}} = \int d^4X \left[\frac{\beta}{2}(\partial_A \varphi)^2 - \frac{\alpha}{2}(n^A \partial_A \varphi)^2 + \dots \right], \tag{4}$$

so that $K^{AB} = \beta\, \delta^{AB} - \alpha\, n^A n^B$.

This tensor has eigenvalues β (triply degenerate, along the three directions perpendicular to n^A) and $\beta - \alpha$ (along n^A). An eigenvalue analysis reveals three distinct regimes:

If $\alpha < \beta$: All eigenvalues are positive. The effective metric remains Euclidean. No time, no causal structure, no wave propagation.

If $\alpha = \beta$: One eigenvalue vanishes. The system is at the Euclidean–Lorentzian boundary, with degenerate, Galilean-like kinematics.

If $\alpha > \beta$ (the "Lorentzian window"): One eigenvalue is negative. The effective metric acquires a Lorentzian signature $(-, +, +, +)$. Perturbations propagate as waves with a finite speed, and a causal cone structure emerges.

ECT operates in the Lorentzian window. In the canonical benchmark $(\alpha, \beta) = (2, 1)$, the kinetic tensor reduces to $K^{AB} = \delta^{AB} - 2\, n^A n^B$, with eigenvalues $(+1, +1, +1, -1)$ — exactly the Lorentzian signature.

The effective propagation speed is:

$$c_* = \sqrt{\frac{\beta}{\alpha - \beta}}, \tag{5}$$

This is the ordered-branch causal speed. Its universality within the native low-energy sectors is structural, while the empirical identification $c_* = c$ is a matching step (*Level A for the one-cone structure; Level B for the empirical matching*).

This is the central result of Part I: the minus sign in the spacetime interval (1) **is not assumed but derived.** It appears because the gradient condensate breaks $O(4)$ in a way that produces an anisotropic kinetic tensor with one negative eigenvalue. The derivation status is *Level A*: it follows from the uniqueness of the kinetic tensor (proved as a theorem) combined with P4.

Comparison with other theories. In Einstein–aether theory (Jacobson, 2004), the anisotropy is introduced by a *postulated* vector field — not derived from a scalar. In Hořava–Lifshitz gravity, Lorentz invariance is broken *explicitly* at the fundamental level. In Causal Dynamical Triangulations (CDT), the Lorentzian structure is imposed as a constraint on the lattice. In the Hartle–Hawking no-boundary proposal, the Euclidean arena is a computational tool, not a fundamental entity. In ECT alone, the Lorentzian signature is *spontaneously generated* from a higher-symmetry Euclidean substrate, and the result is a theorem, not a phenomenological assumption.

6. The equation hierarchy: from Euclidean boundary problem to Schrödinger

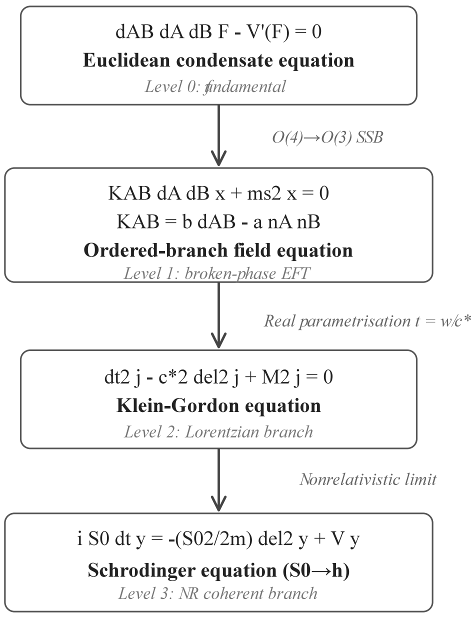


Figure 2: The ECT equation hierarchy. The fundamental Euclidean field equation (a four-dimensional boundary-value problem) generates the Klein–Gordon equation (after $t = w/c_*$), which in turn yields the Schrödinger equation (non-relativistic limit). Each level is a *descendant* of the previous one, not an independent postulate. Standard quantum mechanics occupies the lowest rung. (*Hierarchy structure: Level A; specific forms: Level A/B.*)

The emergence of Lorentzian signature has a direct consequence for the equations of motion (Fig. 2). The Euclidean field equation $\delta^{AB}\partial_A\partial_B\Phi - V'(\Phi) = 0$ is a four-dimensional *boundary-value problem*: its solutions are determined by data on the *entire boundary* of the Euclidean domain, not by initial data propagated forward in time.

Once the ordered branch is established and we introduce the identification $t = w/c_*$, the same equation, restricted to the Lorentzian window, becomes the *Klein–Gordon equation*: $\partial_t^2\varphi - c_*^2\nabla^2\varphi + m_{\text{eff}}^2\varphi = 0$ (*Level A/B*).

In the non-relativistic limit ($E \approx mc^2 + E_{\text{kin}}$, $|E_{\text{kin}}| \ll mc^2$), the Klein–Gordon equation reduces to the *Schrödinger equation*: $i\hbar\,\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi$ (*Level A/B*).

This hierarchy is not merely a formal observation. It encodes a physical claim: the Schrödinger-type coherent equation, which standard quantum mechanics takes as its starting axiom, emerges as the *narrowest, lowest-energy descendant* of a more fundamental Euclidean dynamics. The wave equation is *reconstructed as an effective low-energy descendant*, not postulated; the identification of the emergent ordered-branch equation with the exact textbook Schrödinger equation of non-relativistic quantum mechanics is a Level-B matching step rather than a fully closed Level-A theorem.

The distinction between the Cauchy (initial-value) problem of standard QM and the boundary-value problem of ECT is one of the deepest structural differences between the two frameworks. In standard QM, initial data on one time-slice determine the future uniquely. In ECT, the fundamental equation constrains the field configuration on the *entire* four-dimensional domain. This has far-reaching consequences for entanglement, delayed-choice experiments, and the influence of future boundary conditions on present configurations (see Part III).

7. Causal cone inheritance and what GW170817 tells us

A critical question for any theory claiming to derive Lorentzian spacetime is: why is there a *universal* speed of light? In the real world, photons, gravitons, and all massless particles share the same causal cone to extraordinary precision. A decisive test of any emergent-spacetime framework is whether different low-energy sectors inherit a single causal structure rather than sector-dependent propagation cones—which would destroy universal relativistic kinematics. In ECT the universality of the causal cone is not postulated but inherited from the single ordered condensate background.

In August 2017, the LIGO/Virgo gravitational-wave detectors observed the merger of two neutron stars — an event designated **GW170817**. Just 1.7 seconds later (after a journey of 130 million light-years), the Fermi and INTEGRAL satellites detected a gamma-ray burst from the same source. This simultaneous arrival established that the speed of gravitational waves and the speed of light agree to better than one part in 10^{15}: $|c_{\mathrm{GW}}/c - 1| < 6 \times 10^{-15}$.

In many alternative gravity theories (Einstein–aether, Horndeski scalar-tensor models, massive gravity), such universality requires fine-tuning of multiple parameters. In the Standard Model Extension (SME), each particle sector carries an independent set of Lorentz-violation coefficients.

ECT explains the one-cone structure as a structural result of the ordered branch. The argument is straightforward: (i) the fundamental field is one, Φ; (ii) the ordered-branch kinetic tensor is one, K^{AB}; (iii) the native low-energy sectors retained in the effective closure are built on the same K^{AB}; (iv) no second independent rank-2 tensor is available in the retained low-energy EFT to assign arbitrary unsuppressed cone speeds to different native sectors.

The result is **cross-sector cone universality**:

$$c_{\mathrm{phase}} = c_\gamma = c_{\mathrm{GW}} = c_* \qquad \text{within the retained native low-energy closure.} \tag{6}$$

The GW170817 observation is therefore a structural consistency check of ECT: photons and gravitational waves realise the same ordered-branch cone to very high precision. The existence of the single cone is the structural result; the empirical identification of that cone with the measured vacuum light speed is the matching step.

ECT does admit Planck-suppressed higher-order corrections that can produce tiny sector-dependent speed differences at extreme energies. But arbitrary, unsuppressed, low-energy speed differences between sectors are structurally forbidden. If such differences were ever observed, ECT's one-cone universality would be falsified.

8. Three fundamental scales from one field

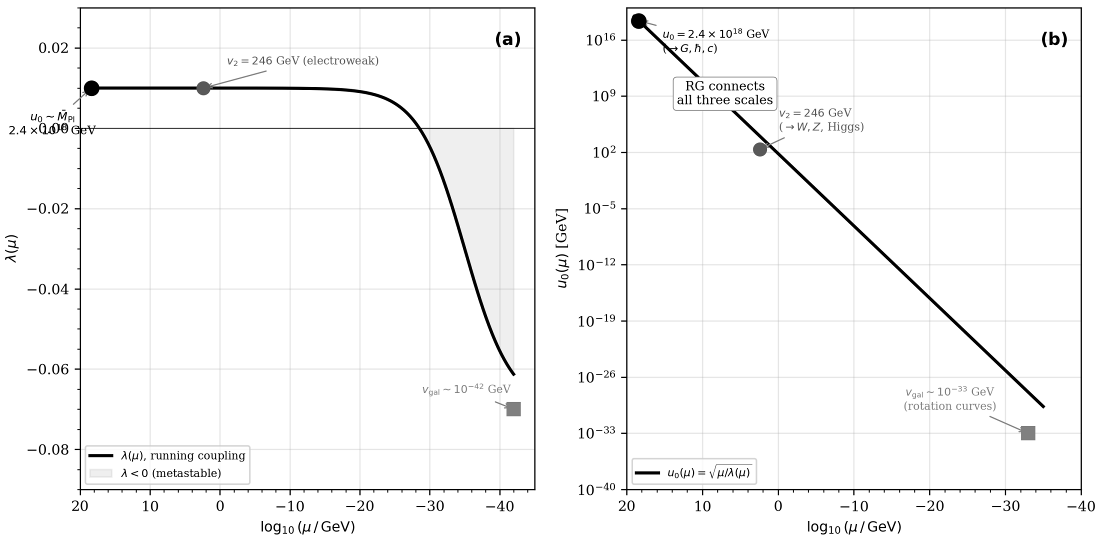


Figure 3: Three characteristic ECT scale regimes. A single condensate architecture organises the UV Planck/gravitational scale ($\phi_0 \sim \bar{M}_{\rm Pl}$), the matched electroweak scale ($v_2 \simeq 246\,\mathrm{GeV}$), and the galactic IR branch-variation regime controlled by the acceleration scale $g^\dagger \sim cH_0/(2\pi)$ and associated kpc-scale lengths. These are not produced by one and the same mechanism: the $\phi_0 \to v_2$ hierarchy is currently reduced to a non-polynomial scale-generation problem, while the galactic scale is an IR/cosmological boundary-response scale. The figure is therefore a structural map of regimes, not a first-principles derivation of all three scales.

A notable feature of ECT is that a single scalar field organises three distinct characteristic scales, not all of the same physical dimension, governing very different regimes of physics (Fig. 3):

The **gravitational scale** $\phi_0 \sim 2.4 \times 10^{18}\,\mathrm{GeV}$ (the reduced Planck mass) arises from the symmetry-breaking amplitude of the condensate. Together with the anisotropy parameters (α, β) and the coupling λ, it enters the matching relations that tie G_N, $\hbar$, and c_* to ordered-branch condensate data (*Level B*):

$$G_N = \frac{c_*^4}{8\pi\phi_0^2}, \qquad \hbar = \frac{c_*\phi_0}{\sqrt{\lambda}}, \qquad c_* = \sqrt{\frac{\beta}{\alpha - \beta}}. \tag{7}$$

These relations do not mean that ϕ_0 alone fixes all three constants. Rather, they show that G_N, $\hbar$, and c_* are tied to the same ordered-branch condensate architecture: c_* is fixed by the kinetic anisotropy parameters (α, β), G_N by the orientation-sector stiffness scale ϕ_0 together with c_*, and $\hbar$ by the coherent loop-action matching involving ϕ_0, c_*, and λ. In standard physics, c, $\hbar$, and G_N appear as three independent empirical constants belonging to different chapters of physics: relativistic kinematics, quantum theory, and gravitation. ECT changes their logical status from unrelated inputs to linked matching data of one condensate medium, with the $\hbar$ identification remaining a Level B closure rather than a stand-alone Level A theorem.

The **electroweak scale** $v_2 \simeq 246\,\mathrm{GeV}$ is treated in the current formulation as the matched secondary scale associated with the electroweak sector. In a completed closure it would correspond to a secondary ordering/locking structure responsible for the $W^\pm$, Z and Higgs-sector mass scale, but this is not yet derived from P1–P6. The enormous hierarchy $\phi_0/v_2 \sim 10^{16}$ is logarithmically equivalent to

$$\mathcal{I}_{\rm EW} \equiv \ln(\phi_0/v_2) \approx 36.8,$$

of the same order as the analogous QCD logarithm $\ln(M_{\rm Pl}/\Lambda_{\rm QCD}) \approx 44$. The technical preprint therefore reduces OP-EW-scale to a non-polynomial scale-generation problem in a phase–orientation alignment/locking sector. Candidate templates include asymptotic-freedom-like running, walking or near-conformal dynamics, Miransky/BKT critical scaling, and topological/Hopf-like routes. These scenarios require modest dimensionless input rather than a direct fine-tuned coefficient of order 10^{-16} or 10^{-32}, but none has been derived from P1–P6. The broader layered OP-EW programme remains open: OP-EW-scale, OP-EW-locking, OP-EW-gauge, OP-EW-naturalness, and OP-EW-matter.

The **galactic scale** $\sim$ kpc marks the transition between Newtonian and modified gravitational behaviour in galaxies, governed by the critical acceleration $g^\dagger \approx cH_0/(2\pi) \approx 1.1 \times 10^{-10}\,\mathrm{m/s^2}$ (*Level B*).

Beyond the three primary scales, the same condensate pair (ϕ_0, v_2) naturally generates two distinguished intermediate combinations that appear in the neutrino sector: the geometric-mean scale $\sqrt{\phi_0\, v_2} \approx 2.4 \times 10^{10}\,\mathrm{GeV}$ (close to the scale of thermal leptogenesis and type-I seesaw), and the inverse-ratio Weinberg-operator scale $v_2^2/\phi_0 \approx 2.5 \times 10^{-5}\,\mathrm{eV}$ (within a few orders of magnitude of the observed neutrino mass splittings). Both combinations are the two simplest dimensionally consistent products that can be built from the pair (ϕ_0, v_2). They are useful ECT scale anchors for neutrino and leptogenesis model-building, but they are not first-principles derivations of the neutrino spectrum, the Dirac/Majorana character, the PMNS matrix, or the mass ordering.

9. ECT loops, quantisation, and the origin of $\hbar$

A distinctive feature of ECT is how quantisation emerges naturally from the compact-phase structure of the condensate, without being imposed as an axiom.

In the coherent sector, the effective order parameter has the polar form $\Phi_{\rm eff} = \rho\, e^{i\theta}$, where θ is a compact (periodic) phase variable. The single-valuedness requirement (BR1) means that if we transport θ around any closed loop $\mathcal{C}$ in the Euclidean manifold, it must return to its original value:

$$\oint_{\mathcal{C}} \partial_A \theta\, dX^A = 2\pi n, \qquad n \in \mathbb{Z}. \tag{8}$$

This is not a quantum axiom — it is a *topological necessity* for any smooth, single-valued function with compact range.

The consequence is immediate: the action associated with loop configurations is quantised. The minimal nontrivial loop ($n = 1$) carries a definite action:

$$S_{\rm loop,min} = 2\pi S_0, \tag{9}$$

where the scale S_0 depends on the condensate parameters. Through the matching relation (7), S_0 is identified with $\hbar$ (*Level B*).

This means that Planck's constant is not introduced as an unrelated primitive in the ECT programme. The compact phase supplies a structural route to a distinguished action scale S_0, while the identification $S_0 = \hbar$ is a Level B matching relation. The analogy is the flux quantum

$\Phi_0 = h/(2e)$ in a superconductor and the Bohr–Sommerfeld condition $\oint p\,dq = nh$: in each case, single-valuedness of a periodic variable leads to quantisation, but the physical normalisation requires matching to the relevant low-energy sector.

The loop sectors partition the coherent branch into topologically distinct classes indexed by the integer n. Configurations in different sectors cannot be continuously deformed into each other. This is why quantum interference can arise in ECT without operator axioms: paths belonging to different winding sectors contribute to the path sum with different phase factors, producing constructive or destructive combination.

10. The arrow of time, the second law, and why time flows

In the Euclidean picture, the coordinate w is just one of four equivalent spatial directions. There is no "forward" or "backward" in w, just as there is no "forward" in x. The arrow of time is not fundamental: it is *emergent*.

ECT constructs the thermodynamic arrow in two stages:

Stage 1: directional backbone (*Level A*). In the Lorentzian branch, the effective dynamics admits a retarded causal structure: perturbations propagate forward along w inside the emergent light cone. The dissipative kernel of the influence-functional formalism is positive-semidefinite ($\Gamma_{\text{irr}} \geq 0$), meaning that off-diagonal histories are *suppressed* rather than amplified. This gives the *directionality* of decoherence.

Stage 2: entropy growth (*Level B*). Under two additional effective-closure assumptions — an ohmic spectral density and a Markovian regime — the coarse-grained entropy satisfies:

$$\frac{dS_{\text{ent}}}{dw} = N_{\text{eff}}\,\gamma \left(\frac{dq}{dw}\right)^2 \geq 0, \tag{10}$$

where N_{eff} is the effective number of environmental modes and γ is the decoherence rate per mode. The sign is anchored in the Level A retarded structure; the explicit formula requires Level B closure.

The result is a three-regime picture:

At the *cosmological* level, the ordered branch defines the cosmic arrow; the second law holds to overwhelming accuracy.

At the *macroscopic* level ($N_{\text{eff}} \gg 1$), the ratio of backward-to-forward transition probabilities is exponentially suppressed: $P_{\text{back}}/P_{\text{fwd}} = \exp(-2N_{\text{eff}}\gamma\,\Delta w \cdot (\Delta q)^2/\ell^2)$. For a mole of gas at room temperature over a nanosecond, the exponent is $\sim 3.6 \times 10^{33}$ — reverse trajectories are fantastically unlikely.

At the *quantum* level ($N_{\text{eff}} \sim 1$), the exponent becomes tiny ($\sim 10^{-15}$), and effective time-reversibility is recovered — consistent with the well-known fact that isolated quantum systems evolve reversibly.

The second law is therefore not an independent postulate in ECT. Its directional backbone is Level A; its full monotonicity is Level B. What ECT adds beyond Boltzmann is a *structural explanation* of why the initial condition is special: it is tied to ordered-branch selection (P4), not imposed ad hoc.

11. Spectrum, branching, gauge symmetries, and particles

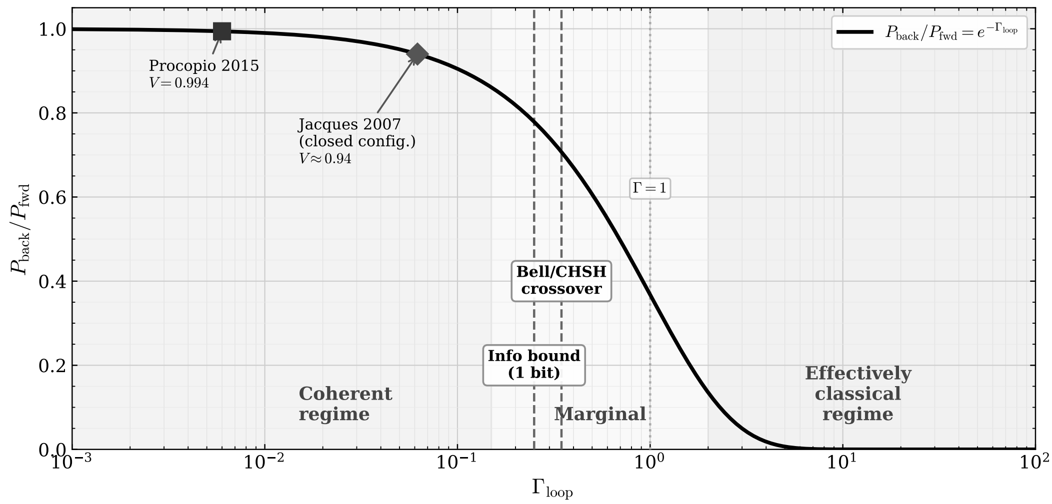


Figure 4: The quantum–classical boundary. The dimensionless decoherence parameter Γ_{loop} organises the reduced description: coherent systems lie in the regime $\Gamma_{loop} \ll 1$, while strongly locked macroscopic systems lie in the regime $\Gamma_{loop} \gg 1$. Different objects — electron, C_{60}, bacterium, dust grain, cat — occupy different positions on this scale. The distinction between coherent amplitudes and decohered reduced sectors is structural; the quantitative crossover criterion $\Gamma_{loop} \sim 1$ and the numerical placement of specific systems are Level B closure results.

The ordered condensate supports three classes of excitations, each playing a distinct physical role:

Amplitude mode (σ): Fluctuations of the gradient magnitude u around u_0. This mode is heavy and short-ranged at the matched ordered point. Its precise u-dependence belongs to the OP-grad/F7 completion problem; the current recorded closure treats the amplitude sector as Planck-scale and non-rolling on cosmological times.

Orientation modes (δn^A): The symmetry breaking $O(4)/O(3)$ naively suggests three Goldstone directions; the scalar-only integrability constraint reduces the independent soft content. The induced effective-metric sector provides a route to a massless spin-2 graviton in the completed linear Fierz–Pauli closure (*Level B/Open*), while exact TT projection, ghost-freedom, and non-perturbative masslessness remain verification tasks. The amplitude mode is too heavy ($m_\sigma \sim M_{Pl}$) for kpc-scale variations, which rules out naive scalar-field dark matter models and motivates the galactic closure used in Part II.

Phase mode (θ): In the coherent sector, phase fluctuations connect to gauge fields: the photon emerges as the gauge boson of the $U(1)$ phase symmetry (*Level A/B after P5*).

The ordered condensate organises into two complementary branches (Fig. 4): the **geometric branch** (Macroscopic Physics, long-wavelength, coarse-grained, $\Gamma_{loop} \gg 1$) and the **coherent branch** (Quantum Sector, short-wavelength, phase-coherent, $\Gamma_{loop} \ll 1$). Both originate from the same ordered condensate action $S_{ord}[u, n]$ (*Level A*). The coherent/decohered-sector distinction is structural, while the sharp crossover criterion $\Gamma_{loop} \sim 1$ is a Level B reduced-state closure estimate.

11.1. Gauge symmetries and the interaction sectors

ECT does not postulate all gauge structures as independent primitive symmetries. Instead, it derives or structurally motivates gauge-sector routes from the internal redundancy structure of the Φ-medium through P5. The compact phase fibre gives the strongest Abelian $U(1)$ route; the oriented condensate selects a chiral weak-sector route; and the colour sector requires additional internal condensate structure beyond the minimal $O(4) \to O(3)$ pattern. Thus P5 supplies the logic of gauge compensation, while the different gauge sectors have different derivational status. In all cases, since pointwise values of internal phase/orientation parameters such as θ and n_A are not observable, local redefinitions must leave physics unchanged—requiring compensating gauge connections. The key insight is that these connections are not added by hand where they appear: they are *forced* by the requirement that the effective description be independent of unphysical local choices.

Abelian gauge route (*Level A/B*): The compact coherent phase $\theta \sim \theta + 2\pi$ supplies a structural route to a principal $U(1)$ bundle over the emergent spacetime: comparing the phase at neighbouring points requires a connection. The Maxwell-type kinetic term is the leading gauge-invariant low-energy closure. This provides a strong route to an Abelian gauge sector, but the precise identification of the compact phase with pre-breaking hypercharge $U(1)_Y$, post-breaking electromagnetism $U(1)_{\text{em}}$, or an intermediate effective phase depends on the still-open electroweak completion.

Weak-sector route (*B/Open*): The residual orientation structure after $O(4) \to O(3)$ has Lie algebra $\mathfrak{so}(3) \cong \mathfrak{su}(2)$, and its spinorial lift is naturally described in $SU(2)$ language. This gives ECT a structural route toward a weak-like sector and toward parity-asymmetric/chiral couplings, but it does not by itself derive the Standard Model $SU(2)_L$ group, hypercharge assignments, the Weinberg angle, or the maximally parity-violating pattern. A reduced $O(3) \to O(2)$ analogue contains only two broken directions. The technical preprint therefore introduces a candidate upgrade: Hopf-fibered phase–orientation diagonal locking, $SU(2)_{\text{orient}} \times U(1)_\theta \to U(1)_{\text{diag}}$, whose pre-gauge vacuum manifold is S^3, matching the fixed-radius Higgs-doublet geometry at the global level. This solves the bare generator-counting deficit only at the level of candidate geometry. The locking dynamics, the identification $SU(2)_{\text{orient}} \to SU(2)_L$, the identification $U(1)_\theta \to U(1)_Y$, local gauge fields, couplings, and protection of the secondary radial mode all remain open.

Strong interaction (*Level B*): $SU(3)$ colour *cannot* arise from the minimal $O(4) \to O(3)$ pattern: the algebra $\mathfrak{su}(3)$ has dimension 8, while $\mathfrak{so}(4)$ has dimension 6, so $\mathfrak{su}(3)$ is simply too large to fit inside $\mathfrak{so}(4)$. This is not merely a technical inconvenience. It means that the colour algebra cannot arise as a direct subgroup or residual factor of the minimal $O(4) \to O(3)$ pattern. The electroweak-type $U(1) \times SU(2)$ architecture is the maximal gauge-like structure naturally suggested by the minimal condensate symmetry pattern; its full chiral local Standard-Model realisation nevertheless remains open. Any colour sector therefore requires additional internal degrees of freedom: a local complex rank-3 module with $SU(3)$-structure. The preprint further shows, at the level of anomaly constraints, that anomaly cancellation alone does not select the colour rank (it imposes only an odd number of colours), so the rank-three input must come from inherited charge data or from the open zero-mode programme. A deeper geometric candidate (G_2 internal fibre) is also identified. Full QCD derivation remains open.

Gravity (*Level A/B*): Gravity arises from orientation stiffness—the condensate's resistance to bending of n^A (Part II).

Director coupling (*Level A/B*): Beyond the four known interactions, ECT identifies a fifth interaction channel: the *director coupling* $\mu_5 \bar{\Psi}\gamma^A n_A \Psi$—the lowest-dimension fermion–condensate operator permitted by the ordered-branch symmetry. Its name derives from the analogy with the director field of nematic liquid crystals. Possible signatures include spin precession in the approximate range $\omega_{\text{dir}} \sim 10^{-10}$–$10^{-15}$ rad/s (depending on the local condensate profile) and composition-dependent effects; the $\eta \sim 10^{-15}$ window is a target sensitivity rather than a sharp prediction.

The colour asymmetry. Among these sectors, colour is the only one that cannot be accessed from the minimal $O(4) \to O(3)$ pattern alone. This qualitative asymmetry motivates the dedicated discussion of colour completion below (Section 54).

Common pre-geometric architecture. ECT places the known interaction sectors within a common pre-geometric framework, while assigning different derivational status to different sectors. Gravity arises from the ordered condensate geometry. The Abelian gauge sector has the strongest route through the compact phase fibre. The weak sector is only a candidate route at present: the orientation-sector $SU(2)$ lift and the Hopf-fibered locking geometry supply a possible pre-gauge structure, but the full chiral $SU(2)_L \times U(1)_Y$ Standard-Model realisation remains open. The colour sector requires additional internal condensate structure beyond the minimal $O(4) \to O(3)$ pattern. Together with the generic director coupling, this yields a unified condensate-based organisation of the interaction sectors. In this sense ECT is broader than conventional GUT schemes, while its notion of unification remains structural rather than simple-group theoretic.

11.2. Elementary particles

Particles as stable patterns. In this picture particles should not be imagined as tiny hard balls placed inside spacetime. They are stable excitation patterns of the ordered condensate. A photon is a gauge-sector excitation; an electron is a stable fermionic excitation with spin, charge, and exchange topology; a massive vector boson, in a completed electroweak closure, would be tied to a secondary ordering or locking scale. This pattern view explains why particles can interfere, entangle, and be created or annihilated: they are configurations of an underlying field architecture, not permanent microscopic beads.

The bosonic content of ECT is best organised hierarchically, by structural level. The description below records the bosonic sectors retained in the present formulation at successive levels of condensate structure; it should not be read as a final unique census of all possible emergent excitations. At the minimal level (Level 0: real scalar condensate with $O(4) \to O(3)$ SSB), ECT contains a spin-2 graviton route in the completed linear closure (*B/Open*), the surviving Goldstone mode χ (ultra-light condensate scalar, *Level A/B*), and the radial mode σ (heavy, *Level A/B*)—no gauge bosons. At Level 1, the compact phase fibre produces the photon ($U(1)$ gauge boson, *Level A*). At Level 2, the matched electroweak scale $v_2 \simeq 246$ GeV anchors a candidate secondary ordering/locking sector associated with $W^\pm$, Z, and Higgs-sector masses. The Hopf-locking geometry gives a possible route to the correct three broken directions, but the full local chiral gauge theory and the dynamical origin of v_2 remain open (*B/Open*). At Level 3, extended internal structure provides a route to the eight gluons and the colour sector (*B/Open*); the full colour-completion construction remains open. Fermions appear as spinorial excitations of the ordered branch; in the emergent Lorentzian regime their dynamics is captured by a reconstructed

Dirac equation, while the chiral structure of the weak sector is tied to the preferred orientation selected by the condensate (*Level A/B*). Topological sectors provide candidate $\pi_3(S^3) = \mathbb{Z}$ states; a linear benchmark mass ansatz $m_n \sim n \times 1.6\,\mathrm{GeV}/c^2$ is phenomenological (*Level C*), not derived from topology alone, and is now retained only as a legacy benchmark after the primary-stiffness objection, with a possible secondary-sector reassignment left open.

Neutrino sector. The geometrically one-chiral weak route that follows from the condensate symmetry breaking structurally selects one chiral branch of the fermion sector (A/B). This constrains the type of viable neutrino mass mechanism: a pure Planck-suppressed Majorana route undershoots the observed atmospheric neutrino scale by three orders of magnitude (B), making an embedded seesaw mechanism with a heavy sterile partner the structurally preferred completion (C). The two simplest dimensionally consistent combinations of ϕ_0 and v_2 populate the neutrino sector: the geometric-mean seesaw scale $M_R^{\mathrm{geom}} = \sqrt{\phi_0\, v_2} \approx 2.4 \times 10^{10}\,\mathrm{GeV}$ and the inverse-ratio Weinberg-operator scale $v_2^2/\phi_0 \approx 2.5 \times 10^{-5}\,\mathrm{eV}$; both are structural scale observations, not predictions (C). Standard PMNS/MSW oscillation phenomenology remains intact at leading order: the ECT preferred-direction correction is suppressed by approximately eight orders of magnitude relative to the standard oscillation Hamiltonian (B). Neutrinoless double-beta decay is the central experimental discriminant of the preferred Majorana/seesaw branch. The full flavour closure (generations, textures, PMNS angles, mass ordering) remains open.

Relative to the minimal Standard-Model field content, the present ECT closure retains an additional condensate scalar sector, represented by the ultra-light χ-mode and the heavy radial mode σ. The detailed dark-sector role of χ remains model-dependent in the current formulation.

What remains open (*Level Open*): the origin of three fermion generations; the Yukawa coupling hierarchy; the fine-structure constant $\alpha = 1/137$. Qualitative structural routes exist (e.g. generations may be related to topological sectors of S^3), but rigorous quantitative derivations do not.

Three logical classes of excitations. ECT does not place all particles on the same logical footing—a fundamental difference from standard QFT. Category (i), the *required condensate excitations* (graviton, radial mode σ, soft mode χ), are analogous to phonons in a crystal: they are inevitable consequences of an ordered medium with broken symmetry. Category (ii), the *structurally embedded Standard Model fields* (photon, $W^\pm$, Z, Higgs, fermions, gluons), are analogous to impurity atoms in a lattice: compatible with the medium but not derived from the minimal scalar action. Category (iii), the *topological sectors* (textures from $\pi_3(S^3) = \mathbb{Z}$; primary-ordering anti-predictions from $\pi_0 = \pi_1 = \pi_2 = 0$: no primary ordering domain walls, strings, or monopoles), are analogous to dislocations: determined by the topology of the primary vacuum manifold. In standard QFT all fields are equally fundamental postulates; in ECT they occupy different logical tiers.

Multi-particle description and the status of Fock space. Ordinary quantum mechanics works with a fixed number of particles N. Processes with variable N (emission, annihilation, pair creation) require quantum field theory, where the Fock space and creation/annihilation operators are additional postulates. In ECT, changing particle number is a reconfiguration of one condensate medium—analogous to vortices appearing and disappearing in a superfluid. ECT does not require Fock space as a fundamental postulate; however, an explicit emergent sector decomposition with variable excitation number remains to be constructed from condensate dynamics (Open; OP-Q11). This is a nontrivial reconstruction problem, not a foundational gap.

Multi-particle description is necessary even in the coherent branch (where no event-like Lorentzian interaction is available) for four reasons: (a) entanglement as non-factorisable common-medium correlation; (b) exchange statistics ($\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$); (c) the decoherence threshold, which depends on N_{eff}; (d) processes with variable excitation number.

Figure 5 summarises the derivation logic of Part I.

12. Conclusions of Part I

12.1. Principal results

From six postulates (P1–P6) and one derived branch rule (BR1), the following structures are obtained:

1. Lorentzian signature $(-,+,+,+)$ from $O(4) \to O(3)$ SSB (A).
2. Universal ordered-branch causal cone: native low-energy sectors share the same leading cone speed c_*; the empirical identification $c_* = c$ is a matching step (*A for cone universality; B for the empirical matching*).
3. Equation hierarchy: Euclidean $\to$ KG $\to$ Schrödinger (A/B).
4. Retarded causal/decoherence backbone of the arrow of time (A); explicit monotonic entropy-growth formula (B).
5. Second-law relation $dS_{\text{ent}}/dw \geq 0$ under the influence-functional closure (*A/B: sign from retarded structure; formula from Level B environmental closure*).
6. Branching into geometric and coherent sectors (A).
7. $U(1)$ electromagnetism from P5 (A/B).
8. $SU(2)$ weak sector (B); $SU(3)$ colour (*B/Open*).
9. Spin-2 graviton route from the orientation/effective-metric sector in the completed linear closure (*B/Open*).
10. $S_0 = \hbar$ from compact-phase winding (B).
11. Quantisation from loop-sector topology (A).
12. Fifth-force coupling $\beta_5 \sim m_f/\bar{M}_{\text{Pl}}$ (B).
13. Discrete-symmetry, anomaly, and flavour architecture of the fermionic sector: P, T, C with conditional CPT via OS reconstruction (A/B); electroweak anomaly conditions verified with inherited content and the $\mathcal{L}_5$ transparency lemma (A); CKM/PMNS structure from overlap geometry with numerical content open under OP-Yukawa (*B/Open*).
14. Neutrino sector: geometric chirality selection (A/B); embedded seesaw as preferred completion with $M_R^{\text{geom}} = \sqrt{\phi_0\, v_2}$ (C); PMNS/MSW phenomenology intact (B); $0\nu\beta\beta$ as central discriminant.

12.2. Key equations and principles of Part I

Effective kinetic tensor: $K^{AB} = \beta\,\delta^{AB} - \alpha\, n^A n^B$ (uniqueness theorem).

Propagation speed: $c_* = \sqrt{\beta/(\alpha-\beta)}$.

Matching relations: $G_N = c_*^4/(8\pi\phi_0^2)$, $\hbar = c_*\phi_0/\sqrt{\lambda}$.

Loop quantisation: $\oint \partial_A\theta\, dX^A = 2\pi n$; $S_{\mathrm{loop,min}} = 2\pi\hbar$.

Cross-sector universality: $c_{\mathrm{phase}} = c_\gamma = c_{\mathrm{GW}} = c_*$.

Second law: $dS_{\mathrm{ent}}/dw = N_{\mathrm{eff}}\gamma(dq/dw)^2 \geq 0$.

12.3. Predictions originating in Part I

GW170817 consistency: $|c_{\mathrm{GW}}/c - 1| < 6\times 10^{-15}$ (satisfied).

Fifth force: $\omega_5 \sim 10^{-10}\,\mathrm{rad/s}$, $\eta \sim 10^{-15}$ (MICROSCOPE-2).

LIV / propagation corrections: photon-sector linear time-of-flight is only a Planck-suppressed benchmark $\Delta t_{\mathrm{LIV}} \propto \zeta_\gamma EL/(M_{\mathrm{Pl}}c^2)$ with sector-dependent matching coefficient already constrained by Fermi–LAT; the GW-sector estimate is quadratic and negligible for current sources ($\Delta t_{\mathrm{GW}} \sim 10^{-64}\,\mathrm{s}$).

Casimir boundary-sensitive vacuum response (conditional; soft-mode counting provisional).

PMNS/MSW oscillation intactness: ECT preferred-direction correction suppressed by $\sim 10^{-8}$ relative to standard oscillation scale at $E \sim 1\,\mathrm{GeV}$ (B).

12.4. Part I derivation logic

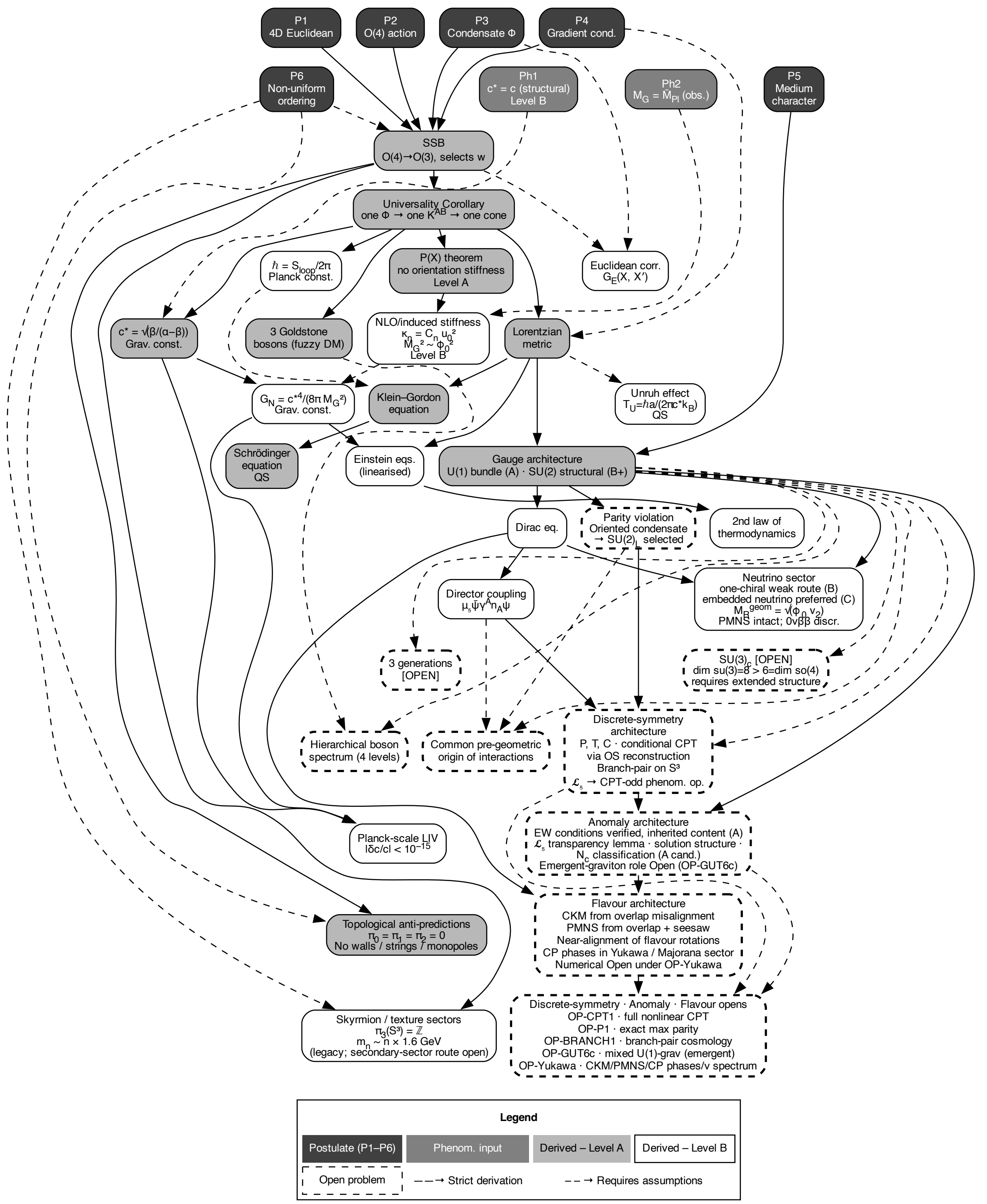


Figure 5: Part I derivation logic. From six postulates and BR1, the theory derives or structurally motivates: Lorentzian signature, universal causal cone, equation hierarchy, the retarded arrow-of-time backbone, the Level B entropy-growth closure, branching into geometric and coherent sectors, gauge-sector routes, particle-sector reconstruction targets, and compact-phase quantisation. Colours indicate Level A, B, and Open status.

Figure 5 shows the complete derivation architecture of Part I.

Part II

Macroscopic Physics: Gravity, Cosmology, and Observational Tests

13. Special Relativity as the homogeneous limit

In the homogeneous ordered phase, where the condensate amplitude and direction are uniform throughout space, the effective metric reduces to the Minkowski form: $ds^2 = -c_*^2 dt^2 + dx^2 + dy^2 + dz^2$. All familiar consequences of special relativity follow as theorems: Lorentz transformations arise as the invariance group of the causal cone; time dilation and length contraction are kinematic consequences; $E^2 = p^2c^2 + m^2c^4$ follows from the effective wave equation. In ECT, special relativity is a *derived limit* (*Level A*), not an independent postulate.

13.1. The universal speed bound

One of the most familiar facts of physics — that nothing can travel faster than light — is, in standard special relativity, a direct consequence of Einstein's second postulate. In ECT, this fact is not postulated: it is a theorem.

The argument rests not on any particular type of excitation but on the *characteristic cone* of the ordered-branch field equation. The principal symbol of the equation of motion for Φ defines a unique causal cone with speed c_*. Through the symmetric-hyperbolic structure of the dynamics and the Leray–Choquet-Bruhat theorem, ECT establishes at Level A that the domain of dependence of any point lies strictly inside its past causal cone, and that no ordered-branch disturbance — linear or nonlinear — propagates outside the cone. This is a theorem about the full nonlinear scalar dynamics, not merely about linearised waves.

For linear plane waves, the bound takes a concrete form. A massive mode has a group velocity strictly less than c_*, with equality only in the massless limit: massive excitations travel inside the cone, massless excitations on its boundary.

An important subtlety deserves mention. The *phase velocity* of massive modes formally *exceeds* c_*. This is a well-known feature of relativistic wave equations and is *not* a violation of causality: phase velocity does not carry information. The causal bound is set by the characteristic cone — the front speed — not by the phase velocity. This distinction highlights the correct ECT formulation: the speed limit is a bound on *causal propagation*, not on every quantity one can label a "velocity".

13.2. Signal speed, not photon speed

A deeper point emerges from the ECT derivation. The fundamental limiting velocity is not "the speed of light" as a photon-specific constant. It is the *signal speed* of the ordered-branch causal cone — the maximum speed of causal influence, front propagation, and information transfer.

The photon merely *realises* this boundary speed. It does so because the emergent Abelian gauge boson is massless (the relevant $U(1)$ symmetry remains unbroken, and gauge invariance forbids a mass term) and because the gauge sector inherits the same causal cone as the scalar branch. Gravitational waves likewise inherit the same cone in the minimal tensor route. The near-equality confirmed by GW170817 ($|c_{\mathrm{GW}}/c-1| < 6\times10^{-15}$) is therefore a structural consistency check, not a coincidence.

Because inertial coordinate changes in the homogeneous ordered branch are precisely the transformations preserving one and the same null cone, the same vacuum signal speed is measured in every inertial frame. What Einstein postulated as the universality of the vacuum light speed is, in ECT, reduced to a structural consequence of the unique ordered-branch causal cone (*Level A*) together with cone inheritance in the minimal gauge route (*Level A/B*).

In this sense, the vacuum speed of light is not fundamental by itself; it is the experimentally accessible manifestation of the deeper ordered-branch causal speed.

14. Emergent gravity and the generalised Einstein equations

In ECT geometry is not the primitive ontology. The effective metric is a derived bookkeeping object encoding how excitations propagate on the ordered condensate background. Gravity is not a fundamental force but the long-wavelength geometric response of the ordered condensate to deformations of its orientation structure n^A. The graviton is not a quantum of fundamental geometry but a low-energy excitation of the condensate order. This inverts the conventional logic of quantum gravity: ECT replaces the question "how to quantise an existing geometry?" with the question "how do collective condensate excitations produce a geometric description?".

The mechanism is analogous to elastic stresses in a crystal lattice. When matter bends n^A away from equilibrium, "orientation stress" acts as an effective gravitational field.

The derivation proceeds through stages. The orientation stiffness $\kappa_n \sim u_0^2/m_\sigma^2 \sim \phi_0^2$ determines the energy cost of bending n^A. Linearising around the flat background, the completed effective-metric closure yields a Fierz–Pauli-like spin-2 equation matching linearised GR (*B/Open*); exact TT projection and ghost-freedom remain open checks.

At the nonlinear effective level, ECT motivates the retained **generalised Einstein equations (GEE)** closure:

$$G_{\mu\nu} + \Lambda_{\mathrm{eff}}\, g_{\mu\nu} = 8\pi G_{\mathrm{eff}}\, T_{\mu\nu} + \Theta_{\mu\nu}[n], \tag{11}$$

where $\Theta_{\mu\nu}[n]$ is the orientation-stress tensor (*Level B*). This term has no analogue in standard GR. It vanishes in homogeneous configurations but becomes dynamically important in three distinct observational domains:

On *galactic scales*, $\Theta_{\mu\nu}$ contributes to the additional response used in the retained galactic closure, allowing flat rotation curves to be addressed without requiring a particle-DM carrier for the galactic mass-discrepancy channel (Section 20).

On *cluster scales*, it provides a Level B route to modified lensing morphology, including Bullet-like offsets, without requiring the galactic mass discrepancy to be carried by particle halos (Section 22).

In *cosmology*, the evolving G_{eff} and $\Theta_{\mu\nu}$ jointly address the Hubble tension and the JWST problem (Sections 18–19).

In the completed linear effective closure, gravitational waves propagate at c_* and are expected to carry only the two transverse-traceless tensor modes. Exact TT projection and ghost-freedom remain verification tasks (*B/Open*). Weak-field and post-Newtonian limits reproduce Newtonian gravity and are compatible with Solar System tests (PPN parameters).

15. Condensate evolution and cosmic history

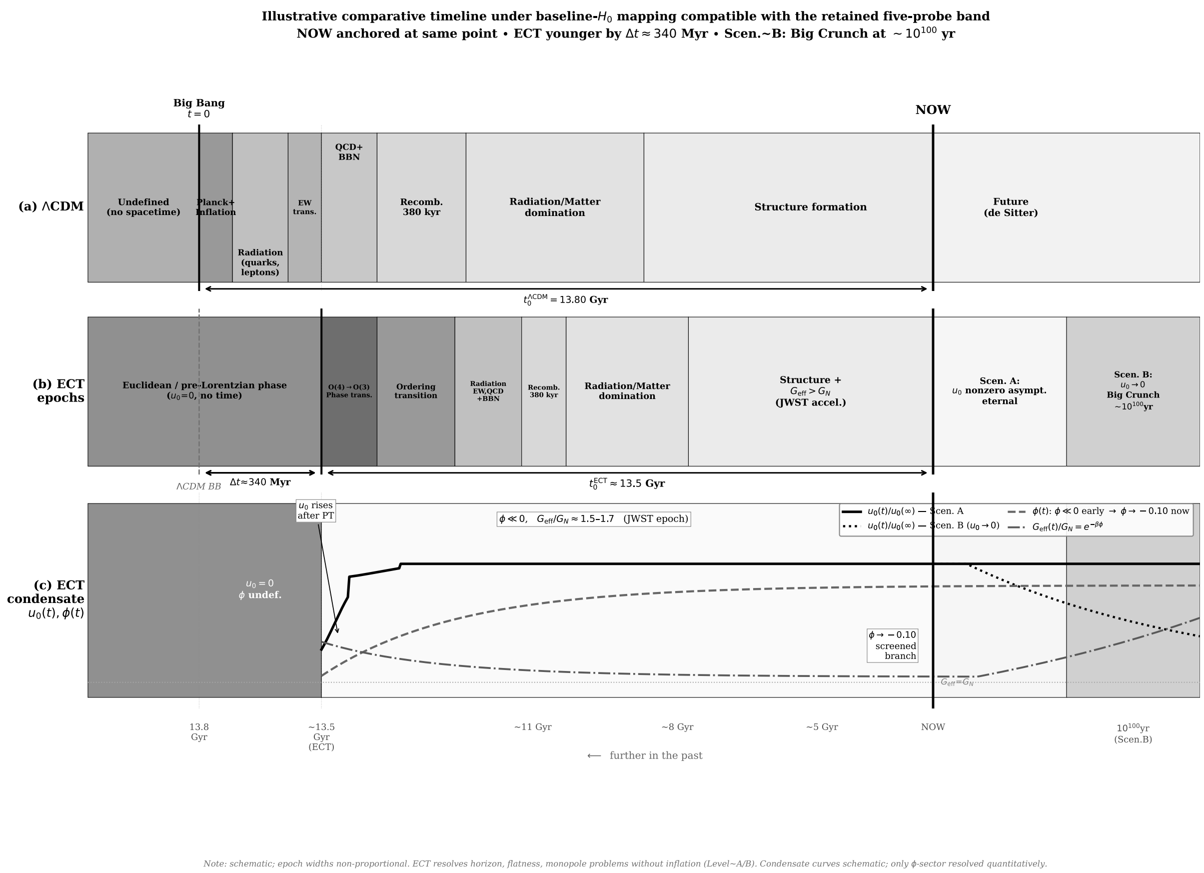


Figure 6: Cosmic timeline comparison: ΛCDM (top) and ECT (bottom). The earliest epoch in ECT is a timeless Euclidean phase; the transition $O(4) \to O(3)$ creates time itself. The lower panel shows the condensate amplitude $u_0(t)$, scale factor $a(t)$, and effective gravitational coupling $G_{\mathrm{eff}}(t)$. (*Level A for the pre-Lorentzian regime; Level B for the detailed timeline.*)

In ΛCDM, the universe begins at $t = 0$ (the Big Bang singularity) and proceeds through inflation, nucleosynthesis, recombination, structure formation, and dark-energy domination. ECT tells a fundamentally different story (Fig. 6).

Before the $O(4) \to O(3)$ transition, there is no time at all. The universe exists as a four-dimensional Euclidean medium with full $O(4)$ symmetry. The concept of "before" has no meaning in the usual temporal sense. The Big Bang singularity is replaced by the ordering transition, at which the condensate amplitude u_0 grows from zero to its equilibrium value $\phi_0 \sim \bar{M}_{\mathrm{Pl}}$. This growth naturally produces a period of rapid ordered-branch expansion, replacing the standard inflationary postulate with a pre-Lorentzian Euclidean ordering picture.

After the ordering is established, the evolution shares familiar milestones (nucleosynthesis, recombination, galaxy formation), but two features distinguish ECT from ΛCDM: the gravitational coupling G_{eff} evolves slowly with redshift, and the dark-energy component arises from residual condensate energy rather than from a cosmological constant.

A new consistency condition follows from the same charge-selective architecture: after the ordered branch has emerged, a neutral relaxing amplitude mode cannot directly populate the gauge/radiation bath unless charged degrees of freedom are available. The allowed ordering is therefore

$$\text{gradient ordering} \to \text{metric}/\Phi\text{-sector matching} \to \text{charged-spectrum population} \to \text{entropy production.}$$

This is only a post-onset routing constraint; it does not specify an efficiency, timescale, or detailed thermalisation mechanism, and it makes no statement at the pre-emergent Euclidean level.

A notable consequence is the **age of the Lorentzian branch**. In ECT, "the age of the universe" means the duration of the ordered Lorentzian branch rather than the time since a fundamental singularity. Because $G_{\text{eff}}(z) > G_N$ at high redshift, the early expansion rate is larger, which tends to shorten the age integral. In the technical preprint this leads to an age-viability ambiguity: under one mapping the resulting branch age remains close to the globular-cluster estimate (about 13.5 Gyr), while under another it becomes noticeably younger (about 12.5 Gyr). For this reason the age channel is treated there not as one of the retained extraction probes, but as a self-consistency note that still requires closure-level clarification. It has consequences for the JWST problem discussed below.

At late times, ECT admits two scenarios: Scenario A ($u_0 \to$ const, eternal expansion) and Scenario B ($u_0 \to 0$, recollapse on a timescale $\sim 10^{100}$ years). Distinguishing these requires data not yet available.

Age as viability discriminator. A separate but related point: the age of the Lorentzian branch is *not* one of the retained cosmological probes used to extract the ECT effective drift parameter ε. It nevertheless provides a distinct *viability discriminator* of the interpretation of that parameter. At the band central value, the baseline-H_0 mapping yields a branch age of about 13.5 Gyr, compatible with globular-cluster estimates; an alternative inferred-H_0 mapping yields about 12.5 Gyr, noticeably below. Improvements in independent age constraints may therefore constrain which mapping is physically correct, independently of the retained cosmological channels.

16. The four classical cosmological problems

The standard Big Bang model faces four well-known puzzles. The conventional resolution invokes cosmic inflation — a brief period of exponential expansion driven by a hypothetical inflaton field. While inflation is remarkably effective, it introduces its own unresolved issues: the inflaton has never been directly observed; inflation requires specific initial conditions to start (the initial homogeneity problem — inflation itself needs a sufficiently uniform region to begin, yet it was introduced to *explain* homogeneity); the measure problem makes probabilistic predictions ill-defined in the context of eternal inflation; and the trans-Planckian problem arises because CMB modes observed today had sub-Planckian wavelengths at the onset of inflation, potentially probing unknown physics.

ECT provides structural non-inflationary routes to all four puzzles, rather than a completed inflation-like dynamical replacement in every detail.

16.1. The horizon problem

The cosmic microwave background (CMB) — relic radiation from 380 000 years after the Big Bang — is uniform to one part in 10^5 across the entire sky. In ΛCDM, opposite sides of the sky were never in causal contact at that epoch: their past light cones do not overlap. The uniformity is unexplained without inflation.

In ECT, the problem is addressed structurally (*Level A for the pre-ordered regime; Level B for its cosmological realisation*). Before the $O(4) \to O(3)$ transition, the Euclidean phase has no causal structure at all: there are no light cones, no causal horizons, no "regions out of contact." All points of $\mathcal{M}^4$ are connected by the Euclidean metric. Uniformity across the CMB is inherited from the symmetric pre-Lorentzian phase.

A useful condensed-matter analogy is a ferromagnet. Below the critical temperature, a sufficiently coherent domain orders as one region, and distant points inside that domain share the same orientation not because they exchanged late-time signals, but because they belong to one ordered phase. ECT uses the same logic at a deeper level: CMB-scale uniformity is inherited from a coherent pre-Lorentzian ordering domain, not produced by causal communication inside an already-existing Lorentzian spacetime.

16.2. The flatness problem

Observations show $|\Omega_k| < 0.002$ (Planck 2018). In ΛCDM, this requires fine-tuning to $\sim 10^{-60}$.

In ECT: the Euclidean manifold $\mathcal{M}^4$ in P1 is flat by construction (δ_{AB}). Spatial flatness of the emergent Lorentzian geometry is a structural consequence of the flat substrate (*Level A*).

16.3. The relic and monopole problem

Grand Unified Theories predict massive topological defects (magnetic monopoles). None observed. In ECT, the primary ordering vacuum manifold S^3 has $\pi_2(S^3) = 0$, so the primary $O(4) \to O(3)$ transition does not produce topologically stable ordering monopoles (*Level A*). This does not rule out monopole-like objects in later gauge-completion sectors, whose topology and abundance remain open.

16.4. The primordial perturbation problem

The CMB contains tiny fluctuations ($\sim 10^{-5}$) with a nearly scale-invariant spectrum ($n_s = 0.9649 \pm 0.0042$, Planck 2018). In ECT, the ordered branch carries exactly one independent scalar perturbation mode at leading order (*Level A*). An ordinary $O(4)$-critical correlator does *not* by itself give this spectrum: it projects to a blue tilt. The ECT-native route is instead a conditional Euclidean Lifshitz window, in which the second-gradient stiffness of the surviving ordering mode produces a flat *local seed* spectrum ($n_\chi = 1$) for any positive second-gradient couplings; the observed tilt $n_s \simeq 0.965$, the amplitude A_s, and the tensor ratio r then depend on the (still open) transfer to the curvature perturbation and on the existence of the window (*Level B/Open*).

16.5. Summary

The four classical puzzles are reformulated without introducing an inflaton field. The pre-Lorentzian Euclidean phase gives a structural route to horizon-scale homogeneity; the flat Euclidean substrate gives the baseline flat branch; $\pi_2(S^3) = 0$ excludes primary ordering monopoles; and a conditional ordering-sector (Lifshitz-window) route gives a near-scale-invariant spectrum. The first-principles probability of selecting the required coherent branch and the full quantitative derivation of (n_s, r, A_s) remain open.

17. The dark sector in ECT

The terms "dark matter" and "dark energy" were introduced phenomenologically to account for observations that standard gravity with visible matter alone cannot explain: flat rotation curves, gravitational lensing, large-scale structure, accelerated expansion. These phenomena are real and well-established. However, the entities postulated to explain them — invisible particles (WIMPs, axions, sterile neutrinos) and a cosmological constant Λ — have never been directly confirmed despite decades of dedicated searches. ECT provides a unified structural reinterpretation of the dark-sector phenomenology from condensate dynamics, without requiring a fundamental particle halo for the galactic mass-discrepancy channel or a rigid external cosmological constant as independent primitives.

17.1. Dark matter: condensate response, not particles

In ΛCDM, the flat rotation curves of galaxies, gravitational lensing in clusters, and the large-scale structure of the universe are attributed to an invisible component — dark matter particles — constituting $\sim 27\%$ of the energy budget. Despite extensive experimental programmes spanning direct detection (LUX, XENON, PandaX, LZ), indirect detection (Fermi-LAT, AMS-02), and collider searches (LHC), no dark matter particle has ever been observed. Large regions of the simplest WIMP parameter space have been excluded or strongly constrained.

In ECT, the galactic mass-discrepancy phenomenology arises from the nonlinear response of the condensate amplitude ϕ in weak gravitational fields (*Level B*). The orientation-stress tensor $\Theta_{\mu\nu}[n]$ in the generalised Einstein equations (11) provides additional effective gravitational response in regimes conventionally associated with dark matter. Flat rotation curves, the BTFR, the RAR, and selected cluster-lensing morphologies are addressed by the present condensate closure without requiring a particle halo as the carrier of the galactic mass discrepancy (Sections 20–22).

17.2. Dark energy: condensate residual energy, not a cosmological constant

In ΛCDM, the accelerated expansion of the universe is attributed to a cosmological constant Λ — a fixed parameter responsible for $\sim 68\%$ of the total energy budget. No fundamental theory explains its value, which differs from quantum field theory estimates by approximately 120 orders of magnitude (the cosmological constant problem).

In ECT, what appears as dark energy is the *residual amplitude energy* of the ordered condensate background — the same medium that produces gravity and quantum mechanics. No independent cosmological constant or dark-energy substance is postulated. The equation-of-state parameter satisfies $w_0 \neq -1$, unlike the fixed $w = -1$ of a cosmological constant; in the present thawing closure the DR1-anchored calibrated benchmark value is $w_0 \approx -0.83$ (a calibration, not a first-principles prediction), numerically comparable to recent DESI-era indications of dynamical dark energy. If $w = -1$ is confirmed with high precision by future surveys (DESI, Euclid), the ECT late-time acceleration closure would be falsified. Conversely, if dynamical dark energy is confirmed, ECT provides a structural explanation while ΛCDM does not.

A key structural result of ECT goes further than a mere reframing: the *direct vacuum-baseline channel* underlying the 120-orders-of-magnitude cosmological-constant catastrophe is blocked at Level A in the leading ordered-branch sector. The reason is kinematic rather than dynamical. In ECT, the determinant of the ordered-branch kinetic tensor is fixed,

$$\sqrt{-\det K^{AB}} = \beta^2/c_*,$$

as a consequence of the unit-norm condition $n^A n_A = 1$ on the ordered direction. A classical vacuum offset $V(\phi_0)$ therefore shifts the Euclidean action by a constant in the leading local gravitational sector and does not source emergent gravity. The direct 10^{120}-level cancellation required in the standard framework is therefore absent in this channel. The extension to the zero-derivative one-loop vacuum contribution is conditional on the spectral condition H_Λ. The observed numerical value of ρ_Λ remains a separate Level C infrared problem.

What remains open is a separate, narrower question: the *numerical value* of the observed cosmological vacuum density ρ_Λ. Its natural ECT scaling $\sim M_{\rm Pl}^2 H_0^2$ already matches observation at order-of-magnitude level, and a possible pseudo-Goldstone realisation exists, but the dimensionless coefficient is not yet derived from first principles. This question should not be conflated with the Level A blocking of the direct vacuum-baseline channel (*Object A: Level A; loop-vacuum channel: H_Λ-conditional; observed value: Level C*).

17.3. ECT reinterpretation of the dark sector

Within the minimal ECT benchmark developed here, the phenomena usually attributed to dark matter particles and to a cosmological constant are reinterpreted as condensate-response effects: condensate-induced gravitational response for galactic mass-discrepancy phenomenology, and ordered-branch background dynamics for late-time accelerated expansion. The terms "dark matter" and "dark energy" are retained as labels for observational phenomena for comparison with the literature; they do not refer to independent physical substances in ECT. The present benchmark does not require a particle-DM carrier for the galactic mass-discrepancy channel, nor an independent rigid cosmological constant for late-time acceleration. This does not exclude hidden-sector particles or weakly coupled condensate excitations; it states only that they are not required as the carrier of the present galaxy-scale discrepancy in the minimal closure.

Important interpretive caveat. The statement "ECT does not need dark matter particles in its minimal benchmark" is not equivalent to "ECT proves that no particle beyond the Standard Model can ever exist." The first is a statement about the present minimal closure. The second would be an overclaim that the theory does not make. Collective condensate excitations and topological sectors remain structurally possible as optional extensions; they are not required by the core benchmark fit.

18. The Hubble tension

The Hubble constant H_0 measures the current expansion rate of the universe. Two independent methods give discrepant results: the Planck satellite, analysing the CMB assuming ΛCDM, infers $H_0 = 67.4 \pm 0.5\,\mathrm{km/s/Mpc}$; the SH0ES team, using the cosmic distance ladder (Cepheids + Type Ia supernovae), measures $H_0 = 73.0 \pm 1.0\,\mathrm{km/s/Mpc}$. The discrepancy exceeds 5σ and is known as the *Hubble tension* — one of the most pressing open problems in cosmology.

Physical meaning of the drift parameter ε. Before describing the ECT alleviation mechanism, it is useful to explain what the parameter ε means. In ECT, the effective gravitational coupling is not a fixed constant but tracks the slow evolution of the condensate background: $G_{\text{eff}}(z)/G_N = (1+z)^{2\varepsilon}$. The single dimensionless number ε controls how much G_{eff} was stronger in the past relative to today. A value of $\varepsilon = 0$ recovers ΛCDM-like constant G; $\varepsilon > 0$ corresponds to a condensate background whose effective gravitational response was slightly less screened at higher redshift than it is today. Because the same drift appears in many cosmological observables, ε simultaneously controls the inferred H_0 shift, the growth of structure, the distance–redshift relation, the time budget available for early galaxies, and the amplitude of the Integrated Sachs–Wolfe cross-correlation. The central technical point of §18 is that many independent observational channels translate into a one-dimensional window on the same ε, and these windows overlap. The overlap is not itself a first-principles ECT prediction; it is a consistency statement within a diagnostic effective layer, and it is in this specific sense that the retained-band result should be read.

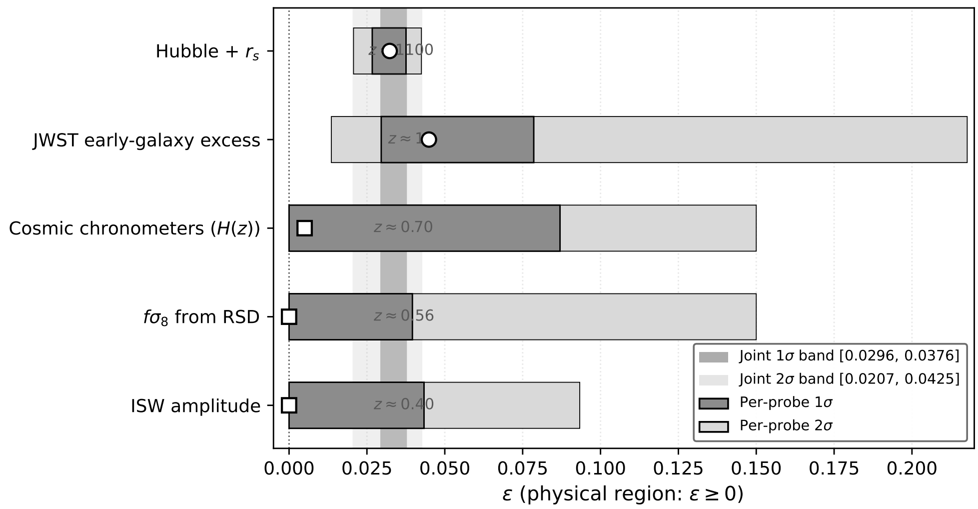


Figure 7: Summary of the five retained ECT diagnostic-layer channels that constrain the effective drift parameter ε: Hubble + r_s (acoustic scale + local ladder), JWST early-galaxy excess (growth + time-budget), cosmic chronometers (passive-galaxy ages), $f\sigma_8$ (redshift-space distortions), and the Integrated Sachs–Wolfe cross-correlation. Each channel yields an independent effective interval; the hatched band shows the common region, $\varepsilon \in [0.0296, 0.0376]$ at 1σ. The lower edge is set by JWST and the upper edge by Hubble + r_s. (*Level B for the diagnostic-layer synthesis; not a first-principles ECT prediction.*)

ECT offers a structural alleviation mechanism (Fig. 7). At high redshift, the condensate was slightly less screened, meaning $G_{\text{eff}}(z) > G_N$. Specifically, $G_{\text{eff}}(z) \approx G_N(1+z)^{2\varepsilon}$, where ε parametrises the late-time effective drift of the gravitational coupling within the uniform-ε diagnostic layer. A five-probe cross-consistency analysis in the technical preprint (Hubble + r_s, JWST, cosmic chronometers, $f\sigma_8$ from RSD, and ISW), performed under a common background module and the physical prior $\varepsilon \geq 0$, yields a retained joint effective band $\varepsilon \in [0.0296, 0.0376]$ at 1σ (centre ≈ 0.033). At the band central value the upward shift of the CMB-inferred H_0 relative to Planck is compatible with an upward shift of the observed Planck–SH0ES size, $\sim 5\,\mathrm{km/s/Mpc}$, at the baseline-background mapping level, with an accompanying age-viability note discussed below. The shift is therefore compatible with the observed tension within this effective diagnostic layer, but the retained band is an empirical cross-probe consistency window, not a first-principles closure-level prediction, and the present preprint does not yet implement the full CMB-inferred early+late H_0 pipeline. The closure-level derivation of $\varepsilon(z)$ from three-stage condensate dynamics (called OP-Hubble-derive in the technical preprint) remains the central open problem for promoting the alleviation mechanism to a derived result.

The *sign* of the shift is a structural prediction (*Level A*): any ordered-branch background with $G_{\mathrm{eff}} > G_N$ at high z necessarily shifts H_0 upward. The magnitude is closure-dependent (*Level B*). The same condensate background simultaneously affects the age budget and early structure formation, producing correlated predictions rather than independent parameters.

It should be emphasised that the retained-band result supports constancy of ε only at the level of effective compatibility across probe-dependent kernels; it is not a proof that the exact closure-level quantity is strictly constant in redshift. The retained band is therefore a common *effective* ε-range across the retained probes, not an established first-principles dynamical law. Moreover, a constructive closure-parameter-space demonstration that the retained band is realised by specific ECT closure configurations has not yet been carried out; benchmark achievability remains a programme target rather than an established result of the present stage.

Why only five retained probes? The retained set is not chosen because it is "convenient", but because these five channels admit the cleanest effective extraction of the same drift parameter within the present methodology. Other cosmological channels were examined in the technical preprint as well, but some are currently only methodologically provisional (for example, BAO or A_{lens}, which require fuller likelihood or Boltzmann-level treatment), while others probe tensions that do not map cleanly onto the present ε-sector at all (notably the S_8 / lensing–growth sector). The retained five-probe band should therefore be read as the common effective window from the channels that can currently be compared most honestly within one diagnostic layer, not as an arbitrary selection of favourable cases.

At the present level of analysis, the five retained channels are therefore consistent with a common effective ε-range. Whether this reflects a truly constant parameter or a slowly varying underlying $\varepsilon(z)$ averaged differently by different probes remains an open question rather than a settled result.

What the retained band is and is not. It is important to be clear about the logical status of the retained five-probe band $\varepsilon \in [0.0296, 0.0376]$. It is *not* a prediction of ECT. It is an empirically extracted effective consistency window from five cosmological channels. What is predictive is its *stability*: more refined future analyses of the same observables should continue to point to the same narrow positive effective range. If they drift toward zero or fragment into mutually incompatible intervals, the present diagnostic layer is in trouble. In the same spirit, the present ε-sector is not expected to close the S_8 / weak-lensing tension by itself; if it appeared to solve every cosmological tension at once within the same simple uniform-ε layer, that would be a warning sign of over-interpretation rather than a success.

Constancy of dimensionless constants. The retained varying-background sector would normally endanger dimensionless constants such as α and μ. In the present ECT closure this is addressed not by assuming those constants fixed, but by a conditional selective-coupling route: the cosmological scalar dressing is recorded in the gravitational channel, while the matter-sector matching point u_0, ϕ_0 remains distinct from the cosmological excursion $u_{\cos}(z)$. Gauge kinetic normalisations are protected only under the recorded charge assignments, anchored charged spectrum, and absence of completion-level charged σ-coupled mediators or explicit portal operators. Thus OP-CONST is a sharpened conditional route, not an unconditional theorem.

19. The JWST early-galaxy problem

The James Webb Space Telescope (JWST), launched in December 2021, has detected galaxies at redshifts $z > 10$, when the universe was less than 500 million years old. Several are unexpectedly massive and mature — containing old stellar populations, organised disk structures, and chemical enrichment that seem to require more formation time than ΛCDM allows. Standard structure formation with fixed G_N cannot produce such objects quickly enough.

ECT addresses this through two complementary mechanisms, both arising from the same ordered-branch condensate dynamics:

First, the evolving $G_{\text{eff}}(z) > G_N$ at high z means gravitational collapse was *more efficient* in the early universe.

Second, the GEE (11) include $\Theta_{\mu\nu}[n]$, which provides additional effective attraction beyond a simple scalar-tensor model.

An important and initially counterintuitive point is that ECT does *not* solve the JWST tension by simply giving early galaxies more time. In the diagnostic layer used here, the ordered Lorentzian branch is actually somewhat younger than in the corresponding ΛCDM reference, so the formation-time budget at $z \sim 10$ is reduced rather than enlarged. The reason the JWST channel still moves in the same direction as the Hubble channel is different: the stronger early gravitational response and the additional orientation-stress contribution accelerate collapse enough to overcompensate for the reduced time budget. This is precisely why the joint Hubble–JWST consistency is non-trivial: the same condensate mechanism pushes both channels in the required direction for different physical reasons.

This is why the co-directedness of the Hubble and JWST channels is a non-trivial consistency check: the same ε that eases the Hubble inference must also be the one that the JWST time-budget requires. If future, more careful high-z analyses stably pulled JWST toward $\varepsilon \leq 0$, or toward a value incompatible with the Hubble-channel range, the present ε-sector would be under serious pressure.

20. Galactic rotation curves

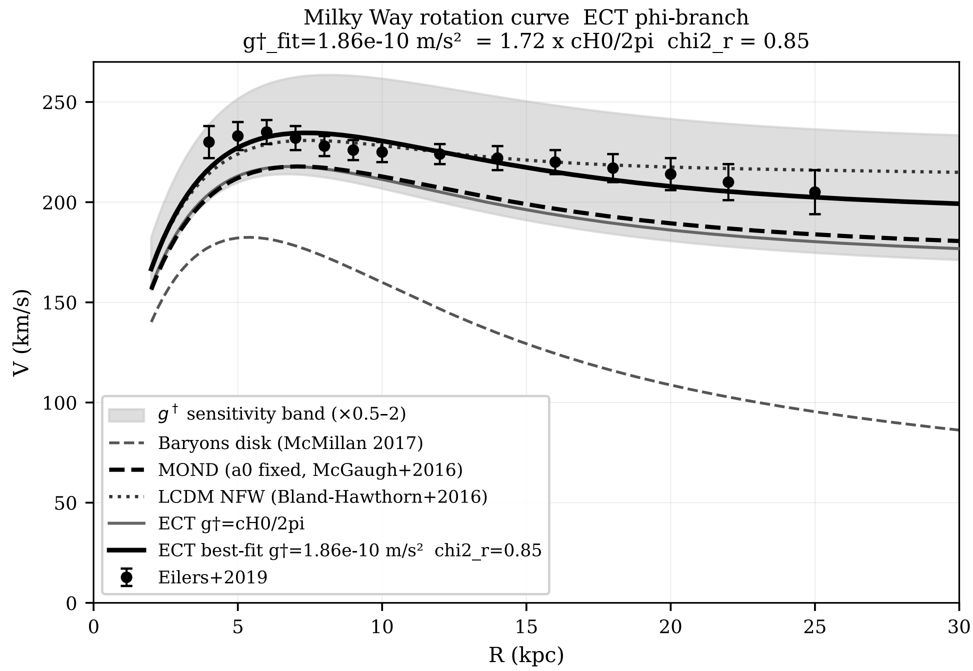


Figure 8: Milky Way rotation curve. ECT (solid) fits the data with one parameter $g^\dagger_{\rm eff} \approx 1.72\, g^\dagger_0$: $\chi^2/N = 2.95$. Newtonian prediction from baryons alone (dashed) falls off at large radii. (*Level B.*)

When astronomers measure the orbital speeds of stars in spiral galaxies, they find that stars in the outskirts orbit much faster than predicted by the visible matter alone. In a Newtonian calculation, the rotation speed should decrease as $v \propto 1/\sqrt{R}$ beyond the luminous disk, but observations show flat rotation curves extending to many disk radii. The standard explanation invokes a halo of dark matter particles surrounding each galaxy.

Despite decades of direct detection experiments (LUX, XENON, PandaX, LZ, and others), no dark matter particle has ever been observed. Large regions of the simplest WIMP parameter space have been excluded or strongly constrained. This persistent non-detection motivates consideration of alternative explanations.

ECT provides one. The nonlinear dynamics of the macroscopic amplitude variable $\phi = \frac{1}{\beta}\ln(u/u_\infty)$, which measures the local "degree of Lorentzianness" of the vacuum, reproduces MOND-like galaxy-scale dynamics without requiring a particle-DM carrier for the galactic mass-discrepancy channel (*Level B*). In dense environments (Solar System, inner galaxy), ϕ is strongly screened and standard gravity is recovered. On galactic outskirts, the ϕ-sector enters a nonlinear regime whose scale-invariant form gives:

$$g(R) = \tfrac{1}{2}\big(g_N + \sqrt{g_N^2 + 4\, g_N\, g^\dagger}\big), \tag{12}$$

with $g^\dagger \approx cH_0/(2\pi)$. In the deep-field limit ($g_N \ll g^\dagger$): $g \approx \sqrt{g_N g^\dagger}$ (flat rotation curves). In the strong-field limit: $g \approx g_N$ (Newton recovered).

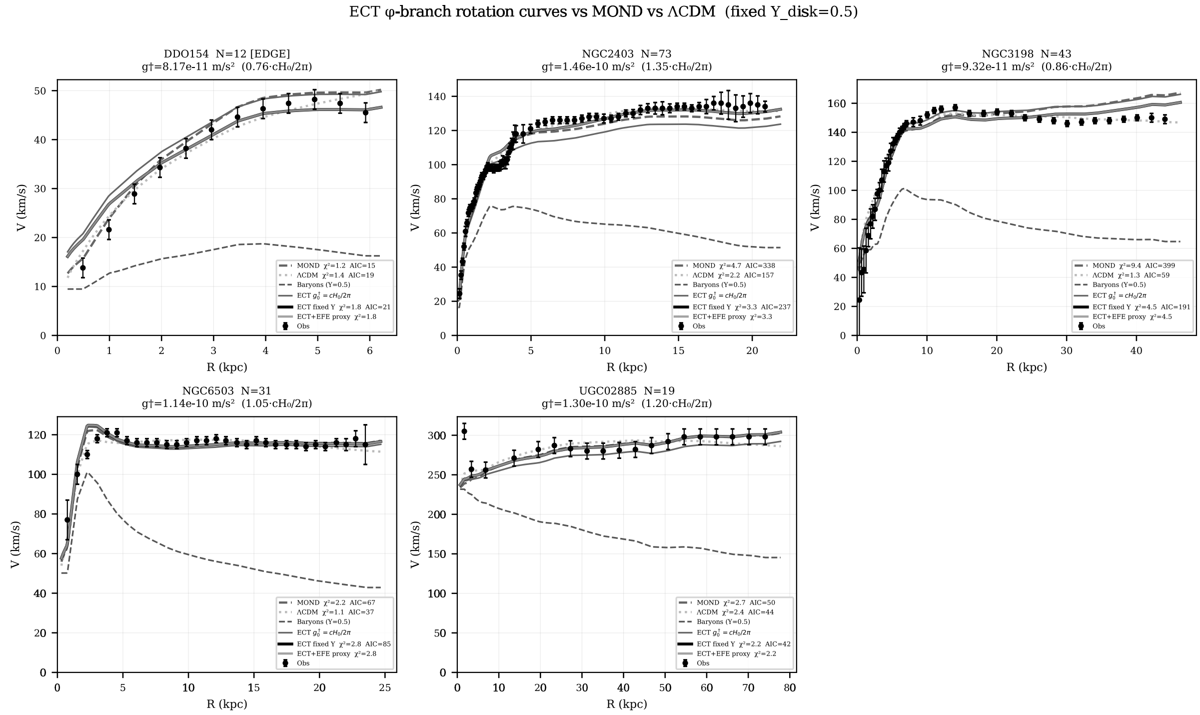


Figure 9: Rotation-curve fits for a sample of SPARC galaxies, spanning a wide range of morphologies: from dwarf irregulars to massive spirals. ECT (solid) with one parameter per galaxy. (*Level B.*)

This has been tested against **165 galaxies from the SPARC database** (Fig. 8, 9). ECT fits each galaxy with a single parameter ($g^\dagger_{\rm eff}$), achieving median $\chi^2/N \approx 2.95$ — substantially better than ΛCDM with NFW dark matter halos ($\chi^2/N \approx 6.55$), which requires more parameters.

21. The baryonic Tully–Fisher relation and the RAR

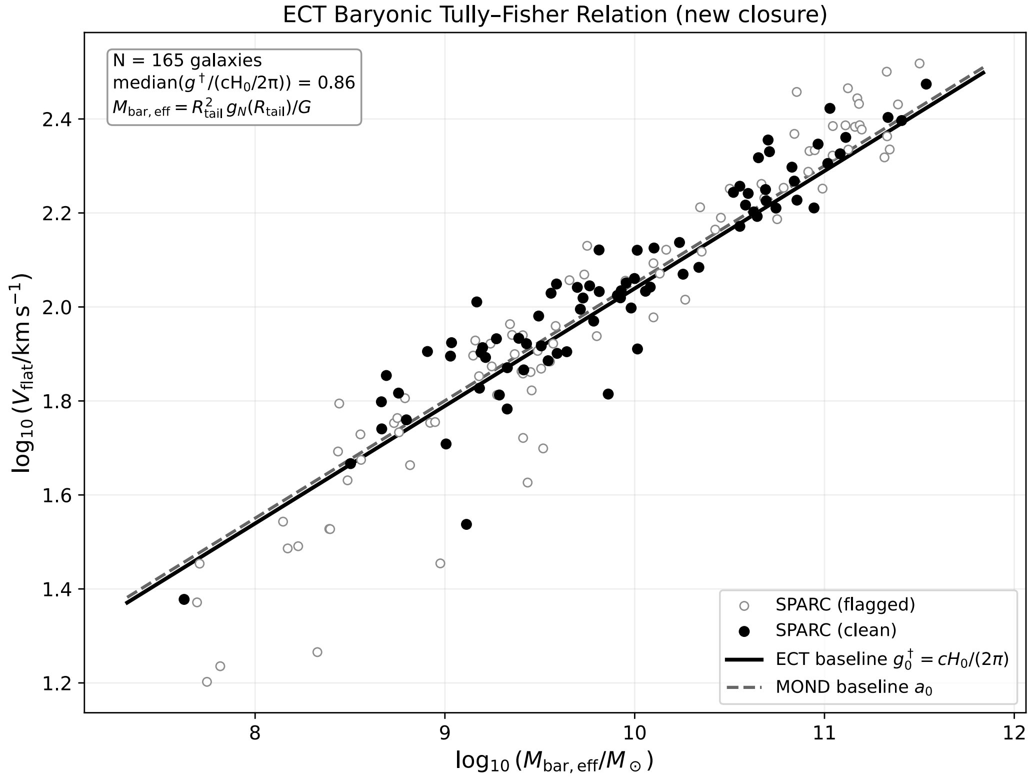


Figure 10: The baryonic Tully–Fisher relation (BTFR) from the SPARC sample. Within the retained galactic ϕ-closure, ECT gives the algebraic slope 4 (solid line): $v_{\mathrm{flat}}^4 = G_N M_{\mathrm{bar}}\, g^\dagger$. (*Slope: Level B, closure-level; $g^\dagger$ value: Level B.*)

The BTFR — $M_{\mathrm{bar}} \propto v_{\mathrm{flat}}^4$ — is one of the tightest empirical correlations in extragalactic astronomy (Fig. 10). In ΛCDM, reproducing it requires careful coordination between baryons and halo structure. In ECT, the BTFR slope is an exact algebraic consequence of the retained galactic ϕ-closure:

$$v_{\mathrm{flat}}^4 = G_N M_{\mathrm{bar}}\, g^\dagger.$$

The slope 4 is therefore not fitted inside that closure, but the closure itself remains Level B.

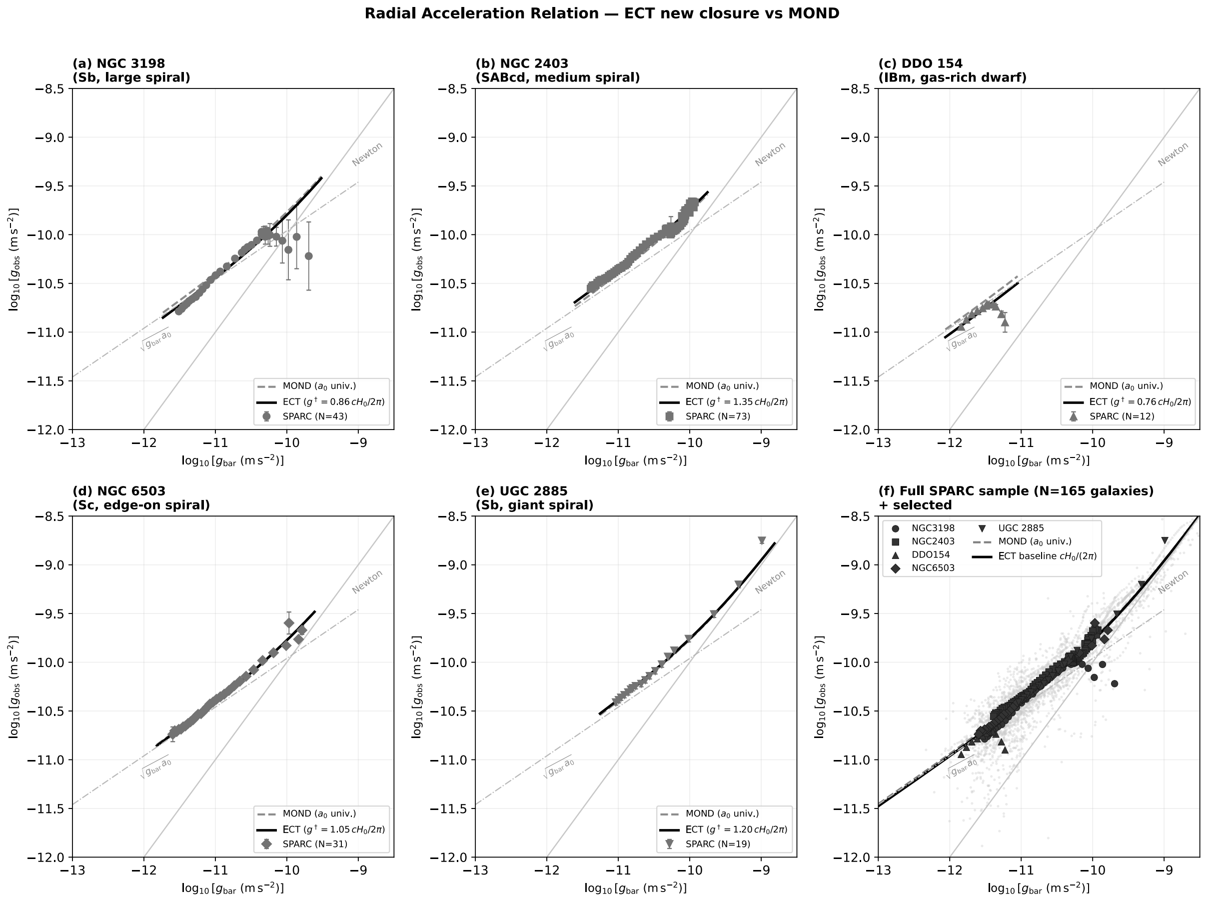


Figure 11: Radial acceleration relation (RAR) diagnostics from the SPARC sample. ECT reproduces the tight observed correlation between total and baryonic gravitational acceleration, with $g^\dagger \approx 1.1\times10^{-10}$ m/s^2 matching the McGaugh et al. (2016) empirical value. (*Level B.*)

The RAR (Fig. 11), discovered by McGaugh et al. (2016), is a direct consequence of the retained ECT galactic interpolation closure (12), not an independent Level A theorem of the bare postulates. Unlike standard MOND where $g^\dagger$ is a universal constant, in ECT $g^\dagger$ depends on the local degree of Lorentzianness (*Level B*), leading to a testable environmental modulation.

Cosmological ε-sector and local galactic fits. It is important not to conflate the two levels. The cosmological retained-band value of the effective drift parameter ε is not used here as a direct fit parameter for today's rotation curves: the galactic sector is a local closure at $z\approx0$, controlled by $g^\dagger$. What the cosmological result does strengthen is the expectation that any genuine ECT redshift dependence of the galactic sector should appear at high redshift as a mild drift of the RAR intercept, and potentially as an environment-sensitive modulation of $g^\dagger$, rather than as a wholesale redefinition of the present SPARC fits. The explicit mapping from the cosmological ε-sector to the local galactic closure remains open.

A sharp stress test: the dark-matter-deficient ultra-diffuse galaxies. The same closure that reproduces the SPARC rotation curves and the radial acceleration relation runs into a sharp and well-known tension on a small, unusual class of observed objects: the so-called *dark-matter-deficient ultra-diffuse galaxies*. The reference cases are NGC 1052–DF4 (often written DF4) and FCC 224 — faint, extended dwarfs of roughly 10^8 solar masses and a few kiloparsecs in size — whose observed internal stellar motions are much colder than the galactic closure of (12) would normally predict at that mass and size. In rough terms, the closure would read a typical line-of-sight velocity dispersion of order 20 $\mathrm{km\,s^{-1}}$ for such objects at the baseline $g^{\dagger}$, while DF4 shows about 6–8 $\mathrm{km\,s^{-1}}$ and FCC 224 shows a similarly low value. A useful matched-mass comparison is provided by NGC 5846-UDG1, a system with nearly identical stellar mass, size, and group-scale environment, which does sit on the standard radial acceleration relation at around 17 $\mathrm{km\,s^{-1}}$. This comparison is structural rather than a perfectly matched same-tracer, same-aperture twin test.

Conceptually, the tension with the present closure is sharpest not in a single number but in the regime-plane picture of where these objects *should* live by their baryonic content versus where their observed kinematics *actually* place them. This is summarised in Fig. 12: the closure applied to the baryonic content of DF4 and FCC 224 would slot them into the MOND-like branch, where an enhanced acceleration $g \sim \sqrt{g_N\, g^{\dagger}}$ is expected at the baseline $g^{\dagger}$, while the acceleration inferred from their observed internal motions places them near the Newtonian branch instead. The matched-mass control NGC 5846-UDG1 shows no such offset and sits on the RAR-normal strip.

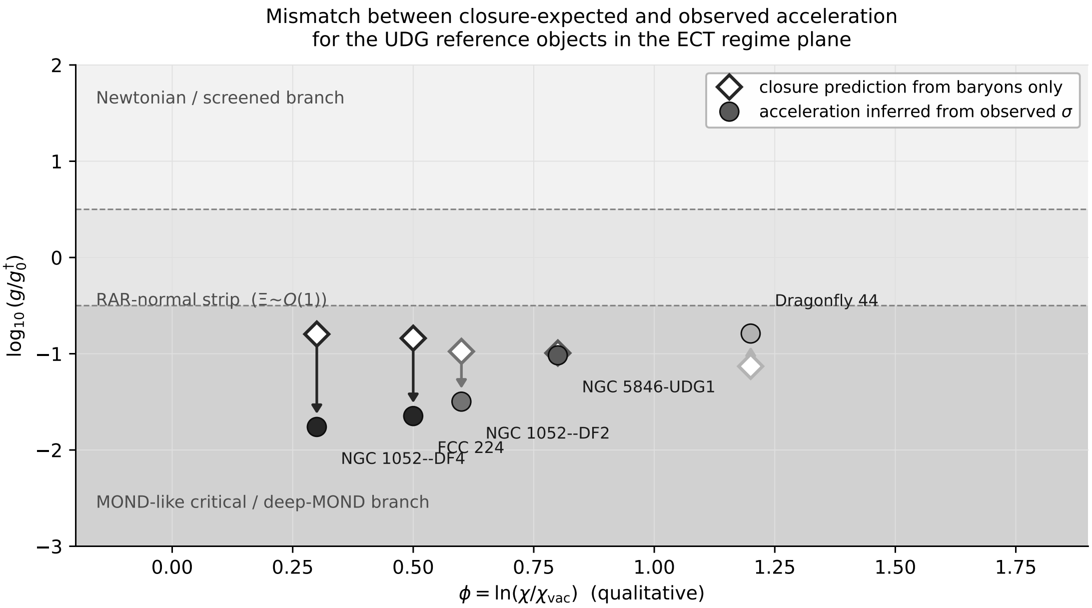


Figure 12: Positioning of the UDG reference objects in the ECT regime plane. Open diamonds show where the present galactic closure puts each object when applied to its baryonic content alone (at the baseline $g^{\dagger}$); filled circles show the acceleration inferred from the observed internal velocity dispersion at the half-light radius. For NGC 1052–DF4 and FCC 224 the closure prediction sits in the MOND-like branch, while the data sit near the Newtonian branch, making the mismatch explicit. The matched-mass DM-rich control NGC 5846-UDG1 shows no such offset; Dragonfly 44 lies at the upper tail of the normal distribution. (*Level B/Open.*)

Quantitatively, one can invert the closure at each object and ask: what value would the environmental factor in $g^{\dagger}$ need to take at that object in order to reproduce the observed dispersion? This is the diagnostic quantity Ξ_{req} shown in Fig. 13. DF4 and FCC 224 sit two to three orders of magnitude below the RAR-normal band, while the matched-mass control NGC 5846-UDG1 falls inside it and Dragonfly 44 sits at the upper tail; DF2 spans an ambiguous bracket. We stress that Ξ_{req} is *not* a prediction of ECT — it is a diagnostic readout of what the present closure would need at each object.

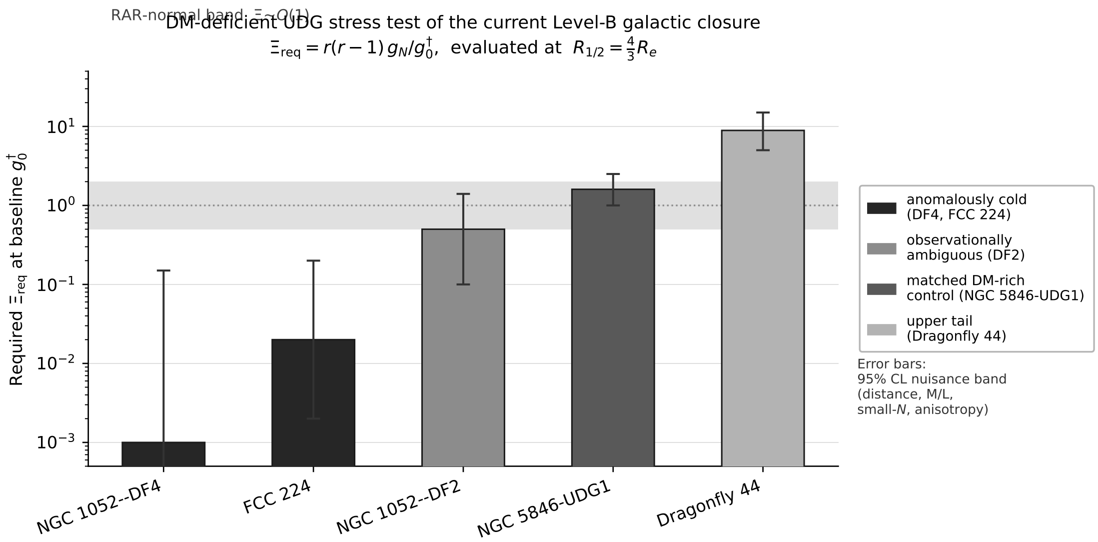


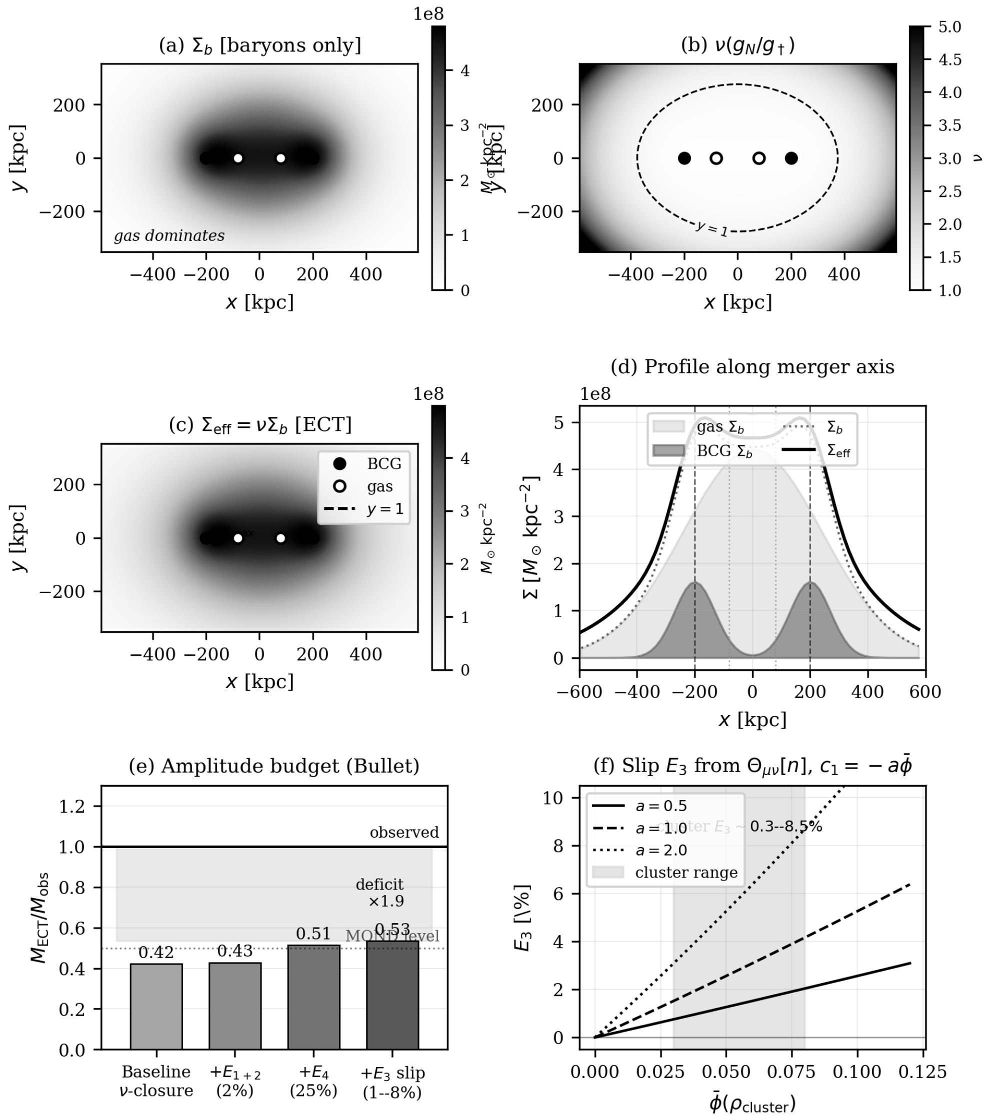
Figure 13: Dark-matter-deficient ultra-diffuse galaxies as a stress test of the current ECT galactic closure. For five reference objects the figure shows $\Xi_{\rm req}$, the value that the environmental factor in $g^\dagger$ would need to take in order to reproduce the observed internal velocity dispersion under the present Level-B closure, evaluated at the half-light radius. The shaded horizontal band marks the ordinary RAR range expected for normal galaxies. Error bars are a 95% nuisance band (distance, mass-to-light ratio, small-N statistics, anisotropy). DF4 and FCC 224 sit two to three orders of magnitude below the normal band, while the matched-mass DM-rich control NGC 5846-UDG1 lies inside it and Dragonfly 44 sits at the upper tail. DF2 spans an ambiguous bracket. The plot is a diagnostic readout of what the present closure would need at each object; it is *not* a prediction of Ξ from ECT. (*Level B/Open.*)

Physically, the observed DF4/FCC 224 kinematics look Newton-like: the stars behave as if there were essentially no enhancement beyond the visible baryons inside the probed radius. In the language of the present closure, the observed kinematics indicate that the deep-MOND branch of (12) is *not effectively realised* at those objects, even though their baryonic surface densities would normally put them in exactly that regime. The minimal environment rule used for $g^\dagger$ depends only on the ambient density, and that alone cannot naturally distinguish DF4 from NGC 5846-UDG1, because their group-scale environments are comparable. This is a sharp empirical discriminant for the present Level-B galactic closure, and it is an honest open question — not a solved consequence of ECT.

It is important to say what this is and what this is not. The baryon-only fits shown in the rotation-curve plots above are not quoted predictions of the theory for these objects: they are *diagnostic readouts* of what the current closure would need in order to reproduce the observed dispersions. A full treatment belongs to the broader open problem of deriving the exact environment law and the broader non-equilibrium sector from the underlying action (see the open-problem registry OP3). Several candidate directions for such a completion — a dynamical effective gravitational coupling $G_{\rm eff}(\phi)$, a history-dependent extension of the environment law, orientation-sector effects, non-quasi-static condensate dynamics — are structurally available in the broader framework but are not derived in the present work. The competing ΛCDM interpretation, in which DF4 and FCC 224 arise as post-collisional remnants whose original dark halos were mostly stripped, remains an honest and viable comparator. The class is flagged explicitly in the main preprint as an open discriminant to be tightened by future data: matched DM-rich ultra-diffuse controls, isolated-field searches, and higher-quality internal kinematics.

22. Cluster-scale gravitational lensing

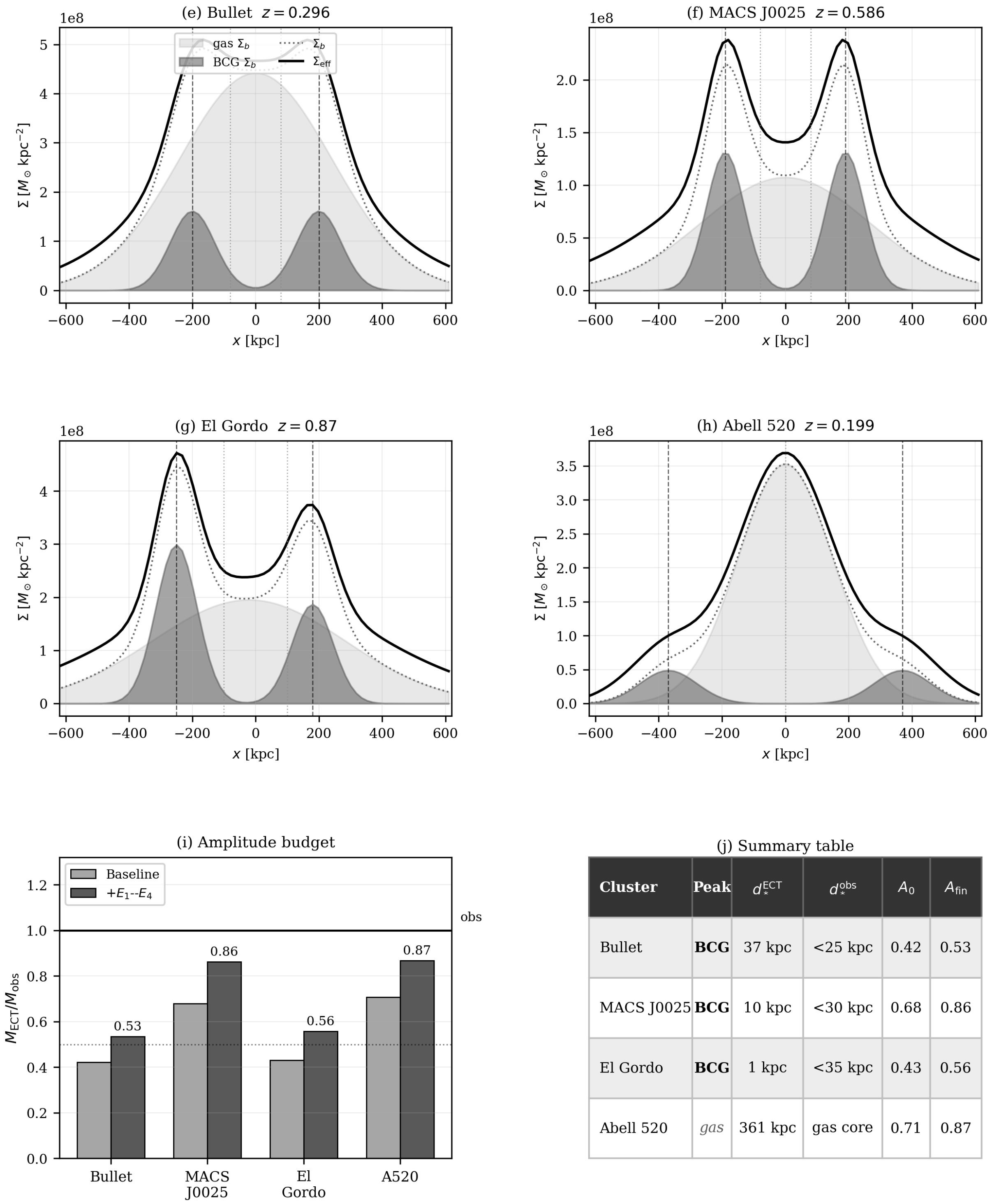
Galaxy-cluster mergers provide a critical dark-sector test. In a collision, the hot intracluster gas decelerates (electromagnetic interaction), while galaxies and any collisionless dark matter should pass through. In the Bullet Cluster, the lensing signal is offset from the gas toward the galaxies — conventionally interpreted as evidence for collisionless dark matter halos (Fig. 14).



Filled circles = BCG (collisionless); open circles = X-ray gas (collisional)

Figure 14: Gravitational lensing in the Bullet Cluster (1E 0657-56). The lensing signal (contours) is offset from the X-ray gas (colour), conventionally attributed to collisionless dark matter. In ECT this is treated as a Level B condensate-response morphology rather than as evidence that the galactic mass discrepancy must be carried by particle halos.

ECT addresses cluster lensing through the ϕ-closure with chameleon screening. The $\Theta_{\mu\nu}$ contribution generates gravitational slip, producing lensing morphology consistent with observations (*Level B*). The same condensate rule, evaluated across a suite of merger clusters, yields the mass ratios and morphology classifications shown in Fig. 15.



Cluster	Peak	$d_*^{\rm ECT}$	$d_*^{\rm obs}$	A_0	$A_{\rm fin}$
Bullet	**BCG**	37 kpc	<25 kpc	0.42	0.53
MACS J0025	**BCG**	10 kpc	<30 kpc	0.68	0.86
El Gordo	**BCG**	1 kpc	<35 kpc	0.43	0.56
Abell 520	*gas*	361 kpc	gas core	0.71	0.87

Figure 15: Cluster-merger gravitational lensing in ECT. Results for the Bullet Cluster, El Gordo, MACS J0025, and Abell 520, including lensing mass ratios and morphology classification. In Abell 520, some reconstructions place an important lensing component near the gas-rich core; this is naturally accommodated by the ECT condensate-response route and is a tension point for the simplest collisionless-halo reading, although reconstruction and modelling uncertainties remain. (*Level B.*)

The case of **Abell 520** is especially useful as a discriminant. In this system, published reconstructions place an important lensing component close to the gas-rich core rather than cleanly on the galaxy concentrations. This is a tension point for the simplest collisionless halo interpretation, although observational and modelling uncertainties remain. In ECT the gas core can naturally contribute strongly because the condensate response is sourced by the compact baryonic component. A single closure can therefore accommodate both galaxy-peaked systems such as Bullet-like mergers and gas-influenced systems such as Abell 520, but the quantitative cluster programme remains Level B.

23. Matter–antimatter asymmetry

The observable universe contains far more matter than antimatter: $\eta_B^{\rm obs} \approx 6 \times 10^{-10}$. This asymmetry requires baryon number violation, C/CP violation, and departure from thermal equilibrium (the three Sakharov conditions, 1967). The Standard Model satisfies these conditions in principle but cannot quantitatively produce the observed asymmetry.

ECT provides a structural framework in which all three Sakharov conditions are satisfied by native ordered-medium mechanisms (*Level A* for the structural possibility). The $O(4) \to O(3)$ transition provides out-of-equilibrium conditions; the fifth coupling $\mathcal{L}_5$ supplies CP violation; defect-mediated topology changes in the condensate allow baryon number violation. The specific leptogenesis mechanism and the estimated $\eta_B \sim 9 \times 10^{-10}$ (factor 1.5 from observation) are *Level B*.

24. Conclusions of Part II

24.1. Principal results

From the ordered-branch EFT, the geometric branch produces:

1. Special Relativity as the homogeneous limit (A).
2. GEE with $\Theta_{\mu\nu}[n]$ (B).
3. Structural non-inflationary routes to the horizon, flatness, primary-monopole, and near-scale-invariance problems (*A/B/Open*, depending on the component).
4. Dynamical dark energy $w_0 \approx -0.83$ (calibrated thawing benchmark; B).
5. Hubble tension *alleviation* (not resolution): a retained five-probe diagnostic-layer analysis (Hubble + r_s, JWST, cosmic chronometers, $f\sigma_8$, ISW) under a common background module yields an effective band $\varepsilon \in [0.0296, 0.0376]$ at 1σ; at the band central value the CMB-inferred H_0 shifts upward by approximately the full Planck–SH0ES gap at the baseline-background mapping level, with an accompanying age-viability note (*sign: A; magnitude: B, diagnostic-layer indicator*).
6. JWST early galaxies via enhanced $G_{\rm eff} + \Theta_{\mu\nu}$ (B).
7. 165 SPARC galaxies: $\chi^2/N \approx 2.95$ vs. ΛCDM 6.55 (B).
8. BTFR slope exactly 4 within the retained galactic closure (*B, closure-level*).
9. RAR: $g^\dagger \approx cH_0/(2\pi)$ (B).
10. Cluster lensing incl. Abell 520 gas-peaked (B).

11. Baryon asymmetry: Sakharov conditions natively (A/B).
12. Galactic dark-matter phenomenology and dark-energy phenomenology reinterpreted through condensate dynamics (B).

24.2. Key equations and principles of Part II

Generalised Einstein equations: $G_{\mu\nu} + \Lambda_{\mathrm{eff}} g_{\mu\nu} = 8\pi G_{\mathrm{eff}} T_{\mu\nu} + \Theta_{\mu\nu}[n]$.

Galactic interpolation: $g(R) = \frac{1}{2}(g_N + \sqrt{g_N^2 + 4 g_N g^\dagger})$.

BTFR: $v_{\mathrm{flat}}^4 = G_N M_{\mathrm{bar}} g^\dagger$ (slope 4, algebraic).

$G_{\mathrm{eff}}(z) \approx G_N (1+z)^{2\varepsilon}$, retained $\varepsilon \in [0.0296,\, 0.0376]$ at 1σ (Hubble shift).

24.3. Observational tests and falsifiers

ECT is falsifiable. Key predictions of the macroscopic sector:

- BTFR slope = 4 within the retained ϕ-closure (B, *closure-level*); $g^\dagger \propto cH_0$ (B).
- Fifth force $\eta \sim 10^{-15}$ (B); dark energy $w_0 \approx -0.83$ (calibrated thawing benchmark; B).
- Environmental $g^\dagger$ modulation (B, *closure-level*); graviton polarisation: two TT modes in the completed linear closure ($B/Open$).
- CMB quadrupole deficit ($B/Open$).

Full list with explanations is given in Part V, Section 53.

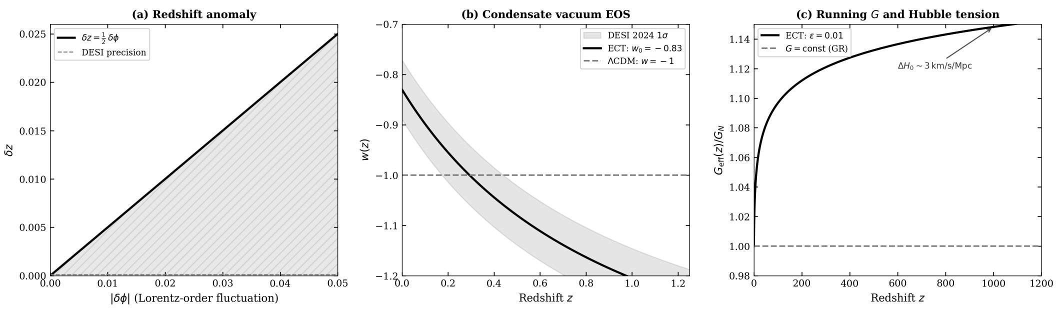


Figure 16: ECT cosmological predictions and comparison with observations and ΛCDM. ECT provides correlated predictions for dark energy, the Hubble tension, the JWST problem, and galactic phenomenology from the same ordered-branch dynamics. (*Level B.*)

24.4. Part II derivation logic

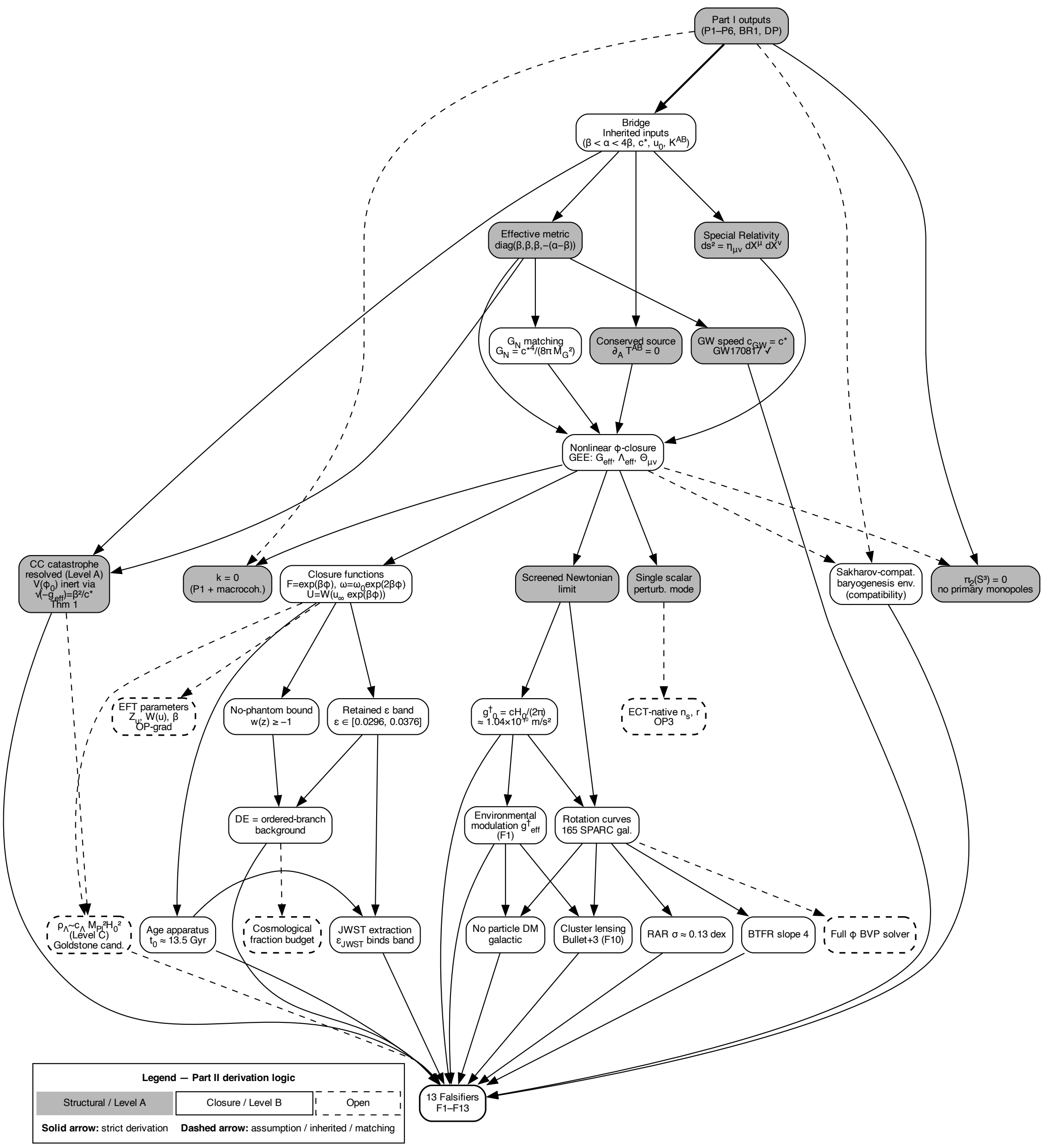


Figure 17: Part II derivation logic. From the ordered-branch EFT, the geometric branch develops the macroscopic sector: strict structural results, closure-level phenomenology, diagnostic retained-band constraints, and open completion targets are kept distinct by the colour coding. The updated Part II derivation logic separates the Level A blocking of the direct vacuum-baseline channel from the distinct Level C problem of the observed value of ρ_Λ.

The full derivation logic of Part II is shown in Fig. 17. The cosmological predictions are summarised in Fig. 16.

Part III

Quantum Sector: Coherent Branch

Why does quantum mechanics exist? In ECT, the answer has three layers.

First: the Euclidean functional integral. Because the fundamental arena is a four-dimensional Euclidean manifold with no built-in time, the natural mathematical object is the functional integral $\int \mathcal{D}\Phi\, e^{-S_E[\Phi]}$, which sums over *all* field configurations — including "closed histories" where the field returns to itself along the fourth coordinate. There are no events, no temporal ordering, no cause-effect sequences at this level. Where reflection positivity and the Osterwalder–Schrader conditions hold, the OS reconstruction maps the Euclidean functional measure to a Lorentzian Hilbert-space theory with unitary evolution. In ECT this is established for the explicitly positive Gaussian/free quadratic coherent subsectors and remains a Level B/Open constructive programme for the full interacting coherent branch.

Second: the action scale $S_0 = \hbar$. The compact phase θ of the condensate ($\Phi_{\text{eff}} = \rho\, e^{i\theta}$) imposes single-valuedness on closed loops: $\oint \partial_A\theta\, dX^A = 2\pi n$. This topological constraint sets the minimal loop action $S_{\text{loop,min}} = 2\pi S_0$, where S_0 is identified with $\hbar$ through the condensate matching relation (*Level B*).

Third: the physical interpretation of the Euclidean–Lorentzian boundary. In the Euclidean regime, all configurations contribute coherently to the functional integral — including closed histories that would be "forbidden" in the Lorentzian sense. After the emergence of the Lorentzian causal structure, these closed histories continue to contribute to quantum amplitudes: they are the source of interference, superposition, and tunnelling. For macroscopic systems with many effectively unresolved environmental modes, the decoherence functional Γ_{loop} can become large and suppress the coherent contribution of these closed Euclidean histories, and classical behaviour emerges. For microscopic systems it can remain small, and the closed histories remain coherent — this is quantum behaviour. The quantum–classical distinction is therefore organised by the reduced decoherence functional rather than by a separate postulate. The mapping from microscopic mode counts such as N_{eff} to a sharp criterion $\Gamma_{\text{loop}} \sim 1$ is a Level B closure estimate, not a stand-alone Level A theorem.

25. ECT versus standard quantum mechanics

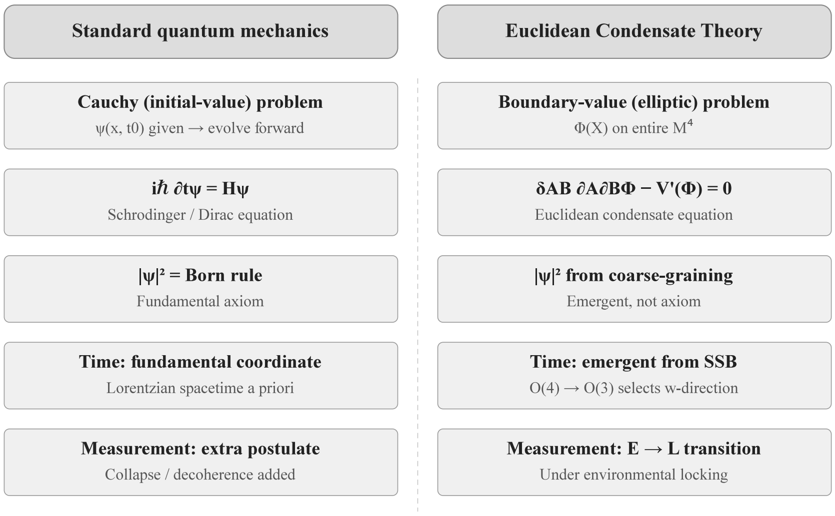


Figure 18: Structural comparison of standard quantum mechanics (left) and ECT's coherent branch (right). Standard QM starts from axioms (Hilbert space, operators, Born rule, Schrödinger equation) and computes predictions. ECT reconstructs or structurally motivates the corresponding ingredients from Euclidean condensate dynamics, with different status levels for different entries. The central column shows the proposed structural origins for each axiom. (*Level A/B for the structural routes; Level B/Open for Born-type weighting and measurement-status closure.*)

Standard quantum mechanics answers the question "how?": given a Hamiltonian and an initial state, it prescribes how to compute probabilities. ECT aims to answer the additional question "why?": why do Hilbert spaces, operators, superposition, Born-type weights, and decoherence structures arise at all? Some parts of this programme are structurally established; the completed Born rule and measurement-update theory remain open (Fig. 18).

The deepest structural distinction lies in the type of mathematical problem. In standard QM, the fundamental problem is the *Cauchy problem* (initial-value problem): $i\hbar\,\partial_t\psi = H\psi$. Given the state at $t = 0$, time evolution uniquely propagates it forward. In ECT, the fundamental problem is a *boundary-value problem* on the Euclidean manifold: $\delta^{AB}\partial_A\partial_B\Phi - V'(\Phi) = 0$. The solution is constrained by data on the *entire boundary* of the four-dimensional domain, not just on one time-slice.

This has profound consequences. The Cauchy problem generates *one-directional* evolution: past determines future, but not vice versa. The boundary-value problem generates a *four-dimensionally self-consistent* configuration: future boundary conditions constrain the present just as past boundary conditions do. This is not retrocausality in the Lorentzian sense — it is the normal functioning of a boundary-value problem. The probabilistic character of quantum mechanics, in ECT's view, arises not from fundamental indeterminacy but from coarse-graining: when untracked environmental modes are integrated out, the effective description of a subsystem becomes inherently statistical.

25.1. Detailed structural comparison: ECT vs standard QM

The following examples show how ECT's coherent branch differs from standard quantum mechanics on specific physical situations. The format is the same in each case: what the standard formalism does, what the ECT condensate picture says instead, what the ECT reading explains beyond the standard account, and the present status of the reformulation.

The hydrogen atom.

Standard QM.
One solves the time-independent Schrödinger equation with a Coulomb potential and obtains the discrete levels $E_n = -13.6/n^2\,\mathrm{eV}$ together with the probability densities $|\psi_{nlm}|^2$.

ECT (physical picture).
The atom exists not in a fundamentally given Lorentzian spacetime but in the symmetry-broken condensate background that supports both the emergent spacetime and the quantum dynamics. The electron is a spinorial field compatible with that background; the proton is a localised composite source in the emergent gauge sector; the electromagnetic binding is interpreted within the emergent $U(1)$ sector rather than introduced as an extra fundamental ingredient. The atom as a whole is a self-consistent bound configuration of these condensate structures. At the structural level, the discrete levels are understood as arising because only certain regular, phase-consistent, single-valued solutions are admissible—in harmony with the winding/single-valuedness conditions of the coherent branch.

What this explains.
Why the spectrum is discrete at all: discreteness appears as a self-consistency condition on admissible condensate configurations rather than as the eigenvalue condition of a fundamentally probabilistic wave equation. The familiar Schrödinger orbital, with its $|\psi_{nlm}|^2$ densities, is recovered as the non-relativistic envelope limit of the deeper spinorial and gauge-coupled description.

Status. Structural interpretation (*Level B*): a completed first-principles numerical derivation of the full hydrogenic spectrum, including fine structure, Lamb shift, hyperfine splitting, and the value of $\alpha_{\mathrm{fs}} = 1/137$, has not yet been carried out within ECT and is an open derivational target (*Level Open*).

The double-slit experiment.

Standard QM.
The incoming wave packet is evolved through the apparatus as a Cauchy (initial-value) problem; the observed fringe pattern is computed from $P = |\psi_1 + \psi_2|^2$, where ψ_1 and ψ_2 are the amplitudes associated with the two slits.

ECT (physical picture).
One solves a global boundary-value problem for the condensate configuration, with source, screen, slits, and detector geometry entering as boundary and

constraint data. The fringe pattern is the reduced Lorentzian imprint of a single globally admissible Euclidean configuration.

What this explains.
Why interference exists at all. The pattern is not the coexistence of two realised trajectories: it is the visible trace of one globally constrained four-dimensional configuration on a reduced Lorentzian slice. Loss of interference when which-path information leaks into the environment is the same condensate–environment response that drives decoherence: $\Gamma_{\text{loop}} \ll 1 \to \Gamma_{\text{loop}} \gtrsim 1$, with off-diagonal cross-slit terms suppressed as the medium ceases to remain phase-coherent.

Status. Structural reformulation (*Level A/B*); not a first-principles derivation of the full laboratory setup from the microscopic condensate action.

The Aharonov–Bohm effect.

Standard QM.
A charged particle acquires an observable phase shift $\Delta\varphi = (e/\hbar) \oint A_\mu \, dx^\mu$ even in regions where the electromagnetic field strength vanishes, provided the path encircles a non-trivial gauge flux.

ECT (physical picture).
The compact phase $\theta(X)$ of the condensate is already a primary structural variable of the coherent branch, and the gauge connection A_A arises as its local compensating field. The AB phase shift is then a direct manifestation of a non-trivial holonomy of the emergent $U(1)$ connection around a closed loop.

What this explains.
Why the effect is natural rather than anomalous. Phase and loops are fundamental ingredients of the ECT quantum sector, so non-trivial holonomy is exactly the kind of phenomenon one should expect when quantum dynamics emerges from phase-sensitive loop structure; the observational $\hbar$ appears once the identification $S_0 = \hbar$ is adopted.

Status. Interpretive structural match (*Level B*); quantitative agreement with standard QM is inherited through the identification of the two phases.

Measurement and outcome selection.

Standard QM.
Outcome selection is described operationally through the projection/update rule: upon measurement the state is updated to an eigenstate of the observable, with probabilities given by the Born rule.

ECT (physical picture).
The detector is treated as part of the full physical system and therefore belongs to the global boundary-value problem. In the coherent regime ($\Gamma_{\text{loop}} \ll 1$), multiple winding sectors can coexist coherently. Environmental coupling to

many modes drives the decoherence functional $\Gamma[q]$ to grow; when $\Gamma_{\text{loop}} \gtrsim 1$, off-diagonal interference terms are suppressed and the reduced subsystem behaves as an effectively classical branch. Effective "collapse" is therefore read as a reduced-sector decoherence transition rather than as a fundamental external projection.

What this explains. ECT gives a physical route to the loss of interference and to the emergence of stable pointer-like classical records. It also explains why the statistical weights inherit a Born-type quadratic form once the OS/Hilbert reconstruction and the standard Gleason-type closure assumptions are admitted.

Status. Structural measurement reading only. A complete detector-level theorem for single-outcome selection, pointer-basis selection, and exact Born-rule weights remains open (*Level Open*, OP-Q8).

Determinism at the fundamental level.

Standard QM. Unitary evolution is deterministic for the amplitude, but the measured outcomes are intrinsically probabilistic; the Born rule is a foundational axiom.

ECT (physical picture). The fundamental Euclidean equation is deterministic at the field level: specify boundary conditions on the four-dimensional domain, and one global field configuration $\Phi(X)$ follows. No probabilities, no collapse, no stochasticity at the deepest level. Statistical quantum mechanics appears because realistic observers never have access to the full condensate state: untracked environmental modes are integrated out, the subsystem picks up an influence functional, a reduced density matrix, and effective Born-type probabilities.

What this explains. Why quantum probabilities are operationally unavoidable even though the underlying field theory is globally deterministic. The relation of ECT to standard QM is the same as the relation of microscopic Newtonian mechanics to classical statistical mechanics: deterministic underneath, probabilistic after coarse-graining.

Status. Structural reading (*Level A* for the determinism of the Euclidean equation; *Level B* for the reduction chain to operational Born-type probabilities).

"Future affects past" and delayed-choice phenomena.

Standard QM. The theory is formulated as a forward-time Cauchy problem; retrocausal language is not part of the fundamental description and appears only as an interpretive remark about delayed-choice experiments.

ECT (physical picture).

For an elliptic boundary-value problem, changing the boundary condition on one side of the domain changes the solution throughout the domain, including the part that the Lorentzian reading identifies as "past". This is not retrocausality in the signalling sense; it is the normal behaviour of a boundary-value problem. Apparent retrocausality reflects the mismatch between global boundary conditions and a local temporal slicing of the reduced Lorentzian branch.

What this explains.

Why delayed-choice-type phenomena can be accommodated naturally within ECT without invoking mysterious backward influences in Lorentzian time.

Status. Structural accommodation (*Level A/B*); not a completed first-principles derivation of any specific laboratory delayed-choice setup.

Unified origin of the coherent and geometric branches.

Standard physics.

General relativity describes gravity; quantum mechanics describes microscopic coherence; how the two fit together is the canonical problem of quantum gravity.

ECT (physical picture).

Both branches arise from the same ordered condensate. The long-wavelength sector is read geometrically (generalised Einstein equations with orientation-stress corrections); the shorter-wavelength coherent sector is read wave-mechanically (through the Klein–Gordon and Schrödinger-type limits). They are not separate ingredients glued together but two regimes of one mode decomposition around the same ordered background.

What this explains.

Why it is structurally consistent at all to describe gravity and quantum coherence in one framework without "quantising gravity" in the usual sense: ECT embeds both sectors in one common substrate. Coherent fluctuations back-react on the geometric side through the same condensate dynamics.

Status. Core structural claim (*Level A* for the mode decomposition itself); a completed quantitative back-reaction programme remains open.

What ECT already reproduces.

(i) The Schrödinger-type coherent sector as the non-relativistic limit of the ordered branch;

(ii) the Klein–Gordon sector from the ordered-branch equation after the identification $t = w/c_*$;

(iii) uncertainty relations together with a structural route to phase quantisation through phase compactness, with the identification of the universal action scale S_0 with $\hbar$ remaining a Level-B matching rather than a fully closed Level-A theorem;

(iv) decoherence and the quantum–classical boundary through the influence-functional formalism;

(v) a structural Euclidean-to-Lorentzian reading of measurement as a regime transition, while a complete measurement theorem remains open;

(vi) a structural route linking spin-sector discreteness to the residual symmetry and topology of the ordered branch after $O(4) \to O(3)$, rather than a fully closed derivation of the complete spin sector;

(vii) the Principle of Euclidean Stationarity, which selects the persistent sector of admissible configurations;

(viii) a structural derivation route for gravitational decoherence from the condensate as a universal bath, without inserting the Diósi–Penrose formula as an independent postulate;

(ix) a common Euclidean structural basis for tunnelling and entanglement, understood at present as a structural bridge based on Euclidean connectivity rather than as an identity theorem reducing one phenomenon to the other;

(x) cross-sector persistence of a common action scale S_0 across the effective sectors discussed in Part III.

What remains open. The Born rule from first principles beyond the present closures; a complete theorem of measurement beyond the current Γ_{loop}-based framework; Bell correlators directly from the condensate theory beyond toy visibility proxies; the detailed origin of three generations; the exact value of $\alpha_{\text{fs}} = 1/137$; the full Yukawa hierarchy. These are presented as open derivational targets, not as failures of the framework.

26. Hilbert-space structure from reflection positivity

One of the central routes in Part III is the reconstruction of Hilbert-space structure from Euclidean data, using the mathematical tool of **reflection positivity** (RP). Where the Euclidean correlators satisfy RP and the Osterwalder–Schrader conditions, the OS reconstruction theorem maps the Euclidean measure to a Lorentzian Hilbert-space description with a positive inner product and unitary time evolution. In ECT this route is explicit for the Gaussian/free quadratic coherent subsectors. The full interacting coherent branch, the complete canonical operator algebra, and the exact recovery of all standard quantum structures remain Level B/Open reconstruction targets.

Standard QM uses the Osterwalder–Schrader theorem in the *opposite* direction: starting from a quantum theory, it checks that the Euclidean continuation is well-defined. ECT uses it *constructively*: where the RP/OS conditions are verified, Euclidean condensate data build the Hilbert-space sector rather than assuming it as a primitive.

27. Spin as a spinorial orientation mode

In standard quantum mechanics, spin is an intrinsic quantum number defined by the representations of the Lorentz group. It is usually introduced as an axiom: some particles carry half-integer spin, period.

ECT offers a deeper geometric backdrop. Before symmetry breaking, the condensate is invariant under the four-dimensional Euclidean rotation group $O(4)$, whose connected covering group is $Spin(4) \simeq SU(2)_L \times SU(2)_R$. This group admits spinorial representations—two-component objects that require a 4π rotation rather than 2π for identity return. After $O(4) \to O(3)$ symmetry breaking and emergence of the Lorentzian sector, these spinorial representations reduce to the standard Dirac spinors of relativistic quantum mechanics.

On this reading, spin is not the literal mechanical rotation of a tiny ball in ordinary three-dimensional space. It is the spinorial orientation degree of freedom of a fermionic field compatible with the condensate symmetry structure—a geometric inheritance from the pre-Lorentzian Euclidean rotational symmetry of the ordered medium. The magnetic moment is then the response of this spinorial degree of freedom to the emergent $U(1)$ gauge sector.

Important caveat. In the current formulation, fermions are introduced as fields compatible with the condensate symmetry, not yet derived as collective condensate excitations. The electron g-factor, the fine-structure constant α_{fs}, and the hydrogenic fine structure are not yet calculated from condensate microdynamics. These remain open derivational targets (*Level Open*).

Superfluid analogy. A useful but limited analogy: in superfluid ^{3}He-A (Volovik 2003), Weyl fermions emerge as low-energy excitations near topological Fermi points—the spinorial character is a property of the quasiparticle spectrum, not an external input. ECT does not yet reach this level of derivation for fermions. The structural compatibility route developed in the main text should be understood as the present limit of what the framework can establish. Nonetheless, the superfluid analogy illustrates how an ordered background can naturally support spinorial quasiparticle structures, and therefore helps motivate why such a route is conceptually plausible for ECT even though the derivation is not yet complete.

28. Superposition, interference, exchange statistics, and the Pauli exclusion principle

Superposition (*Level A/B*): The linearised fluctuation equation in the coherent branch is linear. Solutions can be added, producing superposition without any axiom.

Interference (*Level A*): Arises from the compact-phase structure. Paths with different phase accumulations contribute to the functional integral with different factors $e^{iS/\hbar}$, producing constructive or destructive combination — the same mechanism that produces interference in a superfluid.

Indistinguishability of same-species particles: In ECT, particles of the same species are quanta of the same emergent mode sector of the same ordered medium. There is no hidden individuation label; exchange reassigns occupation rather than permuting individuated objects. The many-particle configuration space is therefore already the quotient by exchange.

Exchange statistics (*Level A/B*): Connected to the topology of path configurations in the Euclidean manifold. Paths that exchange identical particles acquire a factor $+1$ (bosons) or -1 (fermions) depending on the topological class of the exchange. The spin-statistics connection (integer spin $\leftrightarrow$ bosons, half-integer $\leftrightarrow$ fermions) is structurally motivated (*Level A/B*); a fully rigorous derivation from bare P3 remains open. Recently the main preprint has sharpened this picture through a natively Euclidean route: because ECT is built on a Euclidean arena from the outset, the Euclidean (Osterwalder–Schrader) reconstruction framework provides a natural test for the spinor sector. For a free minimal Euclidean Dirac kernel, the preprint gives a convention-fixed calculation, verified against the original Euclidean Fermi-field constructions, showing that reflection positivity — the key condition for recovering a physical quantum theory — is compatible with the antisymmetric (fermionic) quantisation of half-integer spinors and *fails*, with explicit negative-norm modes, for the symmetric alternative. This is a Level A mathematical result for the free kernel; its application to actual ECT fermions remains conditional on deriving or explicitly verifying the ECT spinor kernel and its spinorial reflection-positive structure and is tracked as part of the same open programme.

Pauli exclusion principle (*Level A/B*): For fermions, the -1 exchange factor means that configurations with two identical fermions in the same state contribute with opposite signs and cancel. Pauli exclusion is a consequence of exchange topology, not a separate axiom.

Smooth backgrounds do not weaken Pauli exclusion in ECT (*Level B / provisional no-go*): Because ECT aims to derive, rather than simply assume, the fermionic sector, one might naively expect strong gravity or dense astrophysical backgrounds to induce small violations of exclusion. The opposite is true in the smooth coherent branch. On a smooth coherent ordered branch with a local first-order fermionic description, any condensate-dependent prefactor in the kinetic term can be absorbed into a *local canonical normalisation of the fermionic field*, leaving the underlying exclusion algebra unchanged. Condensate-dependent masses, the ECT-specific preferred-direction coupling $\mu_5\bar{\psi}\gamma^A n_A\psi$, and local fermionic interaction terms all modify the dynamics of fermionic modes without altering the canonical anticommutation relation that enforces Pauli exclusion. Consequently, if the ECT fermionic reconstruction eventually closes to the standard local Grassmann/CAR form, Pauli exclusion remains exact throughout ordinary smooth regions of the ordered branch, including the interiors of compact objects where the same local fermionic EFT still applies. Any genuinely nonstandard fermionic behaviour that might eventually arise in ECT is therefore expected to come not from ordinary strong gravity, but from more exceptional settings where the coherent-branch description itself may cease to apply: condensate branch transitions, topological or defect sectors of the ordered phase, and nonlocal sector mixing beyond the ordinary coherent branch. A physical consequence worth emphasising is that an equivalence-principle argument already constrains the form of any such residual effect: uniform acceleration can be locally removed by a change of frame, so the physically irreducible parameter is the *tidal curvature* $\mathcal{E} \sim GM/R^3$, not the acceleration itself. Even as a coarse-grained packet-level mismatch estimate (which is not itself a Pauli-violation observable), the resulting scale $(\mathcal{E}\,L^2/c_*^2)^2$ with L the fermionic overlap size remains enormously suppressed in all regimes of interest, including compact-object interiors. At the present stage the no-go statement is conditional on the completion of the ECT fermionic reconstruction programme from the exchange-topological and representation-theoretic sector to the full operator-level canonical anticommutation structure; this is tracked as an open problem in the main text (OP-Q21).

In standard QM, these properties are either postulated directly or derived from operator axioms that are themselves postulated. In ECT, they are assigned structural routes through the geometry, topology, and representation theory of the ordered condensate. The exchange-topology route is strong, but the full operator-level fermionic reconstruction, spin–statistics theorem, and Pauli-exclusion closure remain marked by their stated Level A/B or Open status above.

29. The Principle of Euclidean Stationarity

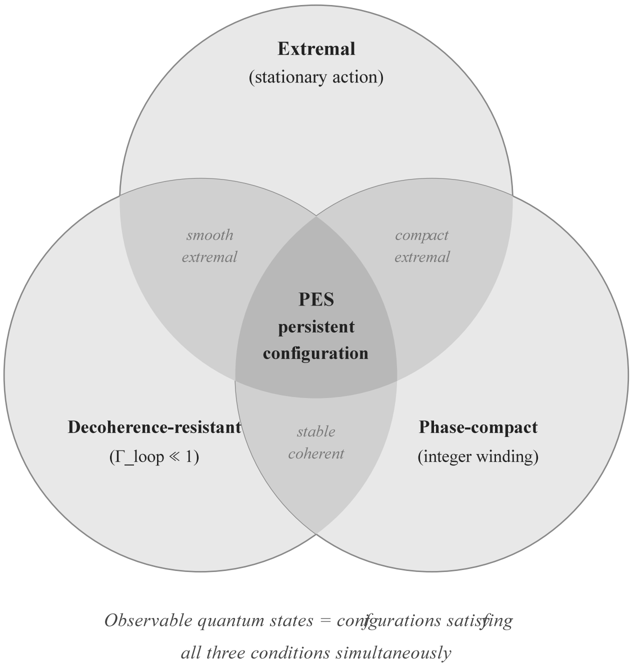


Figure 19: The Principle of Euclidean Stationarity (PES). Three conditions must be simultaneously satisfied: the configuration is an extremal of the Euclidean action (stationary), it is resistant to environmental decoherence ($\Gamma_{\text{loop}} \ll 1$), and it belongs to a well-defined phase-topological class (compact phase with integer winding). Their intersection defines the persistent configurations — the ECT counterpart of observable quantum states. (*Level B.*)

The Principle of Euclidean Stationarity (PES) is a key organising principle of ECT (Fig. 19). It is not a new postulate but an organising principle that emerges as a consequence of three already-established ingredients: the Euclidean-first ontology (P1), the compact-phase topology (BR1), and the influence-functional decoherence framework.

Statement. The physically persistent and observable configurations are those that simultaneously: (i) are stationary or quasi-stationary extremals of the Euclidean action; (ii) maintain $\Gamma_{\text{loop}} \ll 1$ (resist decoherence); (iii) belong to an admissible phase-topological class (integer winding). Configurations failing any of these conditions decohere and are absorbed into the classical branch.

Physical consequences. PES explains *why* quantum systems settle into energy eigenstates: these are precisely the Euclidean-stationary configurations that survive environmental coupling. After the Wick rotation $w \to -ic_* t$, a Euclidean-stationary configuration with action S_E corresponds to an energy eigenstate with energy $E = S_E/\Delta w$. The ground state corresponds to the lowest-action Euclidean extremal.

Relation to the principle of least action. PES provides a candidate physical underpinning for why near-stationary configurations dominate the persistent observable sector. In the strongly decohered limit, where non-stationary alternatives are rapidly suppressed, this reduces toward the classical extremal-action selection rule. This should not yet be read as a completed theorem deriving the principle of least action from first principles; a rigorous bound relating Euclidean non-stationarity to $\Gamma_{\rm loop}$ remains part of the PES completion programme.

Relation to discreteness. PES gives a structural route to the discreteness of persistent coherent configurations: stationary Euclidean sectors, compact-phase winding, and decoherence resistance select discrete surviving modes rather than arbitrary continuous configurations. This helps explain why bound-state spectra and phase-winding quantum numbers are discrete. It should not be read as a complete derivation of every integer quantum number in particle physics: charge quantisation, non-Abelian representation labels, flavour structure, and the full gauge-sector quantum-number architecture remain part of the gauge/topology completion programme.

30. Probability, the Born rule, and measurement

In standard quantum mechanics, the Born rule $P = |\psi|^2$ is a foundational axiom. In ECT it is not postulated as a primitive rule; rather, the theory provides a *Born-type weighting route*. Where reflection positivity and the OS construction apply, the physical inner product is quadratic in amplitudes. Once decoherence suppresses interference between effectively classical alternatives, branch weights inherit this quadratic structure. Conditional uniqueness then follows through Gleason-type reasoning only after the Hilbert/projector structure and the relevant additivity assumptions are admitted. This is a Level B/Open route, not yet a first-principles theorem of the exact Born rule.

More broadly, ECT does not merely reinterpret the axioms of quantum mechanics; it changes their logical status from foundational postulates to reconstruction targets: Hilbert space is constructed rather than postulated, unitarity follows where reflection positivity is available, the Born rule is approached through the OS/Gleason/decoherence route, and effective "collapse" is read as a reduced-sector decoherence transition at $\Gamma_{\rm loop} \sim 1$ rather than as a fundamental observer-induced process.

The measurement problem is therefore not declared solved. ECT provides a structural route to interference suppression and classical record formation, but a complete detector-level theorem for single-outcome selection, pointer-basis selection, and exact Born-rule weights remains open (*Level Open*).

31. Quantum entanglement and Bell correlations

In standard QM, entanglement is a primitive property of the tensor-product Hilbert space. In ECT, it is a structural feature of the Euclidean condensate (*Level A/B*).

In the coherent branch, subsystems are not fundamental isolates — they are reduced sectors of a single ordered medium. Two subsystems sharing a common condensate with $\Gamma_{\rm loop} \ll 1$ are correlated through the Euclidean substrate without requiring Lorentzian signalling. Four-dimensional Euclidean coherence produces non-local (in the 3D Lorentzian sense) correlations: correlations that would be surprising if one thinks in terms of 3D local causality, but are entirely natural as correlations within a single 4D Euclidean field configuration.

ECT does not violate Bell's theorem: it is not a local hidden-variable theory. The "non-locality" of entanglement is reinterpreted as common-medium correlation in the Euclidean arena, consistent with the no-signalling theorem. Two subsystems are entangled not because a hidden superluminal signal connects them, but because both are partial descriptions of one and the same coherent condensate configuration. Measurement of one part does not "send an influence" to the other; it converts the local subsystem into a decohered branch, while the other was already part of the same global configuration. The observed correlation reflects prior inseparability, not signal transfer. A full first-principles derivation of Bell-inequality-violating correlators and Tsirelson bounds remains open (*Level Open*).

Delayed-choice experiments (Wheeler) receive a natural interpretation. In the boundary-value problem of ECT, future detector settings are part of the boundary data that constrain the Euclidean field configuration globally. This is not retrocausality — it is the standard behaviour of a boundary-value problem.

Single-pattern intuition. A helpful intuition is that an entangled pair is not two independent objects connected by a mysterious faster-than-light wire. It is one non-factorised condensate pattern seen through two reduced laboratory descriptions. Alice's and Bob's systems are distant in the emergent Lorentzian branch, but their reduced states belong to the same coherent Euclidean configuration. A measurement at Alice's side does not send a signal to Bob; it conditions Alice's local reduced description within the shared configuration. Bob's local statistics remain unchanged, which is why no signalling is possible. But the joint correlations can still violate Bell inequalities, because the relevant structure is not a set of independent local hidden variables assigned separately to Alice and Bob — it is a single non-factorised four-dimensional configuration whose reduced descriptions appear in two distant laboratories. This is the sense in which ECT is "local" for signals but not "local" in the Bell-hidden-variable sense.

32. Tunnelling

In standard quantum mechanics, tunnelling is the ability of a system to cross a region that is classically forbidden in Lorentzian language. For a particle with energy E facing a barrier $V(x) > E$, the Lorentzian kinetic energy would be negative, so no classical trajectory exists through the barrier.

ECT gives this a Euclidean interpretation. After Wick rotation the forbidden Lorentzian segment is represented by a Euclidean saddle contribution. The barrier does not disappear. It is re-expressed as an action cost:

$$S_E \sim \int dx \sqrt{2m\,[V(x) - E]}.$$

The tunnelling probability is then exponentially suppressed,

$$P_{\text{tunnel}} \sim e^{-2S_E/\hbar},$$

as in the standard WKB result. The Euclidean picture therefore does not say that the particle freely rolls through the barrier. It says that the forbidden Lorentzian process is represented by a classically allowed Euclidean contribution whose weight is small because its Euclidean action is large.

This is the same logic used in standard instanton and WKB methods, but ECT gives it an ontological reading: tunnelling is not a mysterious violation of classical motion, but a Lorentzian projection of a Euclidean saddle contribution. The quantitative suppression remains the standard one at the present level; the gain is interpretive and geometric, not yet a distinct numerical prediction.

Wick rotation: when does it apply. Wick rotation, $t \to -i\tau$, is a standard formal tool when the analytic continuation and boundary conditions are well defined. Its physical content depends on the question being asked. For ordinary WKB tunnelling it converts the classically forbidden Lorentzian segment into a Euclidean action cost. For false-vacuum decay, when two classically distinct configurations are separated by a barrier and a suitable Euclidean saddle exists, it gives the bounce / instanton exponent controlling the semiclassical transition. For a system with a single classical minimum it does not create a bounce saddle; it instead yields the Euclidean structure underlying ordinary stationary states such as ground-state localisation around the minimum. The Euclidean machinery is therefore a powerful reconstruction tool, not a recipe for turning every Lorentzian problem into a barrier-crossing problem.

33. Interactions belong to the Lorentzian branch

A conceptually important distinction in ECT is that *interaction* — in the sense of one physical system exchanging energy, momentum, or information with another — is fundamentally a Lorentzian concept (*Level A*).

In the coherent branch ($\Gamma_{\text{loop}} \ll 1$), what exists is not interaction but *common-medium correlation*: subsystems share a coherent condensate configuration, and their effective descriptions are correlated through the Euclidean substrate. No energy exchange occurs; no signal is transmitted; no "event" in the Lorentzian sense takes place.

Interaction requires decoherence: one subsystem must couple to another strongly enough that Γ_{loop} crosses the threshold ~ 1. At that point, the coherent correlations are converted into classical, Lorentzian, causal events — energy is exchanged, momentum is transferred, information is communicated. Measurement, in this picture, is the process that converts Euclidean common-medium correlation into Lorentzian interaction.

This does not mean there is no coupling at all. The influence of the medium enters through the influence functional; the decoherence kernel, medium response, and suppression of off-diagonal elements are all consequences of the medium, not of "forces" in the Lorentzian sense. The correct distinction is not "interaction versus no interaction" but "decohered Lorentzian event-like interaction versus Euclidean common-medium correlation."

This separation explains why quantum systems can exhibit non-local correlations (entanglement) without violating causality: the correlations belong to the coherent Euclidean branch, while causal signalling belongs to the decoherent Lorentzian branch.

34. Influence of future boundary conditions

Because the fundamental ECT equation is a boundary-value problem on the Euclidean manifold (not a Cauchy problem), boundary data on *all* sides of the domain constrain the solution. After the Wick rotation to Lorentzian signature, this means that future boundary conditions can constrain the present configuration.

This is not backward-in-time causation in the Lorentzian sense. It is the standard mathematical property of boundary-value problems: the solution at any interior point depends on data everywhere on the boundary. Global Euclidean boundary-value dependence is not retro-signalling within Lorentzian spacetime; no Lorentzian causality violation is implied. In ECT, this property provides a natural framework for understanding delayed-choice experiments, quantum eraser phenomena, and the apparent "retrocausal" features of the two-state vector formalism (*Level A/B*).

35. Quantum vacuum effects

The ordered condensate is not an empty backdrop but a physically active medium. Its response to boundaries, acceleration, and cosmological expansion produces three classes of observable effects:

The Casimir effect (*Level B, conditional*): In standard QED, the Casimir force between conducting plates arises from modified vacuum fluctuation modes. In ECT, the ordered vacuum can in principle respond to boundaries that reflect its soft orientation sector, not just the photon sector. Within a provisional multi-component soft-orientation EFT with idealised condensate-reflecting Dirichlet boundaries, a naive counting argument yields an enhancement factor of 3/2 relative to QED. However, this provisional factor depends on boundary conditions that standard metallic plates do not implement (the neutral soft sector couples to matter only gravitationally), and on a three-component coordinate counting that the integrability constraint of the scalar-only basis reduces at the physical level ($N_G^{\mathrm{phys}} = 1$ at linear order). The 3/2 ratio is therefore conditional, boundary-sensitive, and model-dependent—not a universal laboratory prediction. What is structurally robust (Level A) is the claim that a nontrivial ordered vacuum must be boundary-sensitive.

The Unruh effect (*Level A/B*): An accelerated observer in the ECT vacuum perceives a thermal bath with temperature $T = \hbar a/(2\pi c k_B)$, identical to the standard prediction. The derivation traces this to the analytic continuation from the Euclidean frame to the Rindler frame.

Cosmological particle production (*Level B*): At the $O(4) \to O(3)$ phase transition, the abrupt change in the condensate background creates real particle–antiparticle pairs. In ECT this is not the mysterious "appearance of something from nothing": modes that were "quiet" in the symmetric phase cease to be quiet in the ordered phase because the background against which "silence" is defined has changed—analogous to the production of phonons when a superfluid is rapidly cooled.

36. Decoherence and the quantum–classical boundary

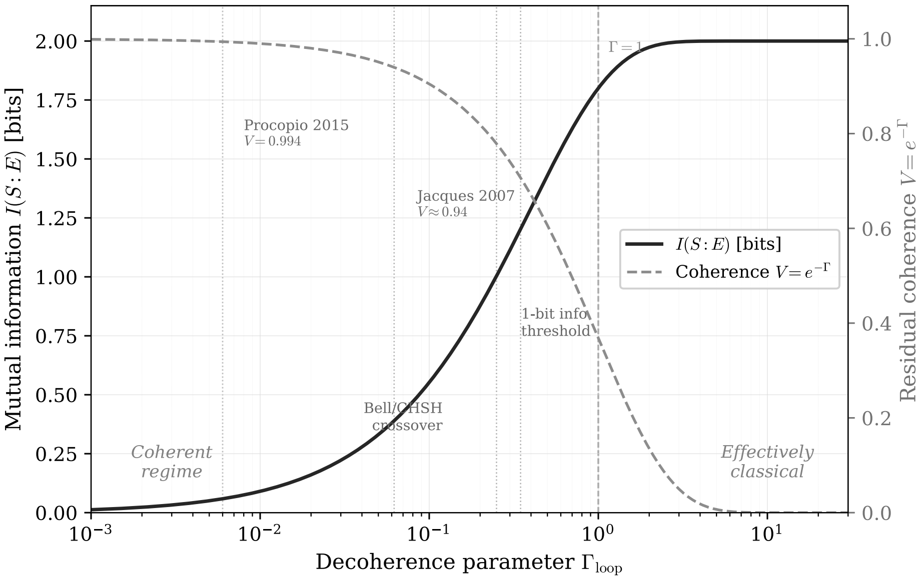


Figure 20: Decoherence regimes in ECT. The reduced description is organised by the decoherence functional Γ_{loop}. In simple estimates one may write $\Gamma_{\text{loop}} \sim N_{\text{eff}}\gamma\,\Delta w$, so weak effective coupling preserves coherent alternatives while strong effective coupling suppresses off-diagonal density-matrix elements. The coherent/decohered-sector distinction is structural; the numerical mapping from $N_{\text{eff}}\gamma$ to a sharp crossover is a Level B closure estimate.

ECT provides a unified influence-functional framework for decoherence (Fig. 20), treating environmental and gravitational channels within the same reduced-state architecture. The structural framework is Level A/B, while quantitative regime placement is closure-level.

Environmental decoherence: interaction with thermal photons, air molecules, phonons, and other environmental degrees of freedom.

Gravitational decoherence: interaction with the condensate orientation field n^A. In the present closure, ECT realises a DP-type leading gravitational-decoherence channel inside the condensate reduced-state architecture. The ECT-specific content is the condensate-origin interpretation, the absence of a freely fitted collapse scale, and possible subleading condensate-sector corrections.

The decoherence time is $\tau_{\text{dec}} = 1/(N_{\text{eff}}\gamma)$, where N_{eff} is the number of effective environmental modes and γ is the coupling rate. Numerical estimates: free electron in vacuum, $\tau_{\text{dec}} \gg 10^{10}$ yr; fullerene C_{60}, $\tau_{\text{dec}} \sim 10^{-6}$ s (consistent with observed interference); dust grain in air, $\tau_{\text{dec}} \sim 10^{-15}$ s.

Physical picture. In ECT, decoherence is not a postulate about observers destroying wave functions. It is a physical reduced-sector process: the subsystem loses coherent integrity because it couples to condensate/environmental degrees of freedom that it cannot track. No fundamental collapse postulate is introduced; however, a complete measurement-update theorem, including single-outcome selection and pointer-basis selection, remains open.

Recoherence. The reverse process—partial reassembly of coherent integrity—is not forbidden in principle but exponentially suppressed for macroscopic systems: $P_{\text{back}}/P_{\text{fwd}} = e^{-\Gamma_{\text{loop}}}$. For mesoscopic or engineered systems, partial recovery of coherence is a physically meaningful possibility, consistent with echo-type and quantum-control protocols.

37. Quantum gravity in ECT

In standard approaches, quantum gravity — the unification of GR and QM — is one of the central unsolved problems of physics. Loop quantum gravity, string theory, causal dynamical triangulations, and other programmes attempt to quantise the gravitational field on a pre-existing spacetime or to construct spacetime from more fundamental elements.

ECT approaches quantum gravity from a different direction (*B/Open for the full closure*). Since both gravity (geometric branch) and quantum mechanics (coherent branch) are emergent from the same ordered condensate, their relation is architecturally unified rather than imposed by quantising a pre-existing metric. The graviton route and the quantum-amplitude route originate in the same medium and are governed by the same ordered-branch kinetic structure. This reframes the quantum-gravity problem, but it does not by itself constitute a completed UV or spin-2 quantum-gravity theorem.

A concrete structural advantage over standard perturbative quantum gravity already follows from the condensate architecture. In perturbative Einstein gravity, Newton's constant has dimension $[G_N] = M^{-2}$, forcing each loop order to generate new curvature counterterms (R^2, R^3, $\ldots$) that cannot be absorbed into a finite parameter set — the theory is non-renormalizable. In ECT, G_N is a condensate parameter, and the natural dimensionless gravitational coupling $g(\mu) = \lambda(\mu)/(8\pi) \approx 4 \times 10^{-5}$ at the Planck scale remains parametrically small throughout the ordered-branch EFT. The Landau pole of the scalar closure lies at $\mu_{\mathrm{LP}} \approx M_{\mathrm{Pl}}\, e^{5.26\times 10^4}$ — far above the physical EFT threshold $m_\sigma \sim M_{\mathrm{Pl}}$ where the low-energy description already ceases to apply. Above that threshold, the correct variables are the microscopic condensate degrees of freedom, not a graviton EFT extrapolated to infinite energy. This is a structurally different starting point from quantizing a pre-existing Lorentzian metric.

The remaining open tasks: explicit graviton-loop power counting (OP-UV1), two-loop RG of the condensate–graviton system (OP-UV3), and establishing reflection positivity for the spin-2 orientation sector (OP-OS1). If the last is achieved, graviton states appear in the Hilbert space automatically from the Osterwalder–Schrader reconstruction — no separate quantisation act needed. These are open problems, not solved theorems (*Level Open*). Thus ECT reframes the question "how do you put gravity and quantum mechanics together?" through a single-medium architecture, while the full UV/spin-2 closure remains open.

38. Reduced-state reformulation of the gravity–matter problem

A long-standing difficulty, often called the *hybrid-consistency problem*, is that combining quantum matter with any non-quantised or emergent description of gravity runs into a characteristic set of obstructions: reduced states must remain positive; no classical channel should transmit quantum information; energy–momentum bookkeeping must stay controlled; relativistic compatibility and diffeomorphism invariance must be preserved. Historical semi-classical proposals by Møller and Rosenfeld already exhibited these tensions; later thought experiments and effective-field-theory analyses sharpened them further. Modern hybrid frameworks, such as post-quantum classical gravity (PCCG), accept these tensions as part of a fundamentally stochastic and irreversible law of gravity (*Level B for the general landscape*).

ECT takes a different architectural route (*Level B structural*). None of these costs is postulated as a fundamental law of the theory. Instead, they are displaced from the fundamental action to the *reduction map* between the full Euclidean condensate and its effective descriptions. The central thesis is simple: irreversibility, decoherence, effective thermality, and apparent stochasticity are not taken as fundamental features of gravity, but appear only in reduced descriptions of a single globally unitary Euclidean condensate.

Two-level ontology. To make this architecture concrete, ECT distinguishes two ontological levels, with the reduced level split into two operational regimes. *Level I* is the full condensate description: one globally unitary Euclidean condensate, no independent quantised metric field, no fundamental collapse, and non-factorisable correlations persisting as a natural feature of a collective medium. *Level IIa* is the reduced *coherent-effective* regime: partial tracing over unresolved condensate modes still leaves coherent common-medium structure intact, mesoscopic interference survives, and the description is reduced but not yet classical. *Level IIb* is the reduced *macroscopic classical-looking* regime: under strong locking and decoherence, the effective gravitational sector behaves as ordinary Einstein/Newton gravity, with effective irreversibility, reduced thermality, and apparent stochasticity.

Crucially, "classicality" in ECT is not a separate ontological sector but a regime of reduced description; the same Level I physics produces different Level II effective descriptions in different regimes. An important warning: denying that the metric is a fundamental quantised field is *not* the same as denying that gravity has any quantum content at all. What ECT denies is the need to quantise an independent metric field as the fundamental starting point, not the quantum character of the underlying medium.

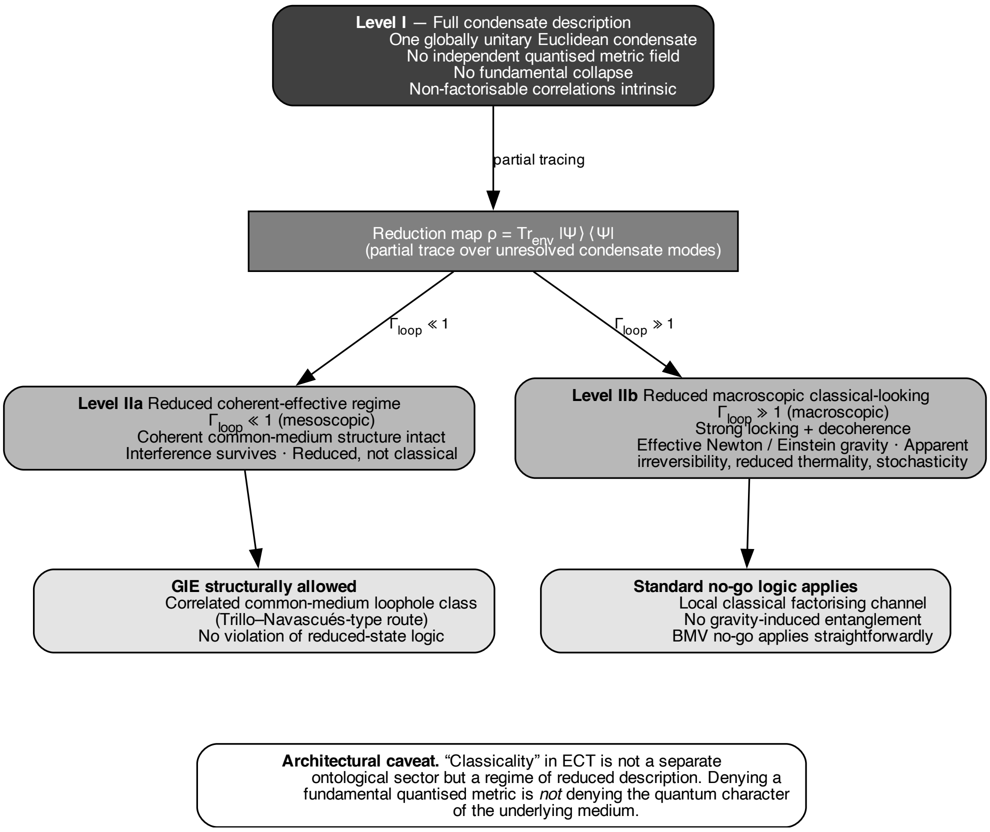


Figure 21: Two-level ontology of the hybrid-consistency architecture. Level I is the full Euclidean condensate: globally unitary, with non-factorisable correlations. The reduction map $\rho = \mathrm{Tr}_{\mathrm{env}}|\Psi\rangle\langle\Psi|$ produces two operational regimes. Level IIa is the reduced *coherent-effective* regime (mesoscopic, $\Gamma_{\mathrm{loop}} \ll 1$): coherent common-medium structure persists, interference survives, and the regime is reduced but not yet classical — this is where gravity-induced entanglement is structurally allowed through the Trillo–Navascués-type loophole class. Level IIb is the reduced *macroscopic classical-looking* regime ($\Gamma_{\mathrm{loop}} \gg 1$): strong locking and decoherence produce ordinary Einstein/Newton behaviour and the standard Bose–Marletto–Vedral no-go logic applies. Crucially, "classicality" in ECT is a regime of reduced description, not a separate ontological sector. (*Level B, structural.*)

39. Gravity-induced entanglement: can gravity be "classical"?

One of the sharpest near-term tests of the quantum character of gravity is the proposal, due independently to Bose, Mazumdar, Morley and to Marletto and Vedral, to look for entanglement between two small mesoscopic masses held in spatial superposition and interacting only through their mutual gravitational field. In its standard reading, *Gravity-Induced Entanglement* (GIE) in such a setup would force the conclusion that the gravitational mediator itself carries non-classical degrees of freedom, because local classical channels cannot transmit quantum information. A refinement due to Christodoulou and collaborators showed that the effect admits a consistent locally mediated reading within linearised quantum gravity, with proper causal retardation.

This standard inference has more recently been stress-tested. A proposal by Aziz and Howl argued that, once matter is treated at the level of quantum field theory, a locally classical-looking gravitational coupling might still generate entanglement; several critical responses argued that in the specific nonrelativistic limit employed, the relevant unitary factorises and no genuine gravity-mediated entanglement is produced. On the constructive side, Trillo and Navascués established that a Diósi–Penrose-type classical gravity, treated as a *correlated-noise* mechanism, *can* generate gravity-induced entanglement in that specific model class. In other words, once one admits common-medium or correlated-noise structure between the probe masses, the slogan "classical gravity cannot entangle" is no longer generally valid.

Three disciplined claims about ECT. (A) ECT does not take the observation of gravity-induced entanglement in a mesoscopic protocol, by itself, to force a fundamentally quantised independent metric field; once common-medium loophole classes are admitted, the standard no-go inference applies most directly to strictly local factorising classical channels. (B) ECT structurally allows gravity-induced entanglement to arise whenever the experimentally relevant reduced dynamics still retains common-medium non-factorisable structure — that is, when the protocol probes a Level IIa regime rather than a Level IIb regime. (C) ECT does not yet establish, from first principles, which exact experimental protocols realise Level IIa, nor does it yet derive the explicit reduced mediator channel through which entanglement would be generated (*Level Open*).

Operational criterion. The architectural position of ECT in the BMV debate can be stated sharply. ECT falls under the standard no-go logic only if, in the actual regime realised by the experiment, the effective reduced mediator is well approximated by a strictly local classical factorising channel with no residual common-medium correlations. ECT *escapes* that logic if the experimentally relevant reduced dynamics still retains correlated common-medium structure — i.e. belongs to the Trillo–Navascués loophole class. Different protocols probe different reduced regimes of one underlying condensate theory; ECT does not claim both descriptions for one and the same operational regime.

Where the crossover lives. A useful order-of-magnitude guide is obtained by equating the scalar-sector reduced decoherence timescale, $\tau_{\rm dec}^{\rm grav} \sim \hbar R/(G_N m^2)$, with the experimental coherence time $\tau_{\rm exp}$, yielding a heuristic mass scale

$$m_{\rm guide} \sim \sqrt{\hbar R/(G_N \tau_{\rm exp})}.$$

For typical near-future levitated-nanoparticle protocols this lies in the broad mesoscopic range 10^{-16}–10^{-14} kg, depending on the setup. However, $m_{\rm guide}$ is only a scalar-sector orienting scale inherited from the distinguishability channel; it is *not* a first-principles crossover mass for the entangling channel itself. A genuine entangling-channel crossover requires derivation of the orientation-sector characteristic scale, which remains an open item. The availability of a loophole class, as shown by Trillo–Navascués, is not the same as demonstrating that a specific experimental realisation of ECT occupies that class.

40. How gravity talks to matter: two channels of the condensate mediator

To make the previous discussion concrete, it is useful to reorganise the effective gravitational mediator of ECT into two physically distinct channels. This taxonomy does not introduce a new sector; it makes explicit what was already implicit in the condensate graviton programme and clarifies which channel does what.

The ϕ-channel: scalar distinguishability. A matter distribution acts on the condensate amplitude through the quasi-static scalar perturbation $\delta\phi$; when a matter system is placed in a superposition of positions, the two branches source two distinguishable condensate configurations. The characteristic timescale at which those two reduced configurations decohere through this scalar channel is the Diósi–Penrose-like estimate $\tau_{\rm dec}^{\rm grav} \sim \hbar R/(G_N m^2)$ discussed above. In the minimal reduced-state reading developed here, the ϕ-channel is treated primarily as a *distinguishability / decoherence channel* rather than as the primary entangling channel.

The δn_A-channel: orientation / tensor candidate mediator. The orientation fluctuations δn_A project onto the symmetric effective-metric channel; they are the ECT-native orientation sector from which the massless spin-2 excitation (the graviton) is extracted at the linearised ordered-branch level. A tensor channel that couples simultaneously to two spatially separated mass distributions is the natural candidate within ECT for the common-medium correlations relevant to gravity-induced entanglement. However, not every δn_A fluctuation automatically plays the role of the physical entangling mediator in a given protocol; what is required is the retarded, common-medium component of the orientation response that survives the reduction map without collapsing to a merely factorising classical channel.

Why the distinction matters. One of the most useful lessons of this taxonomy is retrospective: the heuristic $m_{\rm guide}$ of the previous section is inherited from the *scalar* channel, not from the entangling channel. This explains why it cannot be quoted as a first-principles prediction for gravity-induced entanglement: the scalar and tensor channels are different physical beasts with different characteristic scales. Beyond the minimal split into ϕ- and δn_A-channels, a mixed scalar–orientation sector is expected in general, where decoherence and entangling capacity dress one another. The two-channel taxonomy should therefore be read as a minimal organising skeleton, not as a complete diagonalisation of the reduced mediator. Deriving the tensor/orientation channel from the condensate influence functional and extracting its characteristic retarded timescale remain key *open tasks* (*Level Open*).

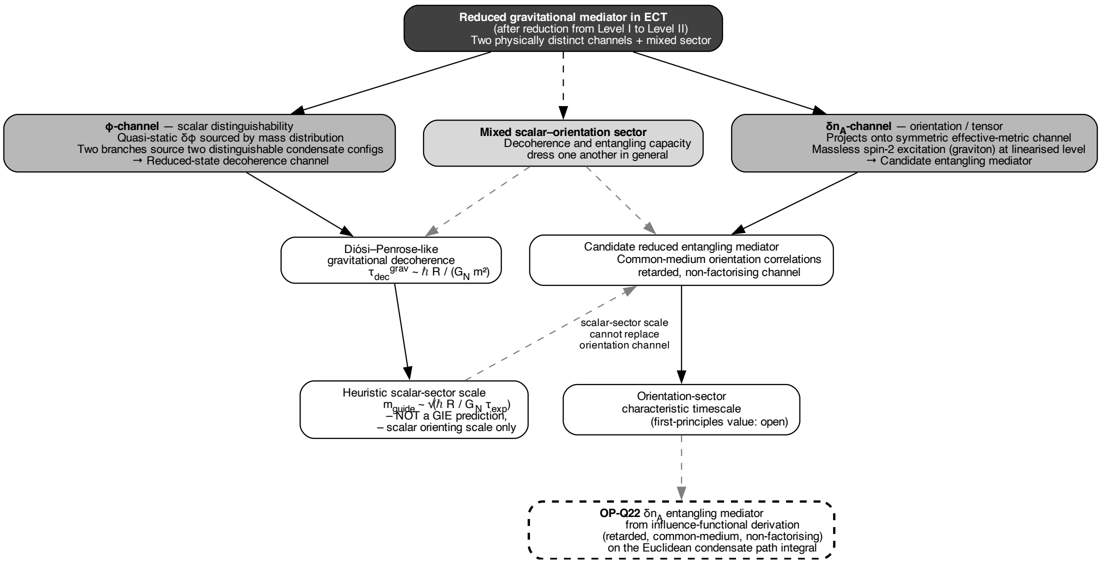


Figure 22: Channel taxonomy of the reduced gravitational mediator in ECT. The reduction from the full Level I condensate to the effective description yields two physically distinct channels. The ϕ-channel is the scalar distinguishability channel: two matter branches of a superposition source distinguishable condensate configurations, and the resulting Diósi–Penrose-type reduced-state decoherence sets the scalar-sector timescale $\tau_{\rm dec}^{\rm grav} \sim \hbar R/(G_N m^2)$ and the heuristic orienting mass scale $m_{\rm guide} \sim \sqrt{\hbar R/(G_N \tau_{\rm exp})}$. The crucial point is that this scale is inherited from the scalar distinguishability channel and *cannot* be quoted as a first-principles gravity-induced-entanglement prediction. The δn_A-channel (orientation/tensor) is the ECT-native candidate entangling mediator, whose characteristic retarded common-medium timescale is left open as OP-Q22. A mixed scalar–orientation sector dresses both channels beyond the minimal split. (*Level B; entangling-channel derivation: Open.*)

41. The infrared sky: asymptotic symmetries, memory, and soft gravitons

Modern general relativity has sharpened an old intuition: at very long wavelengths, near null infinity, there lives a rich structure of *asymptotic symmetries*, *gravitational memory*, and *soft gravitons* that goes far beyond the naive Minkowski-at-infinity picture. The asymptotic symmetry group discovered by Bondi, van der Burg, Metzner, and Sachs (BMS) is an infinite-dimensional extension of Poincaré symmetry; gravitational memory is the permanent shift left behind in the metric at null infinity after a burst of gravitational radiation passes; soft graviton theorems connect zero-energy gravitons to both BMS supertranslations and memory through the modern "asymptotic symmetry / soft theorem / memory" triangle.

In ECT, this infrared sector has a natural reading (*Level B/C, structural*). The relevant discussion is restricted to asymptotically flat ordered-branch settings, where null-infinity language is meaningful. In that arena, the asymptotic ordered-branch data $(\phi_0, n_A^{(0)})$ provide the natural stage on which any BMS-type asymptotic symmetry analysis of ECT would be formulated; structural compatibility is claimed, but a first-principles derivation of the BMS group from the condensate action remains an open programme-level item. Gravitational memory has a direct ECT-native interpretation: a passing gravitational-wave burst leaves behind a *persistent shift* $n_A^{(0)} \to n_A^{(0)} + \Delta n_A$ in the asymptotic orientation of the ordered branch. The long-wavelength δn_A modes introduced in the previous section are the natural ECT-native soft gravitons: their identification with the massless spin-2 mode, already established in Part I, is consistent with their role as the soft excitations whose emission leaves the Hawking–Perry–Strominger "soft hair" on a horizon.

Caveat. The existence of soft modes does not, by itself, solve any information problem; it only identifies a potentially relevant low-energy bookkeeping sector. Whether the long-wavelength δn_A modes suffice to store and recover the information content required by black-hole evaporation is not established here and is not claimed.

42. Black-hole thermodynamics and the information problem

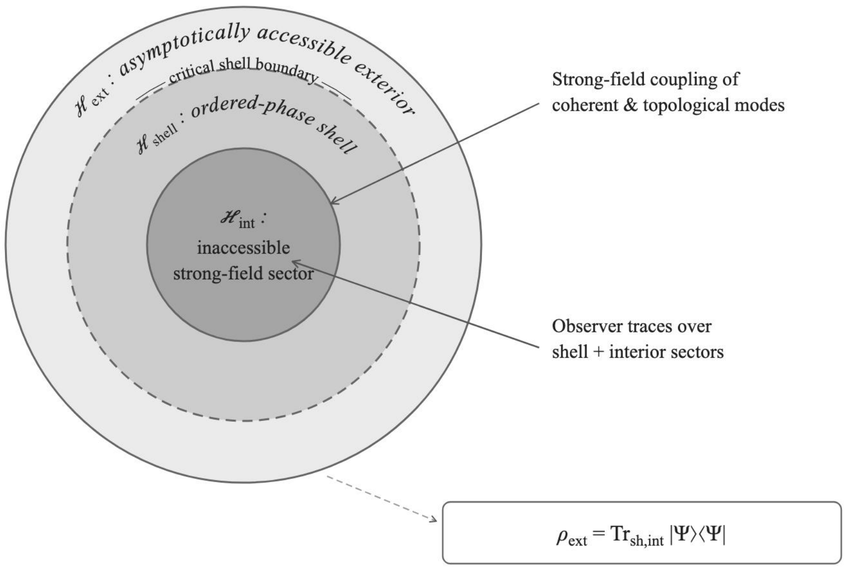


Figure 23: The ECT picture of a black hole. The classical event horizon is replaced by a critical shell where the condensate amplitude reaches a critical value and the effective geometry transitions between regimes. Information is encoded in shell correlations. (*Structurally fixed at postulate level: no fundamental information loss. Level B: shell closure and near-horizon construction.*)

ECT reframes the black-hole horizon not as a fundamental singularity of causal structure but as an emergent interface in the condensate medium. This provides a microscopic language for why semiclassical Hawking thermality may survive while allowing controlled corrections. The present section organises the mechanism and its observable channels; it does not yet provide a complete first-principles microtheory of black holes.

If information falls into a black hole and the black hole evaporates via Hawking radiation, is the information lost? This is the information paradox — one of the deepest problems in theoretical physics.

Ontological reframing. The information paradox assumes, often implicitly, that Lorentzian spacetime is the fundamental arena, that the horizon is a fundamental geometric boundary, and that the inside/outside split is a fundamental division of degrees of freedom. ECT changes the status of all three assumptions. The primitive object is the four-dimensional Euclidean condensate Φ; Lorentzian geometry, horizons, and the inside/outside distinction arise only in the ordered-branch effective description. They are physically meaningful for exterior observers, but they are not primitive features of the underlying ontology. This reframes the paradox. Hawking thermality is read as reduced-state thermality,

$$\rho_{\text{ext}} = \text{Tr}_{\text{shell,int}} \, |\Psi\rangle\langle\Psi|,$$

not as fundamental destruction of the full condensate state. The question "is information lost?" therefore splits into two questions. At the structural level, fundamental information destruction is not part of ECT's law structure. At the observer-accessible level, the recovery of correlations in the outgoing radiation remains the Page-curve problem, to be addressed by the critical-shell closure.

ECT addresses it at two levels (Fig. 23).

Level A (structural): The total condensate state is never fundamentally mixed. In ECT, the apparent mixedness (thermal character) of Hawking radiation is a *reduced-state* phenomenon: the external observer's density matrix $\rho_{\text{ext}} = \text{Tr}_{\text{shell,int}}|\Psi\rangle\langle\Psi|$ is mixed because it traces over the shell and interior degrees of freedom, not because information has been destroyed. A conditional theorem establishes that exact thermality of the radiation is incompatible with unitary evaporation: if the condensate evolves unitarily (as posited), the radiation *must* carry subtle correlations encoding the infalling information. This is a structural consequence of the postulate architecture: fundamental information destruction is not part of the basic law structure of the theory.

Level B (shell closure): The event horizon is replaced by a **critical shell** — a surface where the condensate amplitude reaches a critical value and the effective ordered-phase geometry transitions between regimes. Shell entropy scales with area: $S_{\text{shell}} \sim A/\ell_*^2$ (structural area law; the exact coefficient 1/4 is an open matching condition, not yet derived from first principles). A finite-memory shell model reproduces a Page-curve-like behaviour. These results are model-based (Level B), not yet a full first-principles derivation of black-hole evaporation. The classical singularity at $r = 0$ is replaced by a Euclidean core where the condensate transitions from the Lorentzian ordered phase back to the Euclidean disordered phase.

Three-regime structure. The near-horizon condensate separates into three regimes: (i) the exterior ordered Lorentzian regime, where standard effective geometry is valid and deviations from GR are parametrically small ($\rho_c/r_s \sim \ell_{\text{Pl}}/r_s \ll 1$); (ii) the critical shell / interface regime, where the ordered-branch amplitude transitions from its vacuum value to zero; and (iii) the pre-Lorentzian core, where the metric language ceases to be fundamental.

Topological consistency. The vacuum manifold $O(4)/O(3) \cong S^3$ has $\pi_2(S^3) = 0$, so the critical shell carries no topological charge and can form and dissolve without topological obstruction. The nontrivial sector $\pi_3(S^3) = \mathbb{Z}$ admits possible topological configurations, but the present work does not derive a stabilising Skyrme-type term or a finite-energy gravitating topological black-hole solution.

What is not yet derived. The present shell analysis does not yet derive the full interior profile, first-principles horizon thermodynamic coefficients, or a constructive Page curve.

Comparison with the islands / quantum-extremal-surface language. The modern holographic programme approaches the same black-hole information question through a different language: the *islands formula* and the notion of *quantum extremal surfaces* (QES). In that language, the entropy of Hawking radiation is computed as a generalised entropy $\mathrm{Area}(\partial\Sigma)/(4\,G_N) + S_{\mathrm{bulk}}(\Sigma)$, extremised over bulk regions; before the Page time the minimal QES is trivial, while after the Page time a disconnected "island" inside the black hole dominates and produces the Page-curve turnover. The ECT critical-shell picture and the islands/QES picture make structurally compatible statements (*Level B/C structural*): both read apparent Hawking thermality as a *reduced-state* phenomenon under globally unitary dynamics, the island region plays the same role as the interior sector hidden behind the ECT critical shell, and the QES boundary $\partial\Sigma$ plays the same role as the shell boundary between the exterior and the interior sectors. Crucially, the critical-shell boundary is *not* identified here with a derived quantum extremal surface, and ECT does not derive the generalised-entropy formula or carry out a replica-wormhole-type calculation on its underlying Euclidean condensate path integral; those remain open tasks. The point of the mapping is compatibility of structural logic, not an ECT derivation of the Page curve.

43. Why entropy scales with area: the holographic bound as a medium property

One of the deepest lessons of black-hole physics is that the maximum information content of a region of space scales not with its volume but with the area of its boundary. This observation, first made by Bekenstein and later elevated by 't Hooft and Susskind into the *holographic principle*, has shaped much of modern theoretical physics. In string theory it appears as the AdS/CFT duality; in loop quantum gravity it emerges from a counting of quantum-geometric states. In both cases, however, area scaling is tied to specific features of the gravitational theory.

ECT offers a different perspective. In the condensate picture, area scaling is not a gravity-specific property at all. It is a **structural property of any local medium with finite microscopic resolution**.

The physical mechanism. Imagine dividing a region of the ordered condensate into "inside" and "outside" by drawing a closed surface Σ. An observer outside Σ cannot directly access the interior degrees of freedom. All that reaches the observer is mediated through a thin *boundary layer* of condensate cells straddling Σ.

Why only a boundary layer? Because the condensate is local: each cell talks only to its immediate neighbours, and the massive radial mode σ ensures that correlations decay exponentially over a distance $\xi_{\mathrm{cond}} \sim \ell_* \sim m_\sigma^{-1}$ (the condensate correlation length, comparable to the Planck length under the matching hypothesis). Cells deep inside the bulk have no direct line of communication with the exterior—their influence is screened by the intervening medium.

The number of independent cells in this boundary layer scales with the area of Σ divided by ℓ_*^2. Since each cell carries at most a finite amount of information, the total accessible information is bounded by an area-type formula:

$$S_{\mathrm{red}}(\Sigma) \;\le\; c_0\,\frac{A(\Sigma)}{\ell_*^2}, \qquad c_0 = O(1). \tag{13}$$

This is the **general boundary-layer area bound** of ECT (*Level A for the scaling; Level B for the exact coefficient*).

Why this is not just a restatement of the QFT result. In ordinary quantum field theory, entanglement entropy across a surface also scales with area—but with an ultraviolet-divergent coefficient that depends on a regulator. The result is mathematically correct but physically incomplete: the cutoff is artificial and its value is arbitrary.

ECT upgrades this. The UV regulator is replaced by the *physical* condensate threshold $\ell_* \sim m_\sigma^{-1}$, which is determined by the medium itself. The area law therefore becomes a finite, physically meaningful entropy bound rather than a formal divergence.

Black-hole entropy as a special case. The shell-entropy result discussed earlier, $S_{\text{shell}} \sim A \ln q/\ell_*^2$, now appears as a specific instance of the general area bound. The shell is just the boundary layer closest to the horizon, and its entropy counts the condensate cells at that interface. The exact Bekenstein–Hawking coefficient $1/(4\ell_{\text{Pl}}^2)$ represents a matching condition between condensate parameters and gravitational units—not yet derived from first principles, but consistent.

Strengthening the Page-curve programme. The general area bound has an important consequence for the information paradox. If the shell entropy is area-bounded, then the shell carries only a *finite* information reservoir. This finiteness was previously assumed in the conditional no-information-loss theorem; now it follows from the general area-law mechanism. The theorem's conclusion—that exact Hawking thermality is incompatible with unitary evaporation—is therefore on firmer structural ground. A complete Page-curve derivation still requires additional dynamical ingredients (information return mechanism, purification dynamics), which remain open.

Connection to decoherence. The same boundary-layer logic that bounds entropy also underlies decoherence. When a subsystem is embedded in the condensate medium, the inaccessible degrees of freedom beyond the interface act as an effective environment. Both the entanglement entropy and the decoherence rate are controlled by the same interfacial coupling layer. This is not a coincidence but a structural feature of the reduced-description framework: entropy and decoherence are two manifestations of the same boundary-mediated tracing-out.

From area law to holography. It is important not to overclaim. ECT establishes that reduced accessible entropy scales with area rather than volume—a holographic-*style* entropy bound. What ECT does not yet establish is:

- a covariant entropy bound analogous to the Bousso bound for dynamical horizons;
- a full bulk-boundary reconstruction (a gravity/boundary duality of the AdS/CFT type);
- a complete holographic principle in the strongest reconstruction sense.

The correct summary is:

ECT provides a medium-native structural route to holographic-style entropy bounds without postulating holography as a separate principle. Full holography in the strong reconstruction sense remains open.

How ECT differs from other approaches. In AdS/CFT, area scaling arises from a duality between a gravitational theory in the bulk and a conformal field theory on its boundary. In loop quantum gravity, it arises from counting discrete quantum-geometric states. In ECT, area scaling arises from a much simpler mechanism: boundary-layer locality in a continuous ordered medium with a physical ultraviolet threshold. No extra dimensions, no gravity/gauge duality, no discrete quantum geometry. ECT suggests that the holographic character of nature is not a deep mystery of quantum gravity but a generic interface property of a local ordered medium with finite microscopic resolution.

What remains open. The exact microscopic entropy capacity per UV cell, the detailed treatment of soft and gauge-edge sectors, and the covariant generalisation of the bound to null hypersurfaces all remain open. These are important targets for the programme, but they do not weaken the already established structural area-scaling mechanism.

44. Why electric charge is quantised

One of the oldest puzzles in fundamental physics is why electric charge comes in discrete units. In standard electrodynamics, the gauge group $U(1)$ can in principle be taken either as compact (the circle) or noncompact (the real line). Only the compact choice forces charges onto a discrete lattice; the noncompact choice allows a continuous range of charges. Within the standard framework, which option nature realises is simply an empirical fact.

ECT removes this ambiguity structurally. The Abelian gauge sector is not postulated independently: it arises from the compact phase of the condensate,

$$\theta \sim \theta + 2\pi.$$

There is no consistent way to obtain a noncompact $\mathbb{R}$ gauge group from a field whose phase is topologically a circle. Compactness is therefore not an extra assumption; it is built into the condensate architecture.

The consequence is immediate. Any matter excitation transforming under this compact $U(1)$ must carry a charge that is an integer multiple of the elementary condensate charge unit e_0:

$$q = n\, e_0, \qquad n \in \mathbb{Z}.$$

Continuous charges are simply not representable. In this sense, Abelian charge quantisation is a structural consequence of the compact-phase origin of the gauge sector (*Level A*).

A second, independent route reaches the same conclusion through the canonical structure: the charge operator conjugate to the compact phase has a discrete spectrum for the same topological reason (*Level A/B*).

What this does not yet explain. This compact-$U(1)$ result gives the Abelian charge lattice $q \in \mathbb{Z} \cdot e_0$, but the *observed* charge pattern of the Standard Model is richer: quarks carry charges $\pm e/3$ and $\pm 2e/3$, while leptons carry integer multiples of e. These values arise only after electroweak embedding,

$$Q = T_3 + \frac{Y}{2},$$

combined with the colour sector. Deriving the full observed charge spectrum therefore still requires hypercharge quantisation, anomaly-consistent matter representations, and the global structure of the full gauge group. ECT already provides the Abelian foundation; the full Standard-Model charge pattern remains part of the broader gauge/flavour programme. The preprint now additionally records a colour-interlock consistency layer: for completions with the standard $\mathbb{Z}_6$-type global quotient, hypercharge, colour and weak data must satisfy an explicit congruence, which fixes the fractional electric pattern per colour class ($Q \equiv -t/3 \pmod 1$) and reduces the residual freedom to an explicitly characterised discrete family. This is a consistency requirement on completions, not a derivation of the observed hypercharges.

45. Why the post-Big-Bang Universe developed the way it did

ECT already offers a new answer to one part of the Big-Bang question: the Big Bang is not the beginning of all reality but the boundary at which an ordered Lorentzian branch emerges from a pre-temporal Euclidean regime. Before that boundary there is no physical time in the usual sense.

But this still leaves a second question: once the ordered branch has formed, why does the Universe then evolve in the way we observe?

ECT does not yet solve the full transition dynamics. In the main preprint this open problem is now decomposed into four parts of very different difficulty: the ordering mechanism itself, the orientation of the branch (why expansion), the thermal completion into a hot Big-Bang history, and the observable relics of the ordering event. The orientation question has recently been sharpened in two steps. First, the homogeneous background equations turn out to be blind to the sign of expansion on their own: until a thermodynamic arrow of time is fixed, an "expanding" and a "contracting" description are two parametrisations rather than two physically distinct options, so part of the old question dissolves. Second, once the arrow of time is fixed, a conditional persistence argument applies: under explicit, stated assumptions (positive condensate energy, no ghost modes, spatially flat retained background, no internal bounce mechanism), an ordered branch born contracting cannot turn around and is driven back toward the boundary where the ordered description itself fails, while an expanding branch moves away from that boundary. A long-lived ordered cosmology is therefore expected to be an expanding one. This is a conditional criterion, not a first-principles derivation: the ordering mechanism itself and the thermal completion of the hot Big-Bang history remain open.

What ECT *can* already do is classify the kinds of post-transition histories that are allowed or excluded once the ordered branch exists. The observed history belongs to a particularly natural class:

- ordinary radiation and matter dominate early;
- the condensate background stays nearly frozen during the early epochs, preserving standard cosmology;
- the residual condensate energy is positive and becomes important only late;
- the effective equation of state stays at or above $w = -1$, so the late acceleration is moderate.

In condensed-matter systems, when a disordered phase transitions to an ordered one, ordered domains generically *grow* — they expand into the surrounding disordered medium. In ECT the ordered Lorentzian branch emerging from the disordered Euclidean medium is structurally analogous to such a growing ordered domain. This makes an expanding ordered branch physically natural once ordered-domain formation begins, though it does not yet amount to a first-principles derivation of expanding-branch selection over the contracting alternative.

Other branches are also possible in principle. A contracting branch is kinematically admissible but disfavoured both by the growth analogy and by the conditional persistence argument above. A future recollapse is possible if the condensate support weakens. Strong phantom-like evolution ($w < -1$) is excluded by the no-phantom bound of the healthy ordered branch.

The theory therefore already reduces the question to a much sharper one: instead of asking only "why did the Big Bang happen?", it asks which ordered-branch cosmologies are structurally natural, which are fine-tuned, and which are excluded. The observed history sits in the "structurally natural" class. In other words, ECT already offers a geometric origin of the Big-Bang branch, but not yet a complete thermal origin of the hot radiation bath that must follow it.

46. The grandfather paradox

ECT's emergent view of time dissolves the grandfather paradox structurally.

Classical causal paradoxes of time travel arise from trying to maintain two incompatible assumptions: that one can move backward along the physical time axis, and that causal chains remain linearly irreversible. The contradiction appears because the traveller is imagined as re-entering an already fixed causal sequence from within that sequence.

In the Euclidean picture, the coordinate w is not a fundamental causal axis — it is a spatial coordinate of the underlying substrate. Moving "backward in w" is not literal motion into an already-existing past; it is motion in the deeper Euclidean medium. The segment of w that the agent leaves no longer contains that agent as an independent continuation, and worldlines can cross in the Euclidean picture without producing a contradiction of the classical type. Causal order belongs to the Lorentzian branch, where it emerges through the cone structure and where effectively irreversible macroscopic histories appear when $\Gamma_{\text{loop}} \gg 1$.

The paradox is not "solved" by imposing a rule such as "the universe forbids inconsistencies." It is dissolved by recognising that Lorentzian causal intuitions have a limited domain of validity: they apply within the emergent branch, not at the fundamental Euclidean level.

This is a conceptual statement about the ontology of time in ECT, not an operational claim that macroscopic backward time travel is physically available within the emergent Lorentzian branch.

47. Analogue laboratory programme

If ECT is correct, then superfluid liquids, Bose–Einstein condensates, and other condensed-matter systems are not merely "analogies" but realisations of the same condensate physics at a different scale: ordered medium $\to$ symmetry breaking $\to$ emergent Lorentzian kinematics $\to$ vacuum effects. Experiments in these systems are therefore tests of condensate vacuum physics at accessible scales. Key platforms include:

Superfluid ^{4}He and Bose–Einstein condensates: phase-winding quantisation, vortex dynamics, Goldstone spectrum, and the condensate-mediated correlation structure.

Acoustic black holes ("dumb holes") in BECs: analogue Hawking effect and critical-shell structure.

Condensate-reflecting boundary experiments: if a system can be designed whose boundaries genuinely reflect the soft condensate sector, it would test the boundary-sensitive vacuum response predicted by ECT. Standard metallic-plate Casimir experiments do not probe this channel.

Atom interferometry: gravitational decoherence channel predicted by ECT.

48. Conclusions of Part III

48.1. Principal results

From the coherent branch, ECT derives or structurally motivates:

1. Hilbert-space route via reflection positivity and OS reconstruction in the verified Gaussian/free quadratic coherent subsectors (A/B); full interacting coherent branch and complete canonical operator algebra remain *B/Open*.
2. Schrödinger and KG equations as descendants (A/B).
3. Superposition, interference, Pauli exclusion (A/B).
4. Quantisation from compact-phase winding (A).
5. Abelian charge lattice from compact $U(1)$ gauge origin (A).
6. PES as organising principle (B).
7. Born-type weighting route from RP inner product plus Gleason/decoherence assumptions (*B/Open*; exact Born rule open).
8. Measurement through reduced-sector decoherence (*B/Open*; single-outcome and pointer-basis selection open).
9. Entanglement as common-medium correlation (A/B).
10. Tunnelling as Euclidean passage (A/B).
11. Casimir boundary-sensitive vacuum response (*B, conditional*).
12. Unruh temperature (A/B).
13. Gravitational decoherence channel: DP-type leading scale with ECT-specific condensate-origin interpretation (B).
14. No fundamental information loss in BH (*structural, postulate-level*).
15. BH entropy $\propto$ area; exact coefficient open (B).
16. General boundary-layer area-law bound (*A for scaling; B for coefficient*).
17. Page curve from shell model (B).
18. Quantum gravity structurally reframed by the single-condensate architecture; full UV and spin-2 closure remain open (*B/Open*).

48.2. Key equations and principles of Part III

Winding condition: $\oint \partial_A \theta \, dX^A = 2\pi n$.

Elementary loop action: $S_{\text{loop,min}} = 2\pi\hbar$.

PES: persistent configurations = extremal $\cap$ decoherence-resistant $\cap$ phase-compact.

Decoherence time: $\tau_{\text{dec}} = 1/(N_{\text{eff}}\gamma)$.

Casimir: boundary-sensitive vacuum response; provisional 3/2 factor conditional on soft-mode counting and condensate-reflecting boundaries.

BH entropy: $S = A/(4\ell_{\text{Pl}}^2)$.

General area-law bound: $S_{\text{red}}(\Sigma) \leq c_0 \, A(\Sigma)/\ell_*^2$, $c_0 = O(1)$.

48.3. Part III derivation logic

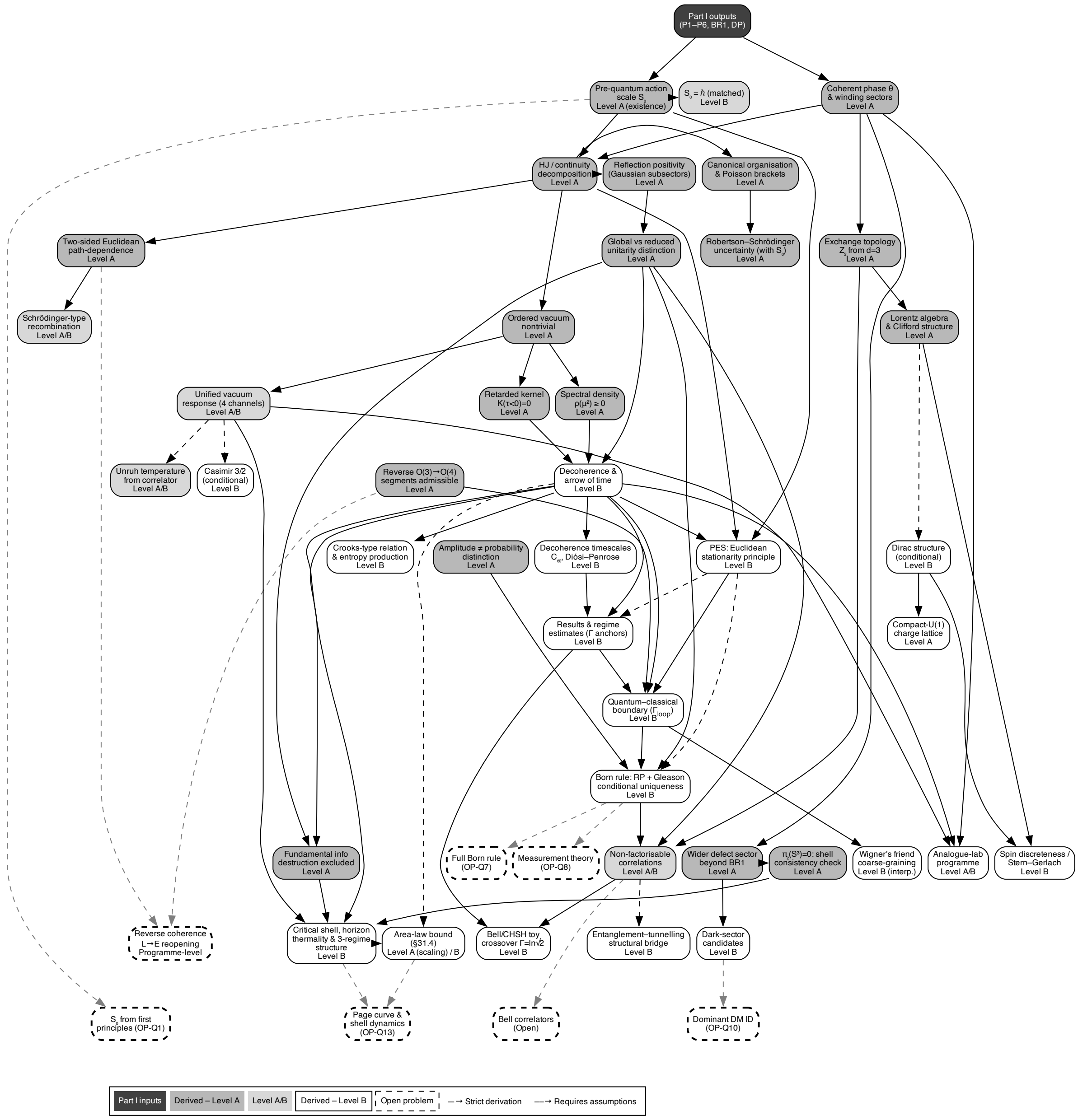


Figure 24: Part III derivation logic, updated with the QG/decoherence upgrade layer. The coherent branch develops structural routes to Hilbert-space structure, wave equations, Born-type weighting, decoherence, PES, vacuum effects, black-hole thermodynamics, and measurement-status analysis. Born weights remain conditional on the RP/Gleason/decoherence route, and a complete measurement-update theorem remains open. The reduced-state upgrade layer comprises the hybrid-consistency architecture and two-level ontology (§38), the gravity-induced-entanglement operational criterion (§39), the ϕ vs δn_A mediator taxonomy (§40), the BMS/memory/soft-graviton infrared sector (§41), the islands/QES structural mapping with the critical shell (discussed within the BH section), and the open orientation-sector entangling-mediator task (OP-Q22). Colours indicate derivation levels.

Figure 24 shows the complete derivation architecture of Part III.

Part IV

Summary, Comparisons, and Outlook

49. Architectural comparison with neighbouring programmes

ECT is not the only attempt to address the foundational questions of physics. The following comparison locates it among its closest architectural neighbours.

General Relativity + Standard Model: Time and Lorentzian structure are fundamental. Gauge group is postulated. Quantum axioms are primitive. Dark sector requires extra particles + Λ. BH information paradox is open.

Einstein–aether theory (Jacobson, 2004): Lorentzian structure is fundamental; a preferred frame is added via a postulated unit vector u^A. In ECT, the orientation n^A is derived from $\partial_A \Phi$, not postulated.

Hořava–Lifshitz gravity: Lorentz invariance is broken *explicitly* at the fundamental level. In ECT, it is broken *spontaneously* from a higher symmetry.

Volovik's "Universe in a Helium Droplet": Spacetime emerges from a condensate, but the analogy remains conceptual. ECT proposes that the condensate mechanism is the actual physical mechanism.

Causal Dynamical Triangulations: Lorentzian structure is imposed as a lattice constraint. ECT derives the Lorentzian signature and ordered-branch causal structure from continuous SSB.

Hartle–Hawking no-boundary proposal: The Euclidean arena is a computational tool for defining path integrals. In ECT, it is the fundamental ontological entity.

MOND (Milgrom, 1983): The critical acceleration $g^\dagger$ is introduced as a universal free constant. In the present ECT galactic closure, the corresponding effective scale is tied to the ordered-branch condensate response and may acquire environmental modulation. This is a Level B closure-level handle, not yet a first-principles derivation of a universal environment law.

Superfluid dark matter models: Spacetime is pre-existing; a superfluid component is added as dark matter. In ECT, spacetime and the "superfluid" dynamics arise from the same condensate.

In all cases, ECT distinguishes itself by relocating structures that other theories postulate into a single condensate framework: some are derived, some are reconstructed, some are reinterpreted, and some remain explicit closure targets.

50. Distinctive features of the framework

Several features distinguish ECT from all other approaches:

Single-condensate architecture: Spacetime, gravitational response, quantum-theoretic structure, gauge-sector routes, and dark-sector phenomenology are all placed within one scalar-condensate architecture, but with different logical status: some parts are strict structural consequences, some are closure-level reconstructions, and others remain open completion targets.

No external time: Time is derived from $O(4) \to O(3)$ SSB, not assumed. This is stricter than most quantum gravity approaches.

"How" and "why": ECT reproduces quantum equations *and* explains why they take the form they do (compact phase structure, reflection positivity, boundary-value vs. Cauchy problem).

Honest epistemology: Every result is classified as Level A, B, or Open. No overclaiming.

Cross-sector role of S_0: The same action scale organises Euclidean phase weighting, wave kinematics, and canonical normalisation across the coherent branch. Born-type weights inherit their quadratic structure only after the RP/Gleason/decoherence route is in place, so they should not be treated as an independent S_0 theorem. If different quantum layers required incompatible action scales, the present single-action-scale closure would be strongly constrained or ruled out.

Unified dark-sector reinterpretation: Galactic dark-matter phenomenology and late-time dark-energy phenomenology are both described through condensate dynamics, without requiring a particle-DM carrier for the galactic mass-discrepancy channel or a rigid external cosmological constant as independent primitives.

Common pre-geometric architecture of interactions: Gravity, the Abelian gauge route, the candidate weak sector, the colour sector, and the predicted director coupling are all organised within the same condensate architecture, but with different logical status. Gravity arises from the ordered condensate geometry. The compact phase fibre gives a strong Abelian bundle route. The weak sector has a candidate orientation/Hopf-locking route, but the full local chiral $SU(2)_L \times U(1)_Y$ Standard-Model realisation remains open. The colour sector requires extended internal structure. This is broader than conventional Grand Unification, which does not include gravity in the same condensate framework. The weak-sector parity asymmetry acquires a structural route from the condensate orientation, but the maximally parity-violating Standard-Model pattern remains an open derivation problem.

Director coupling: A fifth interaction channel—the lowest-dimension fermion–condensate operator permitted by the symmetry content—is a generic consequence of the ordered medium. It provides testable predictions (spin precession, composition-dependent effects) absent from the Standard Model.

Minimal structural parameter count: The primary condensate sector is controlled by the parameters μ^2, λ, α and β; in the canonical benchmark $(\alpha, \beta) = (2, 1)$ the primary sector reduces to two independent bare scales. Many low-energy structures are then linked by matching relations rather than introduced as independent primitives. This economy should not be confused with a fully parameter-free phenomenology: the cosmological diagnostic layer, galactic environmental response, and several particle-sector closures remain Level B/C inputs or open derivation targets.

51. Standard postulates reinterpreted as ECT consequences

A large fraction of structures that standard physics treats as independent axioms, postulates, empirical inputs, or fundamental constants are relocated in ECT into a single scalar-condensate architecture. Some appear as strict structural consequences, some as closure-level derivations, some as matching relations, some as condensate-response reinterpretations, and others as explicit B/Open or Open reconstruction targets. The following compact table collects the most representative cases. A full version with all entries and section references is given in the technical preprint (§51 corresponds to the preprint's "Comprehensive map"; see there for the extended seven-group classification). Status labels: A = strictly derived; B = derived under stated assumptions or closures; C = conditional/target; Open = not yet established.

Standard postulates vs. ECT consequences (compact version).

Standard physics	Status there	ECT origin	Level
Foundations and spacetime			
Lorentzian signature $(-,+,+,+)$	Fundamental input	SSB $O(4) \to O(3)$ with $\alpha > \beta$ selects the hyperbolic ordered branch	A
Time as a distinguished direction	Fundamental	Emergent direction w fixed by the gradient-ordered vacuum	A
Arrow of time / second law	Open / empirical	Retarded propagation + $\Gamma_{\rm irr} \geq 0$ from influence-functional kernel	B
Ordered-branch causal speed / observed c	Fundamental speed constant	$c_*^2 = \beta/(\alpha - \beta)$ from ordered-branch EFT; empirical identification $c_* = c$ is a matching step	A/B
Newton's constant G_N	Fundamental constant	EFT matching $G_N = c_*^4/(8\pi M_G^2)$	B
Planck's constant $\hbar$	Fundamental constant	Elementary-loop action scale S_0; identification $S_0 = \hbar$	B
Quantum mechanics postulates			
Complex Hilbert space / superposition	Axioms	RP/OS reconstruction in verified Gaussian/free quadratic coherent subsectors; full interacting coherent branch remains a reconstruction target	A/B/Open
Schrödinger equation	Axiom	Non-relativistic limit of Klein–Gordon from coherent branch	B
Planck–Einstein relation $E = \hbar\omega$	Quantum postulate / empirical identification	Coherent stationary eigenmodes in the Schrödinger-type reconstruction give $E = S_0\omega$; Level-B matching $S_0 = \hbar$ yields $E = \hbar\omega$	B
Canonical commutator $[\hat{q}, \hat{p}] = i\hbar$	Postulate	Compact-phase and coherent-branch Poisson-bracket route normalised by S_0; complete canonical operator algebra remains open	B/Open
Wavefunction collapse	Postulate	Reduced-sector decoherence transition at $\Gamma_{\rm loop} \sim 1$; not a fundamental projection rule; single-outcome update remains open	B/Open
Born rule $P = \|\psi\|^2$	Postulate	Born-type weighting route from RP inner product plus Gleason/decoherence assumptions; exact Born rule remains open	B/Open

Standard physics (cont.)	Status there	ECT origin	Level
Pauli exclusion principle	Postulate	Exchange topology $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ + OS reflection positivity	A / B
Dirac equation	Postulated fermion dynamics	Minimal first-order closure on Spin(3) spinors of the ordered branch	B
Gauge sector			
Abelian gauge sector	Postulated gauge principle	Compact phase fibre supplies a structural $U(1)$ bundle route and a Maxwell-type leading closure; exact placement within $U(1)_Y$ versus $U(1)_{\mathrm{em}}$ belongs to the EW completion	A/B
Weak-sector structure	Postulated chiral $SU(2)_L$	Orientation-sector $SU(2)$ lift plus Hopf-fibered phase–orientation locking gives a candidate pre-gauge geometry with three broken directions; local chiral $SU(2)_L$, hypercharge, g, g' and θ_W remain open	B/Open
Neutrino masses and mixing	Seesaw / Dirac / Majorana (model-dependent)	Geometric chirality selection (A/B); embedded seesaw preferred (C); $M_R^{\mathrm{geom}} = \sqrt{\phi_0 v_2}$ as scale anchor; PMNS intact	A/B–C
$SU(3)$ colour sector	Postulated colour symmetry	Not accessible from the minimal $O(4) \to O(3)$ pattern; requires extended internal condensate structure and full colour-completion construction	B/Open
Gravity, cosmology, vacuum			
Einstein equations	Postulate	Unique rank-2 tensor compatible with broken $O(4) \to O(3)$; linearised Fierz–Pauli to Einstein–Hilbert	A/B
Anisotropic term $\Theta_{\mu\nu}[n]$	Absent in GR	Derived from ordered-branch direction-dependent closure	A(lin.)/B
Late-time acceleration (Λ)	Postulated constant	Ordered-branch background energy, no separate dark-energy substance	B

Standard physics (cont.)	Status there	ECT origin	Level
Cosmological-constant catastrophe	10^{120}-orders vacuum fine-tuning problem (unsolved)	**Direct vacuum-baseline channel blocked at Level A**: the fixed kinetic-tensor determinant $\sqrt{-\det K^{AB}} = \beta^2/c_*$ makes the classical vacuum offset $V(\phi_0)$ inert for local emergent gravity; the zero-derivative loop-vacuum channel is conditional on H_Λ; the observed value of ρ_Λ remains a separate Level C question	Object A: A; Object B: H_Λ-conditional; observed value: C
Observed value of ρ_Λ	Empirical fitted vacuum density	Natural infrared scaling $\sim M_{\rm Pl}^2 H_0^2$; dimensionless coefficient open; a possible pseudo-Goldstone realisation may exist if an ultralight pseudo-Goldstone sector with $m_G \sim H_0$ and $f_G \sim \phi_0$ is realised in the cosmological regime	C
Particle dark matter (galactic)	Postulated halo	Critical ϕ-branch closure reproduces rotation curves and BTFR slope 4	B
Casimir / Unruh / Hawking sector	QFT vacuum/observer effects	Unified ordered-vacuum response route; Unruh-type response is structural, while Casimir boundary response and Hawking/shell closures carry sector-specific B/Open status	A/B/Open
Long-standing mysteries and targets			
Fine-structure constant $\alpha_{\rm fs} = 1/137$	Fundamental constant	Low-energy $\alpha_{\rm fs} = q^2/(4\pi Z_A)$; UV recast as electroweak (Z_B, Z_W, θ_W) target	B/C
Three fermion generations	Empirical input	Bound-state spectral route at $y = \mathcal{O}(1)$	B/C
Proton stability	Symmetry-protected fact	Pattern III benchmark $\tau_p \sim 10^{41}$–10^{42} yr, conditional	C
Baryon asymmetry η_B	Sakharov + model	Ordering-transition + benchmark leptogenesis closure	B
Quantum gravity (UV/spin-2 closure)	Open problem	Single-medium architecture reframes the problem; graviton route and coherent sector share condensate origin, but graviton-loop power counting, spin-2 RP/OS reconstruction, and full UV closure remain open	Open

Standard physics (cont.)	Status there	ECT origin	Level
Measurement problem	Open problem	Structural route through decoherence, PES, and conditional Born-type weights; exact Born rule, pointer-basis selection, and single-outcome selection remain open	Open

The compact table above is a selective extraction: it highlights the most structurally clean cases. The full preprint table adds reflection-positivity details, Fermi–Dirac/Bose–Einstein statistics, graviton structure, Klein–Gordon derivation, Crooks fluctuation theorem, Hubble evolution, S_8 lensing–growth sector, neutrino geometric masses, and the Level A blocking of the direct vacuum-baseline channel of the cosmological-constant problem, together with the H_Λ-conditional loop-vacuum channel and the separate Level C question of the observed value of ρ_Λ. The reading of the table is not that every standard postulate is already fully derived in ECT, but that each one has been reclassified from "input" to "derived or structurally targeted within a single ordered-condensate framework", with its rigour level explicitly labelled. This is the distinctive scope of the ECT programme.

52. Key equations, laws, and principles of ECT

The following equations and principles define ECT not only qualitatively but in its applied, quantitative aspects:

Foundational action: $S_E[\Phi] = \int d^4X\{\frac{1}{2}(\partial_A\Phi)^2 + V(\Phi)\}$, $V = -\frac{\mu^2}{2}\Phi^2 + \frac{\lambda}{4}\Phi^4$.

Gradient condensate: $\langle\partial_A\Phi\rangle = u_0\,\delta_{Aw}$; breaks $O(4) \to O(3)$.

Effective kinetic tensor: $K^{AB} = \beta\,\delta^{AB} - \alpha\,n^A n^B$ (uniqueness theorem).

Ordered-branch causal speed: $c_* = \sqrt{\beta/(\alpha-\beta)}$; the empirical identification $c_* = c$ is a matching step.

Matching relations: $G_N = c_*^4/(8\pi\phi_0^2)$; $\quad \hbar = c_*\phi_0/\sqrt{\lambda}$.

Cross-sector cone universality: $c_{\text{phase}} = c_\gamma = c_{\text{GW}} = c_*$.

Loop quantisation: $\oint \partial_A\theta\,dX^A = 2\pi n$; $\quad S_{\text{loop,min}} = 2\pi\hbar$.

Second-law closure: $dS_{\text{ent}}/dw = N_{\text{eff}}\gamma(dq/dw)^2 \geq 0$ within the influence-functional/environmental closure; the retarded backbone is structural, while the explicit formula is Level B.

Generalised Einstein equations: $G_{\mu\nu} + \Lambda_{\text{eff}}g_{\mu\nu} = 8\pi G_{\text{eff}}T_{\mu\nu} + \Theta_{\mu\nu}[n]$.

Galactic interpolation: $g(R) = \frac{1}{2}(g_N + \sqrt{g_N^2 + 4g_N g^\dagger})$.

BTFR: $v_{\text{flat}}^4 = G_N M_{\text{bar}}\,g^\dagger$ (slope 4, algebraic).

Hubble evolution: $G_{\text{eff}}(z) \approx G_N(1+z)^{2\varepsilon}$, retained $\varepsilon \in [0.0296,\, 0.0376]$ at 1σ.

PES: Persistent configurations = extremal ∩ decoherence-resistant ∩ phase-compact.

Decoherence time: $\tau_{\text{dec}} = 1/(N_{\text{eff}}\gamma)$.

BH entropy: $S = A/(4\ell_{\text{Pl}}^2)$.

Casimir (conditional): $F_{\mathrm{ECT}} = \frac{3}{2} F_{\mathrm{QED}}$ in the provisional three-component soft-orientation EFT with condensate-reflecting boundaries; not a universal laboratory prediction.

53. Falsifiable predictions

ECT does not make just one headline prediction. Its empirical consequences fall into several distinct categories: hard structural discriminants of the framework, consistency requirements that the theory must and does satisfy, stability tests of the retained effective ε-band, viability discriminators such as the age sector under alternative H_0 mappings, and anti-predictions — outcomes that the present uniform-ε layer is not expected to deliver. With this distinction in mind, the items below are specific, testable, and should not all be read in the same way:

1. **BTFR slope = 4 within the retained galactic closure** (*B, closure-level*). This is an algebraic consequence of the ϕ-closure. A robust systematic deviation would challenge the current ECT galactic closure rather than the entire ECT framework.
2. $g^\dagger \propto cH_0$ (B). Links galactic and cosmological scales. Testable by comparing $g^\dagger$ from galaxies with independently measured H_0.
3. **Fifth force:** $\eta \sim 10^{-15}$ (B). Within reach of MICROSCOPE-2 (current: $\eta < 10^{-14}$).
4. **Dark energy:** $w_0 \approx -0.83$ **(calibrated thawing benchmark)** (B). Distinguishable from $w = -1$ by DESI/Euclid. Confirmed high-precision $w = -1$ would rule out the present ECT thawing dark-energy closure.
5. **Environmental modulation of** $g^\dagger_{\mathrm{eff}}$ (*B, closure-level*). A positive correlation with environment would support the present closure; a null result constrains the calibrated environmental response $\gamma\, 4\pi \bar{\rho}_{\mathrm{m,\,env}} / U_0''(\bar{\phi}_{\mathrm{env}})$ and the density-only environment ansatz rather than falsifying the ordered-branch amplitude mechanism as a whole.
6. **Casimir boundary-sensitive vacuum response** (*B, conditional*). The ordered vacuum must respond to boundaries that reflect its soft sector. The provisional 3/2 factor depends on naive Goldstone-coordinate counting and idealised condensate-reflecting boundaries; it is not a universal laboratory prediction for standard metallic plates.
7. **Graviton polarisation: two TT modes in the completed linear closure** (*B/Open*). If the induced effective-metric sector closes exactly to the Fierz–Pauli form, only the two tensor polarisations propagate; robust scalar/vector GW polarisations would disfavour this closure. Exact TT projection and ghost-freedom remain verification tasks.
8. **CMB quadrupole deficit** (*B/Open*). A structural argument for horizon-scale suppression exists, but the quantitative prediction of $\delta C_2/C_2$ remains open.
9. **No particle-DM carrier for the galactic mass-discrepancy channel** (B). A confirmed particle that is required to account for galaxy rotation curves and RAR/BTFR phenomenology would challenge the current ECT galactic closure; discovery of a particle in a hidden sector would not by itself falsify ECT.
10. $w \neq -1$ (B). In the present ECT thawing dark-energy closure, late-time acceleration is dynamical. A confirmed high-precision result $w = -1$ would rule out this thawing closure, rather than the entire ECT architecture.

11. **Planck-suppressed propagation corrections** (*B/Open*). Photon-sector LIV requires a matching coefficient ζ_γ already constrained by Fermi–LAT; the GW-sector correction is quadratic and negligible for current sources ($\Delta t_{\rm GW} \sim 10^{-64}$ s).

12. **Proton stability** (C). Conditional on the gauge/flavour completion: three selection-rule patterns are possible — full low-energy protection (proton stable), partial protection ($\Delta B = 1$ forbidden, n–$\bar{n}$ oscillations possible), or generic Planck-suppressed $\Delta B = 1$ decay ($\tau_p \sim 10^{41}$–10^{42} yr). All consistent with Super-K limit $\tau_p > 2.4 \times 10^{34}$ yr. This is a Level C closure discriminant, not a strict-core prediction.

Near-future observational discriminants. Beyond the predictions already testable with current data, several observational programmes expected to deliver results within the next decade open especially powerful discriminating channels for ECT. *(i)* Full-sky CMB polarisation missions (LiteBIRD, launch expected in the early 2030s) will test whether the primordial tensor spectrum is compatible with any viable condensate-based early-Universe completion of ECT. *(ii)* Precision ground-based CMB lensing and growth measurements (Simons Observatory, Euclid, Rubin/LSST) can probe whether the condensate-induced modifications through $G_{\rm eff}(z)$ and the anisotropic response $\Theta_{\mu\nu}[n]$ leave a detectable signature in the lensing–growth consistency relation. *(iii)* Space-based gravitational-wave detection (LISA, launch planned for 2035) will test the ECT single-cone prediction for GW propagation, search for additional GW polarisation modes beyond the two transverse-traceless modes expected in the linearised ECT graviton sector, and provide independent standard-siren determinations of H_0 that can test the retained-band condensate-driven Hubble shift. A detailed table of near-future discriminants, with explicit ECT statements and falsification verdicts, is given in the technical preprint.

ISW diagnostic. Late-time evolution of the condensate response modifies the time dependence of the gravitational potentials and hence the integrated Sachs–Wolfe (ISW) channel. Using a representative value within the retained joint band $\varepsilon \in [0.0296, 0.0376]$ of the technical preprint, a semi-analytic ISW–tracer overlap proxy yields a modest enhancement of about 6% in amplitude ($A_{\rm proxy} = 1.060$) relative to ΛCDM. An internal source-power diagnostic gives $\int \mathcal{F}^2_{\rm ECT}\, da / \int \mathcal{F}^2_{\Lambda\rm CDM}\, da = 1.108$, corresponding to an $\sim 11\%$ enhancement of the semi-analytic ISW source kernel. This retained-band representative indicator is not obviously in tension with the broad range of published ISW amplitude measurements, but the comparison remains approximate: the present treatment uses an overlap proxy rather than a full C_ℓ^{Tg} observable, and a non-Limber Boltzmann-level calculation with realistic tracer windows remains future work. Any additional slip contribution from $\Theta_{\mu\nu}[n]$ is illustrated only through phenomenological ansätze and is not yet a derived ECT prediction. The full derivation, diagnostic figures, representative numerical values, and limitations are documented in the ISW subsection and ISW appendix of the technical preprint.

54. Colour completion: a consistency requirement

The current ECT derivations impose a reverse-engineered constraint on the medium: any viable completion must contain an exact unbroken colour sector. A strict obstruction excludes colour-carrying condensates: any condensate transforming nontrivially under an internal group G

breaks G to its stabilizer. Therefore the condensate must remain a colour singlet, and colour must reside in a separate internal sector.

A complementary algebraic obstruction reinforces this conclusion: $\dim \mathfrak{su}(3) = 8 > 6 = \dim \mathfrak{so}(4)$, so the colour algebra cannot even fit inside $\mathfrak{so}(4)$ as a subalgebra. The electroweak-type $U(1)\times SU(2)$ is the maximal gauge-like structure naturally accessible from the minimal condensate symmetry pattern; its full chiral local Standard-Model realisation nevertheless remains open.

The primary candidate completion assumes the medium carries a local complex rank-3 internal module with $SU(3)$-structure: a Hermitian metric and a distinguished antisymmetric trilinear form ε_{abc}. Local non-rigidity of this module (extending the P5 logic) yields $SU(3)$ gauge redundancy. The preserved ε_{abc} has a direct counterpart in observed physics: it is the tensor from which colour-neutral baryonic channels are constructed.

A deeper alternative would be an internal G_2-structured fibre, where $SU(3)$ would arise as the stabilizer of a unit internal direction: $\mathrm{Stab}_{G_2}(u) = SU(3)$. This would explain *why* the colour group is exactly $SU(3)$, but requires exceptional internal geometry not otherwise forced by the postulates.

The postulate P7. In the main preprint the primary completion described above is elevated to an explicit reverse-engineered postulate, called *P7*. Stated in physical language: in addition to the six primary postulates of ECT, the medium carries at each spacetime point a private internal complex "colour cell" of three equivalent degrees of freedom, equipped with two structural data — a Hermitian metric that makes it meaningful to speak about complex lengths and angles in the cell, and a fixed antisymmetric trilinear invariant that distinguishes a definite colour-singlet three-channel combination. Physically meaningless local rearrangements of this internal cell that preserve both pieces of data form precisely the group $SU(3)$; this is the origin of exact unbroken colour. The ordered condensate itself is left untouched by the cell and remains a colour singlet — colour lives in this extra internal sector, not inside the condensate.

P7 is honestly labelled a reverse-engineered postulate: it is extracted from consistency with observed strong-interaction physics rather than derived from the primary postulates alone. It is the minimal structural addition required for a viable exact-colour completion, given the two obstruction theorems above.

What follows structurally from P7. Once P7 is accepted, several things become structurally fixed without any new dynamical input.

First, there is no room for a classical gluon mass term: exact local colour forbids such a term as in standard Yang–Mills. Second, the condensate itself cannot play the role of a colour Higgs field, because P7 requires the condensate to be a colour singlet. Third, a naive "oscillator tower" reading of internal excitations with evenly spaced masses cannot be identified with the physical colour spectrum — glueball masses must come from nonperturbative dynamics, not from quantising an internal mode. These three statements are obstruction theorems at the structural level.

A central negative result. Perhaps the most important outcome of the Part IV analysis is honestly negative: *at the level of constant background couplings, the colour-completion layer does not by itself generate new predictive strong-sector physics.* Everything reduces to standard $SU(3)$

Yang–Mills with a redefined matching convention. Any genuinely new strong-sector content of ECT would have to come from condensate-dependent couplings or from a direct derivation of matching data from the primary postulates; at present neither is available. The *admissible form* of the colour infrared scale under P7 is therefore not a prediction of ECT but a classificatory statement: it records the shape that any P7-compatible strong scale must take, and at $Z_0 \to 1$ it collapses back to the ordinary QCD formula.

The anomaly architecture under P7. Adding coloured matter requires standard quantum-consistency conditions: triangle and mixed anomalies must cancel, and the Witten global $SU(2)$ anomaly must vanish (which it does for three generations and twelve weak doublets). Baryon number, in a quantum-gravity-consistent framework, must appear as a selection rule rather than a fundamental exact symmetry; this is compatible with the companion's proton-stability discussion. Under these constraints, the Standard-Model gauge structure is a *natural* chiral-compatible extension candidate, with hypercharges fixed up to normalisation by anomaly cancellation once the one-generation matter content is assumed; the global-quotient colour-interlock congruence recorded in the preprint further constrains the admissible normalisation choices to a discrete family. ECT does not claim that the SM gauge group is the unique solution.

Strong CP reopened, not resolved. P7 allows a topological term of the $F\tilde{F}$ form, and its constant-background value reproduces the ordinary theta-angle problem constrained by the neutron electric dipole moment to be below about 10^{-10}. ECT with P7 therefore *reopens* the strong-CP problem rather than resolving it. Three possible resolution routes are identified honestly — a CP-symmetric ordered background (would require new structural input), an axion-like relaxation through a new pseudoscalar mode (not available in the current minimal setup), and a geometric obstruction from the complex volume form Ω of the internal cell (conceptually attractive but not yet shown) — all currently open. The η' mass through the Witten–Veneziano relation remains the cleanest strong-sector channel against which any ECT-specific topological physics must be checked.

Three distinct questions about the strong sector. It is useful to keep three different questions apart. First, does the colour sector possess an infrared mass scale Λ_{col}? Yes, inherited from standard QCD phenomenology under the admissible EFT. Second, does the colour sector exhibit a gauge-invariant mass gap — no physical state below a finite threshold? Here the preprint goes further than a bare "not proven": it separates the *physical* question (why the real strong sector is gapped) from the *mathematical* Clay question (constructing a continuum Yang–Mills theory on flat spacetime), and gives a conditional theorem for the physical one. If the coarse-grained pure colour sector lands in the polymer-convergence domain of compact transfer-positive $SU(3)$, then a positive gap of order Λ_{col} follows by standard results; the remaining gap in the argument has been narrowed to a pair of sharply stated technical conditions — preservation of the compact positive local class under blocking, and a polymer-activity bound — and, within a fixed heat-kernel blocking scheme, further to a single one-sided anisotropy budget for the colour cone; in its delayed-entry existence form this budget is satisfied by any finite order-unity anisotropy (a conditional Level B criterion), so the exact matching value of the anisotropy is an optional sharpening rather than a required input. Third, can the full glueball spectrum be derived from ECT? No: this is a quantitative dynamical question inaccessible to reverse analysis. As for the Clay Millennium problem itself, its central object — a fundamental continuum Yang–Mills theory on flat spacetime, with Lorentz invariance built in from the start — is not a fundamental object of ECT at all, since in ECT spacetime and Lorentz symmetry are emergent. ECT therefore does not solve the Clay problem in its literal form and does not dismiss it; it shows that the Clay object is not fundamental here, and it reformulates the physical content as the colour-correlation-length question above.

Non-replacement statement. Part IV of the preprint does not replace the existing forward routes to chirality and to three generations in Parts I–III; it clarifies which exact-colour completions are compatible with those routes and which unresolved items remain blocked even after P7.

Neither P7 nor the deeper G_2 alternative yet derives full QCD: coupling g_s, confinement, the exact glueball spectrum, quark representations as theorems, and θ_{QCD} remain open.

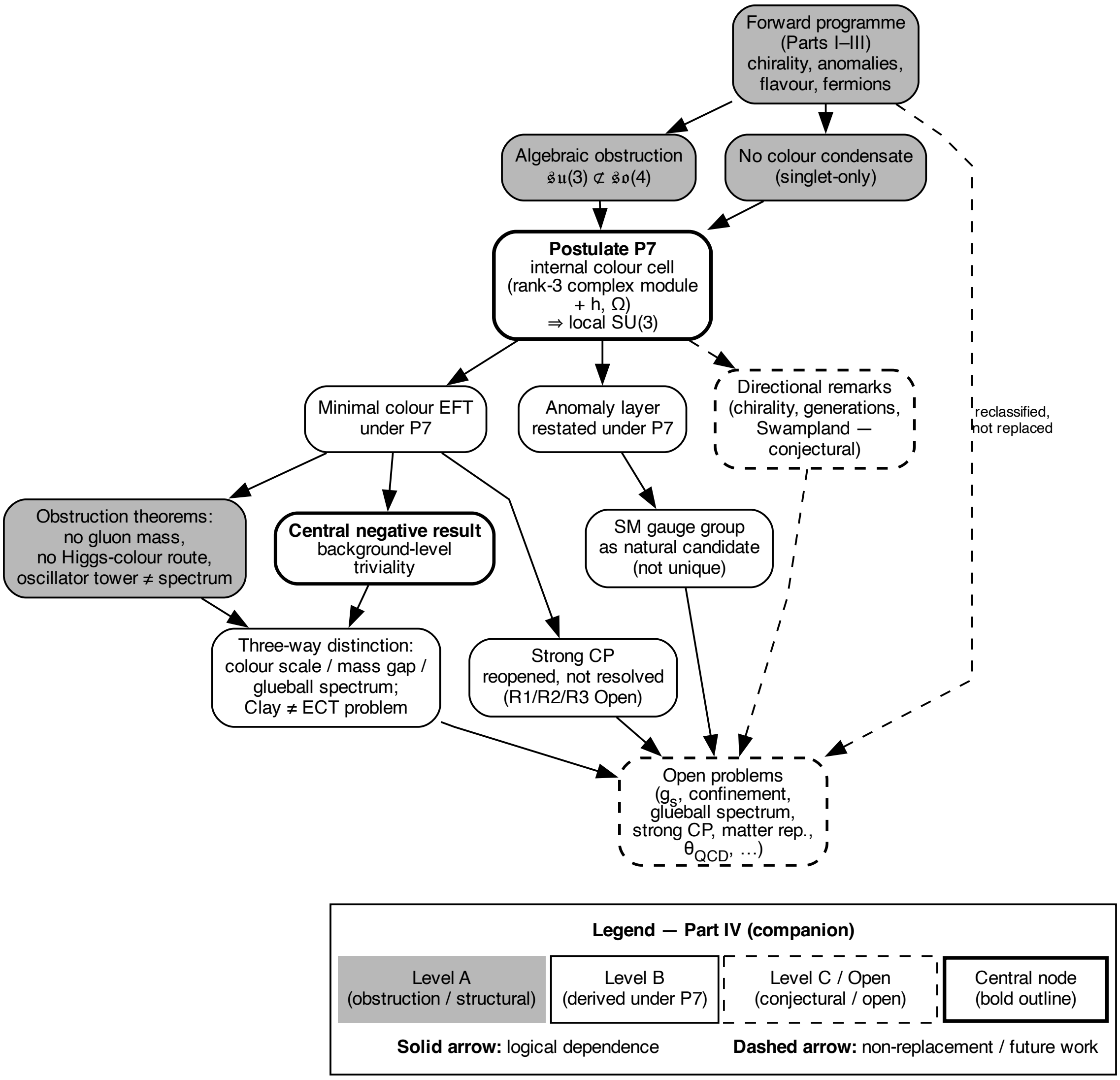


Figure 25: Part IV derivation logic at a glance. Starting from the forward programme of Parts I–III, two obstruction theorems force the externalisation of colour; the reverse-engineered postulate P7 equips the medium with an internal colour cell carrying exact local $SU(3)$. Two branches follow: a minimal admissible colour effective theory (with obstruction theorems and a central negative result — background-level triviality) and an anomaly layer yielding the Standard-Model gauge group as a natural candidate. The three-way distinction (colour scale vs mass gap vs glueball spectrum) is sharpened in the preprint into a separation of the physical mass-gap question, for which a conditional theorem is given, from the mathematical Clay problem, whose continuum object is shown not to be fundamental in ECT; the strong-CP problem is reopened rather than resolved. Apart from one explicitly marked Level A meta-statement (that the Clay continuum object is not a fundamental object of ECT), everything in Part IV is Level B or C.

Table 4: Key status changes under the primary colour completion.

Item	Before	After	Still open
Colour symmetry	Missing	$SU(3)$ available	Dynamical realisation
Gauge bosons	4 of 12	$8+3+1=12$ at symmetry level	Full dynamics
Quark/lepton pattern	Inaccessible	Structurally compatible	Chirality, reps
Meson/baryon channels	Inaccessible	Singlet channels available	Bound-state dynamics
α_s, Λ_{QCD}	Undefined	Definable targets	Not derived
Proton mass ($\sim$938 MeV)	Inaccessible	Accessible in principle	Binding dynamics
α_{fs} normalisation	Quark loops undefined	Z_A strengthened	Full calculation open
Baryogenesis (η_B)	Baryon number incomplete	Colour defines baryon number	Transport open
Proton stability (τ_p)	Selection rules not formulable	Three patterns (I/II/III) formulable	Pattern identification, channels open

55. Open problems

These are stated explicitly as unsolved targets:

1. **Three fermion generations.** Qualitative routes exist (topological sectors of S^3); rigorous derivation of exactly three is not achieved.
2. **Yukawa hierarchy.** Mass differences from $m_e \sim 0.5\,\text{MeV}$ to $m_t \sim 173\,\text{GeV}$ not derived.
3. **$\alpha = 1/137$.** No first-principles prediction.
4. **Renormalisability.** Probable on fixed Euclidean background; not proven.
5. **Nonlinear GEE.** Structurally motivated (Level B); full mathematical rigour not achieved.
6. **Born rule.** Born-type weighting route via RP/Gleason/decoherence assumptions (*B/Open*); exact first-principles Born rule open.
7. **Bell correlators.** Medium-based entanglement established; explicit correlators not derived.
8. **BH evaporation.** Page curve from toy model (Level B); full dynamics open.
9. **Full observed electric-charge pattern.** ECT now supports an Abelian charge lattice from the compact condensate phase, but the derivation of the full Standard-Model charge spectrum, hypercharge quantisation, anomaly-consistent representations, and the global gauge-group quotient remains open; the recorded colour-interlock congruence constrains admissible assignments to an explicitly characterised discrete family (a consistency requirement on completions, not a derivation).

10. **Holographic bound and strong holography.** ECT supports a medium-native area-law entropy bound (Level A for scaling; Level B for coefficient), but the exact microscopic coefficient, the treatment of soft/gauge-edge sectors, a covariant Bousso-type formulation, and full bulk-boundary reconstruction remain open.

11. **High-z background.** Self-consistent $\phi_b(z)$ at $z \gg 1$ not derived.

12. **Gradient condensation (OP-grad).** Dynamical origin of ordered branch from bare P3: the deepest open problem. Recent sharpening localises the remaining gap (OP3.1a) to the constraint-relaxation/coarse-graining bridge, with the EFT-data derivation (OP3.1b) and the measure-concentration selection step (OP3.1c) recorded as separate subproblems; the computed candidate bridges are excluded as completions.

13. **Graviton-loop power counting (OP-UV1).** Explicit one-loop graviton power counting in the ECT condensate background. Structurally expected to be tame (Landau pole far above EFT threshold), but not yet computed.

14. **Two-loop RG of the condensate–graviton system (OP-UV3).** Full RG flow including condensate–graviton mixing; needed to establish the UV behaviour beyond the scalar closure.

15. **OS reconstruction for spin-2 sector (OP-OS1).** Reflection positivity for the orientation (tensor) sector. If established, graviton states appear automatically in the Hilbert space without a separate quantisation act.

16. **Numerical value of ρ_Λ (reformulated OP-Λ3).** The direct classical vacuum-baseline component of the 120-orders cosmological-constant *catastrophe* is blocked at Level A by the emergent-unimodular decoupling theorem: because the kinetic-tensor determinant $\sqrt{-\det K^{AB}} = \beta^2/c_*$ is fixed, the classical vacuum offset $V(\phi_0)$ does not source emergent gravity. The zero-derivative loop contribution decouples only under condition H_Λ. What remains open is the narrower Level C question of the observed *value* ρ_Λ: its natural scaling $\sim M_{\rm Pl}^2 H_0^2$ matches observation at order-of-magnitude, but the dimensionless coefficient is not yet computed from first principles. A possible pseudo-Goldstone realisation may exist if an ultralight pseudo-Goldstone sector with $m_G \sim H_0$ and decay-constant scale of order ϕ_0 is realised in the cosmological regime.

17. **Proton stability and baryon-number selection rules (OP-proton).** Derivation of which baryon-number selection-rule pattern (full protection / partial $\Delta B = 2$ only / generic $\Delta B = 1$) is realised in the completed gauge/flavour sector; if generic violation, identification of decay channels and coefficient structure.

18. **Origin of neutrino masses (OP-neutrino).** The present formulation does not derive the absolute neutrino mass scale, the Dirac/Majorana character, the PMNS matrix, or the mass ordering. Two ECT-specific scale observations ($M_R^{\rm geom}$ and $m_\nu^{\rm geom}$) show that the condensate hierarchy naturally provides scale combinations close to the observed neutrino range, but neither is a first-principles derivation. Any future derivation of neutrino masses from condensate dynamics would simultaneously constrain the leptogenesis benchmark and the electroweak input status of v_2.

19. **Gauge-sector completion.** The compact phase fibre gives a strong structural route to an Abelian gauge sector. What remains open is its exact placement in the electroweak

chain, the full internal derivation of the weak sector, the construction of the colour sector from extended internal condensate structure, and the derivation of the observed matter representations.

20. **Charge quantisation (OP-GUT3).** If the global electroweak structure group is $U(2) = (SU(2) \times U(1))/\mathbb{Z}_2$, representations satisfy $2T + Y \in 2\mathbb{Z}$; with the colour sector included, the recorded $\mathbb{Z}_6$ interlock congruence further fixes the fractional electric pattern per colour class. What remains open is the selection within the resulting explicitly characterised discrete family of charge assignments.

21. **Anomaly cancellation (OP-GUT6).** Any chiral $SU(2)$ sector must be checked for both local triangle anomalies and the global Witten $\mathbb{Z}_2$ anomaly. These become sharp consistency tests once the fermion spectrum is derived. Besides the usual local triangle anomalies, any chiral $SU(2)$ completion must also pass the global Witten $\mathbb{Z}_2$ anomaly test, which excludes an odd number of left-handed $SU(2)$ doublets—a condition that becomes a sharp consistency constraint once the emergent fermion spectrum is fixed. A useful structural payoff of the discrete-symmetry analysis is that the preferred-direction coupling $\mathcal{L}_5$, being a vector bilinear, does not contribute to chiral anomaly coefficients and is therefore compatible with the standard anomaly-consistency requirements of the emergent chiral gauge sector.

22. **Weinberg angle (OP-GUT7).** $\sin^2\theta_W$ from condensate parameters: not yet derived.

23. **Director coupling coefficient (OP-GUT8).** Microscopic derivation of μ_5 from condensate microdynamics and establishment of radiative stability.

24. **Full weak-gauge derivation (OP-GRAV-WEAK).** Whether the oriented connection sector contains independent spin-1 modes beyond those fixed by the metric, which could underlie the weak interaction as a geometric rather than purely internal gauge field.

56. Complete derivation map

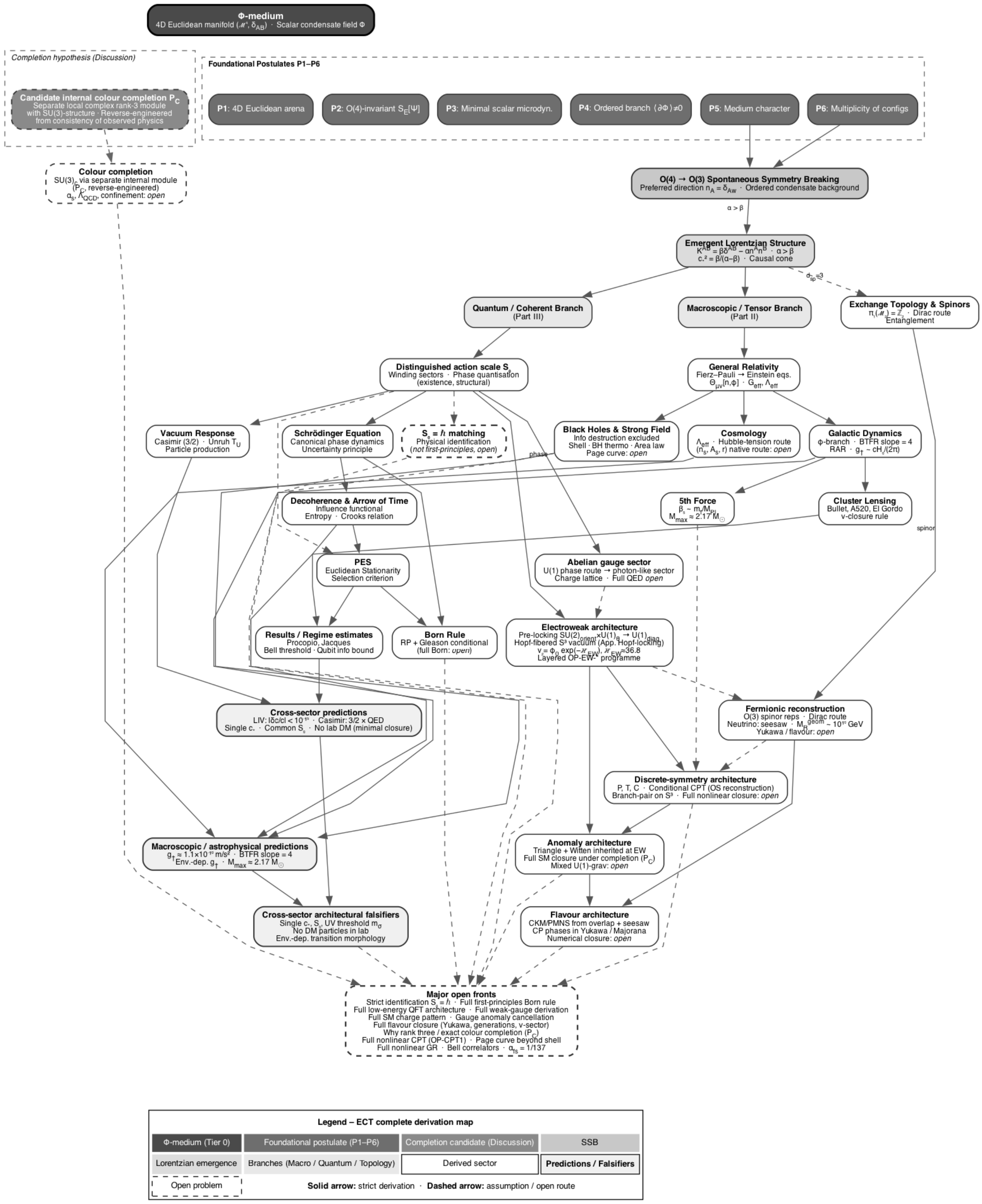


Figure 26: Complete ECT derivation map. The full logical architecture from the Φ-medium and the six postulates through Parts I–IV and the Summary Part. Colours indicate postulates (dark), phenomenological inputs (grey), Level A (light grey), Level B (white), and Open (dashed).

Figure 26 shows the complete derivation architecture of ECT.

57. Open problems of physics addressed within ECT

Within its framework, ECT addresses several well-known open problems of modern physics:

1. **Unification of quantum mechanics and gravity.** Both sectors are traced to the same ordered condensate architecture: gravity to the geometric/orientation branch and quantum mechanics to the coherent branch. This structurally reframes the unification problem, but the full UV/spin-2 quantum-gravity closure remains open (*B/Open*).

2. **The origin of time.** Why does spacetime have signature $(-,+,+,+)$? Derived from $O(4) \to O(3)$ SSB (A).

3. **The dark matter problem.** No particle dark matter has been detected despite decades of searches. In its present closure, ECT reinterprets the galactic mass-discrepancy phenomenology—rotation curves, BTFR/RAR behaviour, and related galaxy-scale observables—as condensate response rather than as a particle-halo carrier. This does not by itself exclude hidden-sector particles; it says that the current ECT galactic channel does not require them as the carrier of the observed galaxy-scale discrepancy (B).

4. **The cosmological constant problem.** Standard quantum field theory suggests a vacuum scale roughly 10^{120} times larger than the observed late-time dark-energy density. ECT separates this into two distinct questions. First, the *direct catastrophe itself* is blocked in the leading ordered-branch sector: because the kinetic-tensor determinant $\sqrt{-\det K^{AB}} = \beta^2/c_*$ is kinematically fixed by $n^A n_A = 1$, a classical vacuum offset $V(\phi_0)$ does not source emergent gravity. Under the spectral condition H_Λ, the same statement extends to the zero-derivative one-loop vacuum term. Second, the *observed value* of ρ_Λ remains open: its natural ECT scaling is of order $M_{\rm Pl}^2 H_0^2$, but the dimensionless coefficient is not yet derived from first principles. In this sense, ECT gives a Level A blocking of the direct vacuum-baseline channel while leaving the loop-vacuum extension H_Λ-conditional and the numerical value as a Level C infrared problem.

5. **The Hubble tension.** A $> 5\sigma$ discrepancy between the local and CMB-inferred values of H_0. In the present diagnostic-layer treatment, a retained five-probe effective band, $\varepsilon \in [0.0296, 0.0376]$ at 1σ, is compatible with the observed upward H_0 shift at the baseline-background mapping level. This should not yet be read as a completed early+late CMB-pipeline resolution; the closure-level derivation of $\varepsilon(z)$ remains open (*sign: A; magnitude: B, diagnostic-layer indicator*).

6. **JWST early galaxies.** Enhanced early gravity + $\Theta_{\mu\nu}$ accelerate structure formation (B).

7. **Baryon asymmetry.** A benchmark leptogenesis closure yields $\eta_B \sim 9 \times 10^{-10}$ versus the observed $\sim 6 \times 10^{-10}$; this is a closure-level result, not yet a strict first-principles theorem (B).

8. **Black-hole information paradox.** Fundamental information destruction is excluded at the condensate-state level; shell closure and Page-curve-like behaviour are Level B; a complete constructive evaporation solution remains open. General area-law entropy bound established (*A/B*).

9. **The measurement problem.** Decoherence and PES provide a structural route to interference suppression and classical record formation without a fundamental observer-induced collapse postulate. However, the exact Born rule, pointer-basis selection, single-outcome selection, and a complete measurement-update theorem remain open (*B/Open*; complete closure open).

10. **The origin of quantum mechanics.** ECT provides a structural route to Hilbert-space structure, superposition, and quantum dynamics from the condensate. The Born rule is conditionally unique only after the RP/Gleason/decoherence assumptions are in place, and is not yet a first-principles theorem (*A/B for the structural route; B/Open for Born-type weighting; full closure open*).

11. **The arrow of time.** The retarded causal/decoherence backbone follows from ordered-branch structure, while the explicit monotonic entropy-growth formula and full second-law relation require the Level B influence-functional/environmental closure (*A for the backbone; B for the full monotonicity formula*).

12. **The need for inflation.** ECT provides structural non-inflationary routes to the horizon, flatness, primary-monopole, and near-scale-invariance problems, while the branch-selection probability and full quantitative spectrum remain open (*A/B/Open*).

13. **The hierarchy problem.** The ratio $M_{\text{Pl}}/M_{\text{EW}} \sim 10^{16}$ is logarithmically equivalent to $\mathcal{I}_{\text{EW}} \approx 36.8$, of the same order as the analogous QCD logarithm. The technical preprint reduces OP-EW-scale to a non-polynomial scale-generation problem with four candidate templates (asymptotic-freedom-like running, walking/near-conformal dynamics, Miransky/BKT scaling, and topological/Hopf-like routes). This is a reduction, not a solution: the broader programme decomposes into OP-EW-scale, OP-EW-locking, OP-EW-gauge, OP-EW-naturalness, and OP-EW-matter. The scale problem is sharply reformulated; the first-principles dynamics and the full electroweak completion remain open (*B/Open*).

14. **Why $3+1$ dimensions and why no fundamental anyons?** Given the P1 postulate of a four-dimensional Euclidean arena, the $O(4) \to O(3)$ transition singles out one temporal seed and three spatial directions. The deeper origin of four-dimensionality itself remains OP0. In the resulting three-dimensional spatial branch, the exchange topology permits only the ordinary boson/fermion split; fundamental anyons are excluded in the primary branch (*A conditional on P1; OP0 open*).

15. **Why no primary ordering monopoles?** $\pi_2(S^3) = 0$ for the primary $O(4) \to O(3)$ vacuum manifold, so the ordering transition does not generate topologically stable primary monopoles. This does not exclude monopole-like objects in later gauge completion sectors, whose topology and abundance remain open (*A/Open*).

16. **Vacuum stability.** In the primary condensate sector, the ordered branch is locally stable at tree level, and the standard one-loop scalar running closes the Standard-Model-type high-field metastability channel in the primary amplitude sector: here $\beta_\lambda > 0$, unlike the Higgs sector of the Standard Model where the top-Yukawa contribution drives metastability. The symmetric point $\Phi = 0$ is a hilltop, not a metastable minimum, so Coleman false-vacuum decay does not apply to the minimal primary potential. The full stability of the complete ordered branch architecture, and the question of whether a secondary embedded electroweak sector inherits this stability, remain open (*B/Open*).

Vacuum decay as Euclidean weighting. In ordinary language, false-vacuum decay sounds like a field sitting in one state and then tunnelling to another state in time. ECT reads this more carefully. The primitive object is a four-dimensional Euclidean condensate configuration, not a field evolving in an external clock time. The Coleman bounce remains the correct semiclassical tool when its assumptions apply, but in ECT it is interpreted as a Euclidean saddle controlling a relative weight or effective transition amplitude between admissible configurations. In the Lorentzian branch this may be reconstructed as a decay rate; at the deeper ECT level it is a statement about Euclidean configuration weighting. For the minimal primary condensate potential, the false-vacuum structure required by Coleman decay is absent.

17. **Quantum gravity.** ECT offers a structural route in which gravity appears as a condensate response and the graviton as an excitation of the ordered medium on a Euclidean background. This reframes the architecture of quantum gravity, but the full UV/spin-2 closure remains open (*B/Open*).

18. **Cosmological coincidence question.** The "why now" question — why Ω_m and Ω_Λ are of comparable order today — is debated in the literature, since in pure ΛCDM the order-unity window of ρ_m/ρ_Λ already spans a sizeable low-redshift range, and its force depends on measure and observer-selection assumptions. In ECT dark energy is supplied by the slow-rolling amplitude of the same ordered condensate that generates G_N, c_* and $\hbar$, which makes the matter-to-dark-energy ratio at least a dynamically defined quantity rather than a strictly rigid external constant. In the current calibrated thawing benchmark, however, the condensate dark-energy density varies by only $\approx$11% between the present epoch and the asymptotic past, and the corresponding order-unity window in ρ_m/ρ_ϕ is only marginally wider than in pure ΛCDM. We therefore do not list coincidence alleviation among current ECT results; it remains an open programmatic question, distinct from both the Level A blocking of the direct vacuum-baseline channel and the separate Level C question of the observed ρ_Λ value (*Level B/C-Open*).

19. **Massless spin-2 sector.** The spin-2 mode emerges from SSB with $m_g = 0$ in the linear Fierz–Pauli closure of the effective-metric channel, consistent with the LIGO bound $m_g < 1.2 \times 10^{-22}$ eV; exact non-perturbative proof open (*B/Open*).

20. **Why are c, G_N, and $\hbar$ not independent primitives?** In ECT these constants are tied to the same condensate architecture rather than introduced as unrelated fundamental inputs: c_* follows from the ordered-branch kinetic structure, the empirical identification $c_* = c$ is a matching step, G_N is fixed through the effective gravitational-stiffness matching, and $\hbar$ is identified with the coherent action scale S_0 through a Level B matching relation (*mixed A/B status*).

21. **Why do Casimir, Unruh, Hawking, and particle-production effects belong to one family?** ECT treats them as different manifestations of a single ordered-vacuum response logic rather than as unrelated effects (*A/B*).

22. **Why does smooth gravity not weaken Pauli exclusion?** In the smooth coherent branch, condensate inhomogeneity changes local fermionic dynamics but not the exclusion algebra after canonical normalisation; any non-standard statistics would have to come from more exceptional sectors (*B*).

23. **Fine-structure constant.** After colour completion the fine-structure problem is recast as a concrete electroweak kinetic/mixing target rather than a purely formal route: external one-loop benchmark running points to moderate ultraviolet electroweak gauge-sector normalisations, structurally tied to the same condensate architecture that also underlies G_N and $\hbar$, while a full first-principles derivation remains open (B/C).

24. S_8 **/ lensing–growth sector.** ECT has a natural route to the weak-lensing / growth sector through the tensor/orientation stress of the ordered condensate, which can produce a mismatch between the dynamical and lensing gravitational potentials. The relevant future target is the completed set of effective functions $\mu(k,z)$, $\eta(k,z)$, and $\Sigma(k,z)$. Whether this route actually alleviates the presently mixed lensing–growth discrepancy remains open; it is a high-priority discriminator rather than a current ECT result (*Level B/C*).

The breadth of problems addressed from a single starting point is a distinctive feature of the framework. The technical preprint provides a comprehensive table of about seventy structures—from superposition, the Pauli principle, and Born-type probability to Einstein's equations, the Casimir effect, dark-energy phenomenology, and rotation-curve phenomenology—that standard physics treats as independent postulates or empirical inputs, while ECT classifies them as derived structures, reconstruction targets, condensate-response reinterpretations, or explicit closure-level/open problems. A complementary table records ECT-specific predictions and observational handles with no direct analogue in standard physics in the same form.

58. Conclusion

ECT is an attempt to place spacetime, gravity, and quantum behaviour within a single pre-spacetime framework. In its current form, some parts of this programme are structural results, some are closure-dependent effective constructions, and some remain open.

Common pre-geometric architecture of interactions. ECT now supports a hierarchical picture of the interaction sectors: a strong Abelian gauge route from the compact phase fibre; a candidate weak-sector route tied to the oriented broken phase and to Hopf-fibered phase–orientation locking; and a colour sector that, being algebraically inaccessible from $O(4)$ alone ($\dim \mathfrak{su}(3) > \dim \mathfrak{so}(4)$), requires extended internal condensate structure. Together with emergent gravity and the generic director coupling, ECT places the known interaction sectors within a common pre-geometric condensate architecture—a structural notion of unification that, unlike conventional GUT programmes, also attempts to include gravity within the same framework. The sectors do not all have the same derivational status. The full Standard-Model matching (matter representations, charge assignments, anomaly cancellation, Weinberg angle) remains an open programme.

Discrete-symmetry architecture. ECT identifies a layered discrete-symmetry structure of the emergent matter sector: a geometric backbone in which Euclidean reflections of $O(4)$ play the role of emergent parity and the Euclidean antecedent of time reversal (Level A); a conditional CPT route for those local sectors where Osterwalder–Schrader reconstruction is available (Level A/B for free/quadratic sectors, Open for the full nonlinear theory); and a branch-sensitive phenomenological layer in which the preferred-direction operator $\mathcal{L}_5$ is re-classified as a CPT-odd, CP-odd, C-odd, P-even, T-even operator. Three notions that are sometimes conflated are kept explicitly distinct in ECT: the fundamental anti-unitary T on the reconstructed Hilbert space, the effective branch-level T-violation carried by $\mathcal{L}_5$, and the thermodynamic arrow of time emerging through decoherence. The antipodal branch-pair structure of the S^3 vacuum manifold admits a speculative cosmological reading (CPT-symmetric pair with matter-dominated and antimatter-dominated partners) that is recorded as a future direction (Level C) rather than as an established result. A useful structural payoff of the discrete-symmetry analysis is that the preferred-direction coupling $\mathcal{L}_5$, being a vector bilinear, does not contribute to chiral anomaly coefficients and is therefore compatible with the standard anomaly-consistency requirements of the emergent chiral gauge sector.

Flavour architecture. The observed "CKM small, PMNS large" dichotomy admits an ECT structural reading: quark mixing is associated with the overlap-based Yukawa hierarchy, while lepton mixing combines the same overlap mechanism for charged leptons with the separate seesaw embedding for active neutrinos, making larger basis misalignment plausible. Complex flavour phases live in the effective Yukawa/seesaw sector, flavour structure is anomaly-decoupled, and the resulting framework does not spoil the standard leading-order PMNS/MSW phenomenology. Quantitative derivation of mixing angles, CP phases, Majorana phases, and the absolute neutrino spectrum remains part of the open flavour programme.

What is established structurally at the strongest level: Lorentzian signature from $O(4) \to O(3)$ SSB; the existence of a universal ordered-branch causal cone shared by native low-energy sectors, with the empirical identification $c_* = c$ treated as a matching step; the equation hierarchy (Euclidean $\to$ KG $\to$ Schrödinger); the retarded backbone of the arrow of time; and the exclusion of fundamental information destruction at the condensate-state level.

What is structurally derived within closure assumptions: the BTFR slope exactly 4 in the galactic closure; structural non-inflationary routes to the horizon, flatness, primary-monopole, and near-scale-invariance problems; and the shell-model route to black-hole area entropy and Page-curve-like behaviour, with constructive evaporation still open.

What is obtained at effective/closure level (Level B): Newton's constant, Planck's constant, and c as condensate matching relations; one-loop vacuum stability of the primary amplitude sector (no Standard-Model-type metastability; no Coleman false vacuum in the minimal potential); generalised Einstein equations with $\Theta_{\mu\nu}$; dark energy as condensate residual energy ($w_0 \approx -0.83$); a retained five-probe effective band for the cosmological drift parameter, $\varepsilon \in [0.0296, 0.0376]$ at 1σ, with cross-channel compatibility between Hubble + r_s, JWST, cosmic chronometers, $f\sigma_8$, and ISW at the diagnostic-layer level; galaxy rotation curves for 165 SPARC galaxies ($\chi^2/N \approx 2.95$); reinterpretation of the dark sector as condensate dynamics; decoherence and the quantum–classical boundary; BH area entropy and Page-curve-like behaviour from the shell model; PES; boundary-sensitive Casimir vacuum response (conditional).

What remains an open programme: Three fermion generations; Yukawa hierarchy; $\alpha = 1/137$; renormalisability; Born rule as first-principles theorem; Bell correlators; complete BH evaporation; gradient condensation from bare P3; proton stability as a UV discriminant (three selection-rule patterns conditional on gauge/flavour completion). This also includes the full interacting closure of the conditional CPT route, the exact derivation of the maximally parity-violating weak-sector pattern, and the flavour-architecture derivation targets (CKM and PMNS mixing parameters, CP phases, absolute neutrino spectrum) under OP-Yukawa. The constancy of the dimensionless constants α and μ is recorded only as a conditional selective-coupling protection route (OP-CONST); the unconditional microscopic proof remains open and belongs to OP-matter plus OP-grad completion.

Distinctive conceptual achievements (not available in standard physics): The effective metric is derived, not fundamental—the condensate is primary, geometry is secondary. Three logically distinct classes of excitations replace the uniform field-postulate structure of QFT. The axioms of QM are relocated from postulates to reconstruction targets. Event-like interaction is emergent; in the coherent branch, subsystems are related by common-medium correlation. Gravitational decoherence realises a DP-type leading scale within the condensate reduced-state architecture; ECT-specific content lies in the condensate-origin interpretation, absence of a freely fitted collapse scale, and possible subleading corrections. For micron-scale nanospheres the corresponding benchmark timescale is $\tau \sim 38\,\mathrm{ms}$. A recoherence law $P_{\mathrm{back}}/P_{\mathrm{fwd}} = e^{-\Gamma_{\mathrm{loop}}}$ is predicted; standard QM has no analogue. Fock space is a reconstruction target, not a foundational axiom.

Table 5: Distinctive ECT outputs and foundational questions addressed.

Problem / Achievement	ECT mechanism	Level	Unique?
Conceptual achievements			
Metric derived	Condensate primary, geometry secondary	A/B	Yes
Three excitation-sector classification	Condensate scalar sector / gauge-sector realisations / topological sectors; full Standard-Model matter content remains a completion target	A/B/Open	Yes, as classification
QM axioms → reconstruction targets	Hilbert-space and wave-kinematic structures are reconstructed or structurally motivated; Born weights, pointer basis, single outcome, and full measurement-update theory remain B/Open targets	A/B/Open	Yes, with open closures
Interaction = Lorentzian	Coherent branch: correlation only	A	Yes
Fock as reconstruction target	Variable N = condensate reconfiguration	Open	Yes (conceptual)
Foundational questions			
Why 3+1 dimensions?	Given P1, $O(4)\to O(3)$ yields one temporal seed and three spatial directions; deeper origin of four-dimensionality remains OP0	A conditional on P1	Structural
No primary ordering monopoles	$\pi_2(S^3) = 0$ for the primary $O(4) \to O(3)$ vacuum manifold; later gauge-sector monopoles remain completion-dependent	A/Open	Anti-prediction
Vacuum stability	Tree-level local stability; one-loop primary amplitude-sector stability; no Coleman false vacuum in minimal potential	B/Open	Yes
Quantum gravity	Metric quantisation is reframed by condensate response; complete UV quantum-gravity closure remains open	B/Open	Architecture
Massless spin-2 sector	Linear Fierz–Pauli closure of the effective-metric channel; exact non-perturbative proof open	B/Open	Structural
Quantitative predictions			
Grav. decoherence	DP-type leading scale $\tau \sim 38\,\mathrm{ms}$ for micron nanospheres; ECT-specific condensate-origin interpretation	B	Distinctive interpretation, not unique leading scale

In its minimal closure, ECT does not require a particle-DM carrier for the galactic mass-discrepancy channel, a rigid fundamental cosmological constant, an inflaton field, separate quantum axioms, or an ad hoc collapse postulate as independent starting ingredients. With at most four bare parameters—reducible to two in the canonical benchmark—ECT addresses a scope that typically requires many independent inputs in the standard framework, although several low-energy closures remain Level B/C or open derivation targets.

Whether ECT ultimately becomes a viable theory of nature or remains a productive conceptual laboratory, it does something valuable: it shows how spacetime, gravitational response, and quantum-theoretic structure can be placed within a common condensate architecture, with some parts derived, others reconstructed, and others left as explicit closure targets. It identifies the specific structures — gradient condensation, compact-phase topology, orientation stiffness, and the Principle of Euclidean Stationarity — that make this unification programme possible.

If even partly correct, the deepest truth about reality is not that the universe *contains* time, but that under the right conditions, the universe *creates* it.

Note. This article is a narrative companion to: *"Euclidean Condensate Theory (ECT): Emergence of Spacetime, Quantum Mechanics, and Gravity from Spontaneous $O(4)$ Symmetry Breaking"* by Valeriy Blagovidov (2026). Full preprint: doi:10.5281/zenodo.18917930.

`https://github.com/chufelo/ECT-preprint-code`

Zenodo: `https://doi.org/10.5281/zenodo.18917930`